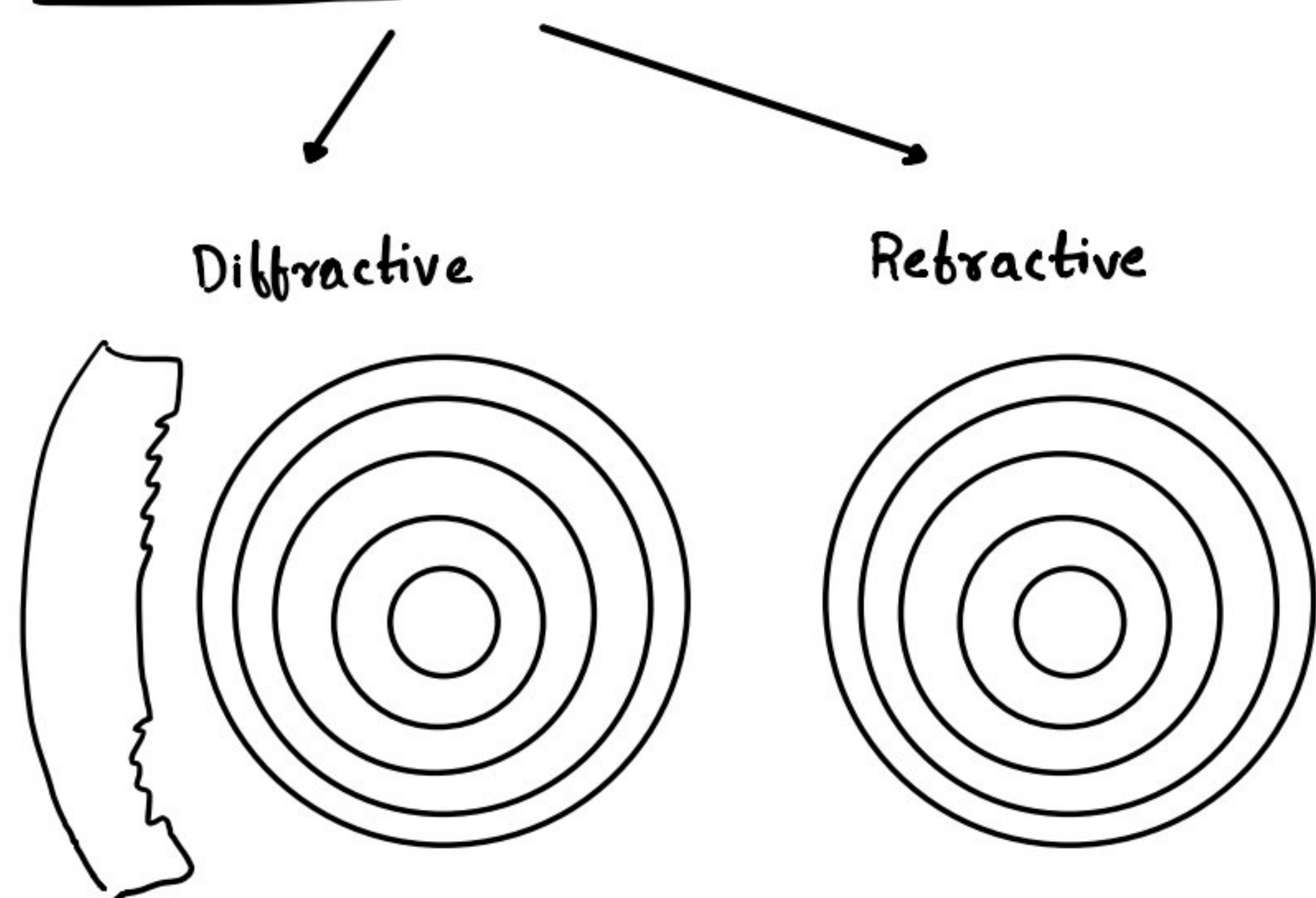
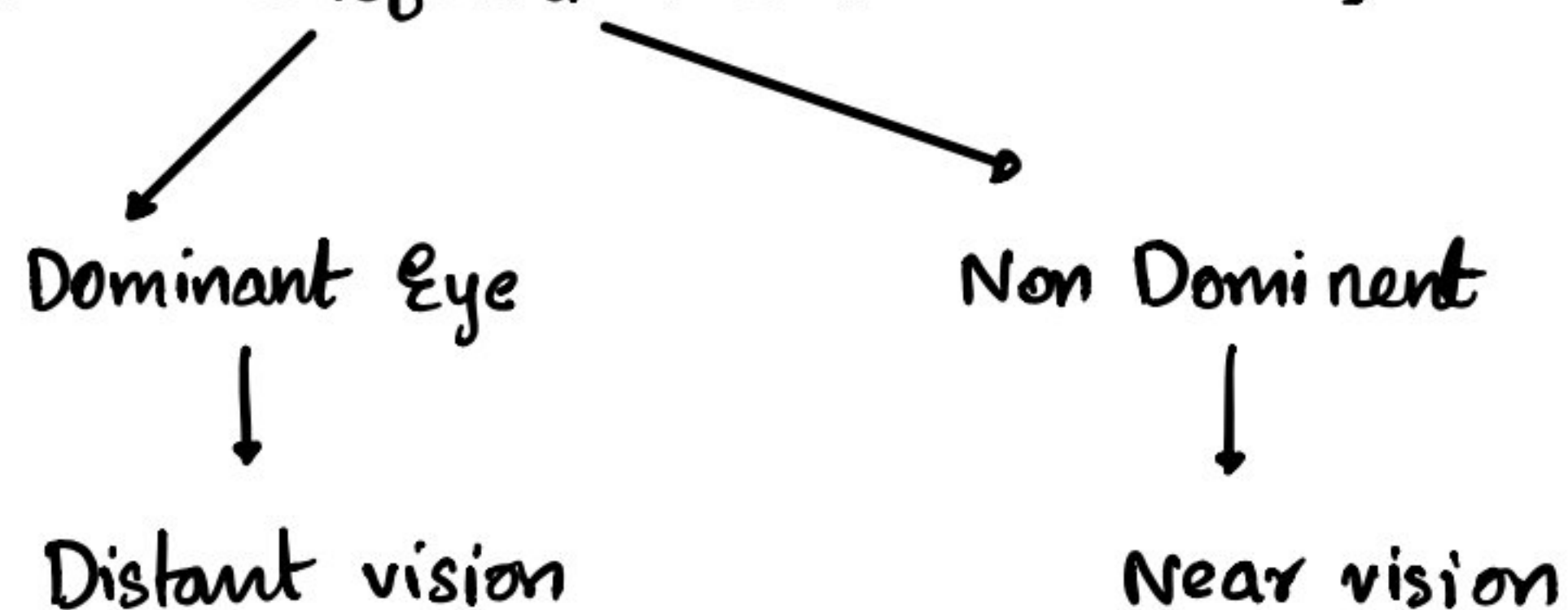


Multifocal IOL:-



Presbyopic correction be done with monofocal IOL & MONOVISION?

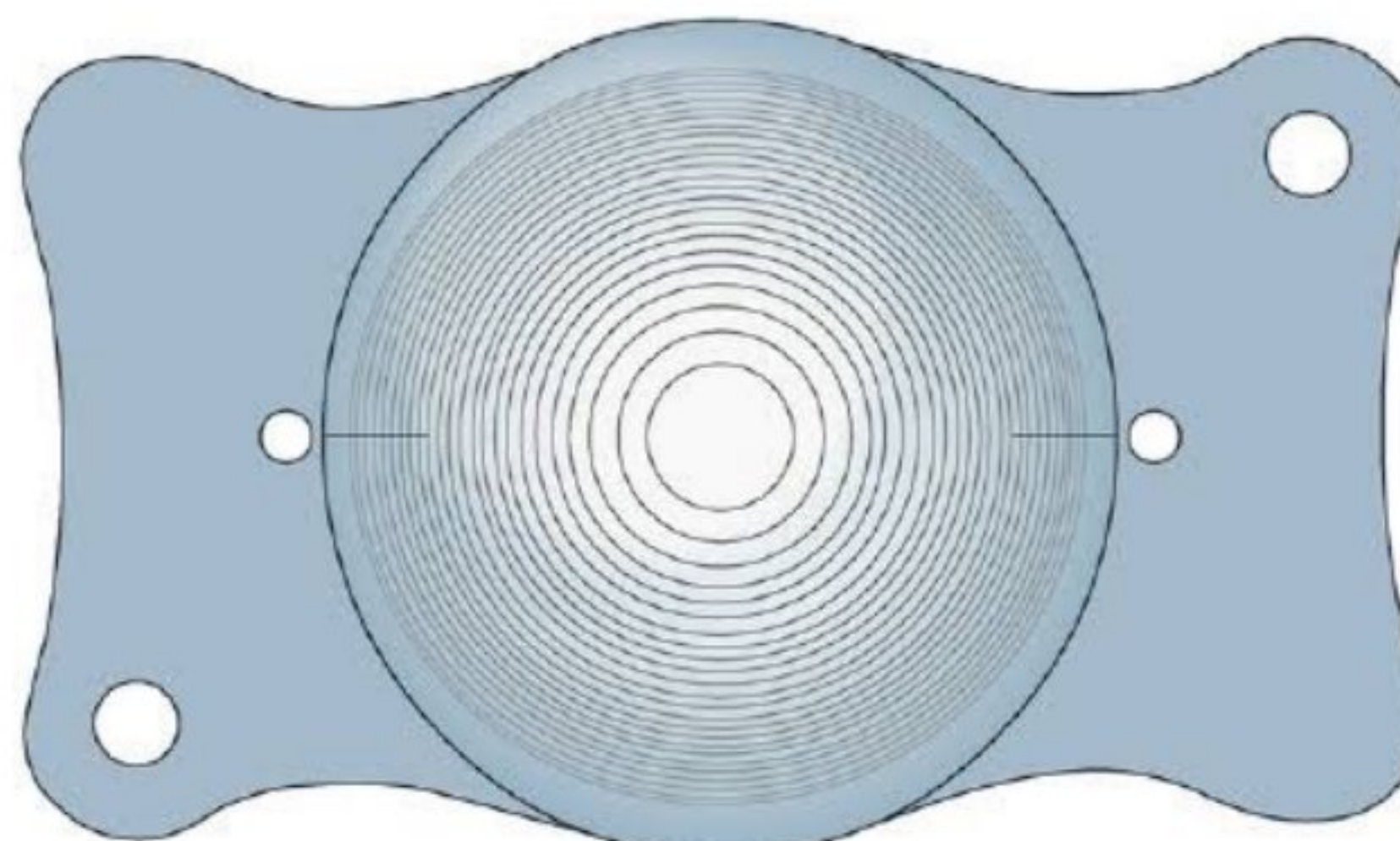


Accommodating IOL ⇒ **CRYSTA LENS**



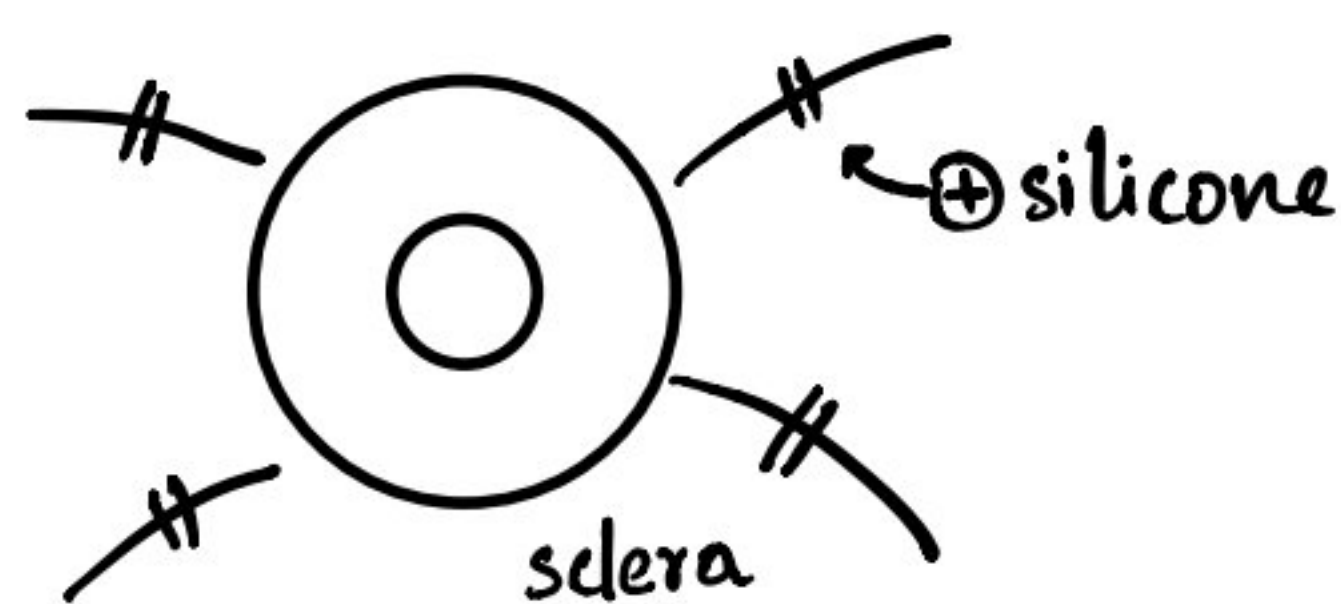
Crysta lens

Toric multifocal IOL → for pre-existing Astigmatism + Presbyopia



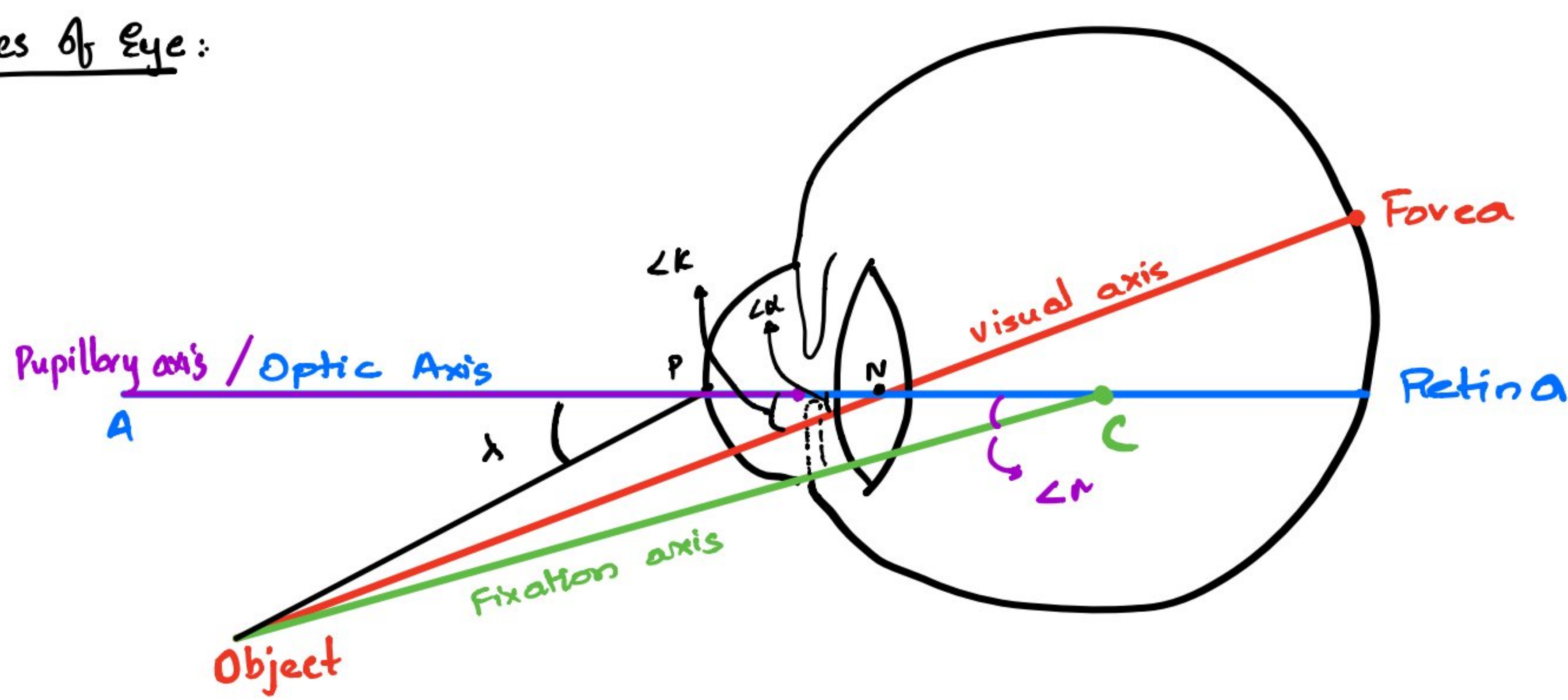
Toric Multifocal IOL

Scleral Expansion Rings → Presbyopia



Laser → Intra Cor Femtosecond laser } Helps in **Bifocal accommodation**

Axes of Eye:



P = Centre of Pupil (cornea)

N = Nodal point

C = Centre of Rotation

O = Fixation point

OP = Pupillary line

Angles of Eye:

$\angle \alpha$ = Angle b/w optical axis & visual axis at nodal point

$\angle \kappa$ {clinical significance} = Angle b/w visual axis and pupillary axis at pupil

Pseudoexotropia

↓
+ve angle kappa

Pseudoesotropia

↓
-ve angle kappa

$\angle \nu$ = Angle b/w fixation axis and optical axis at center of eyeball rotation

$\angle \lambda$ = Line of sight

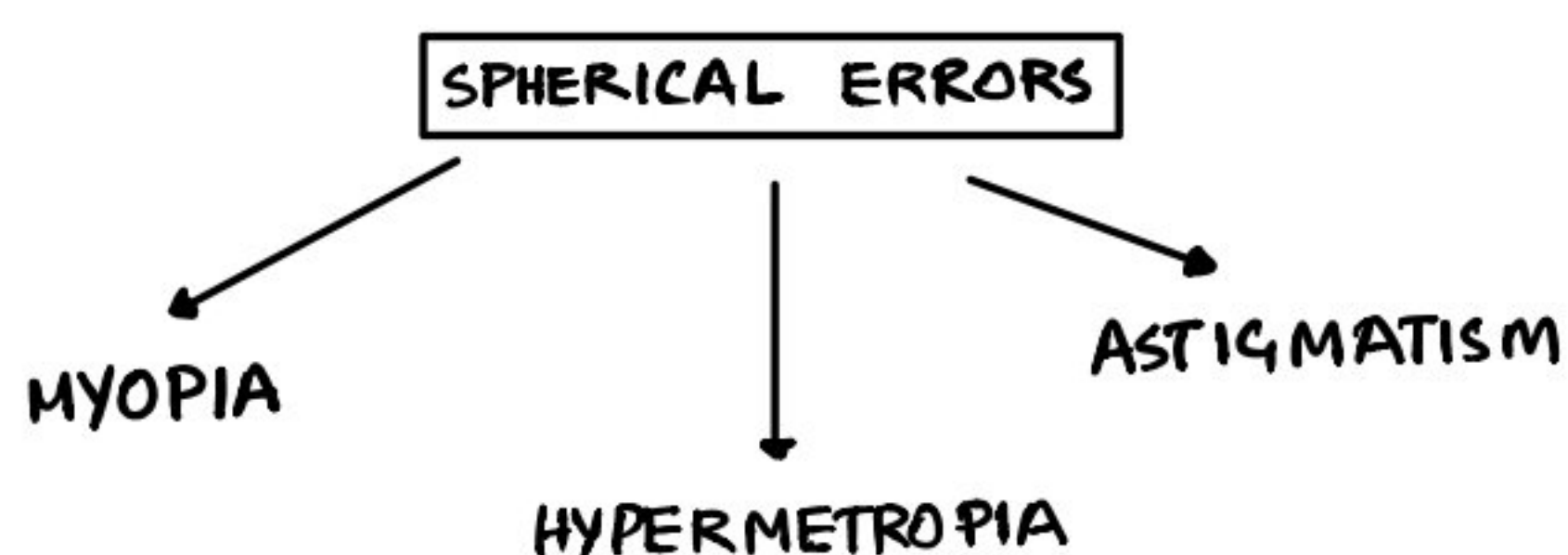
Optic axis: Line passing through the centre of cornea and center of the lens to meet the retina on the nasal side of the fovea.

Visual axis: Line joining the fixation point (object) and the center of rotation

Fixation Axis: Line joining the fixation point {object} and the centre of rotation

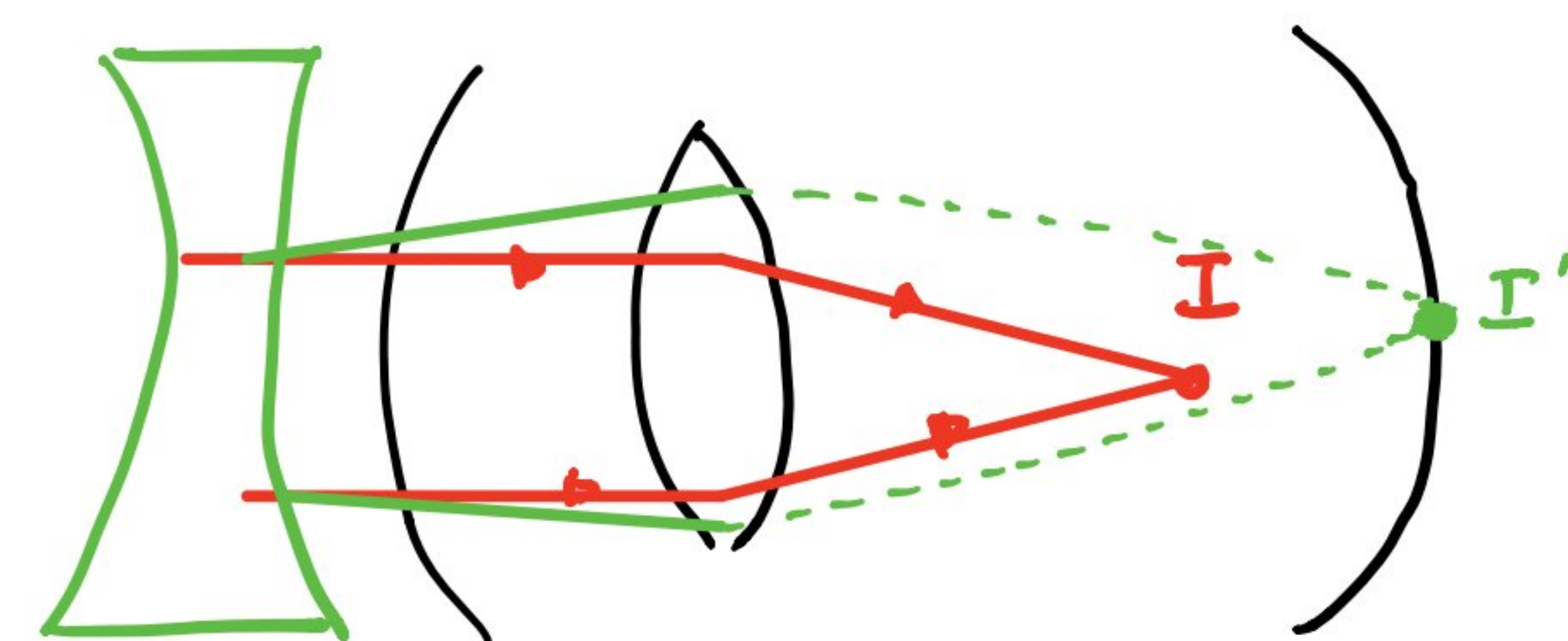
REFRACTIVE ERROR

Ametropia: Image not formed on the Retina.



MYOPIA: {near sightedness / short sight}

• (MC) visually significant Refractive error



concave / -ve sphere $\rightarrow \underline{\underline{-D}}$

NEET PG

Grading of Myopia

Low $\leq -3D$
moderate = $-3D$ to $-6D$
High $\geq -6D$

Properties of Concave sphere:

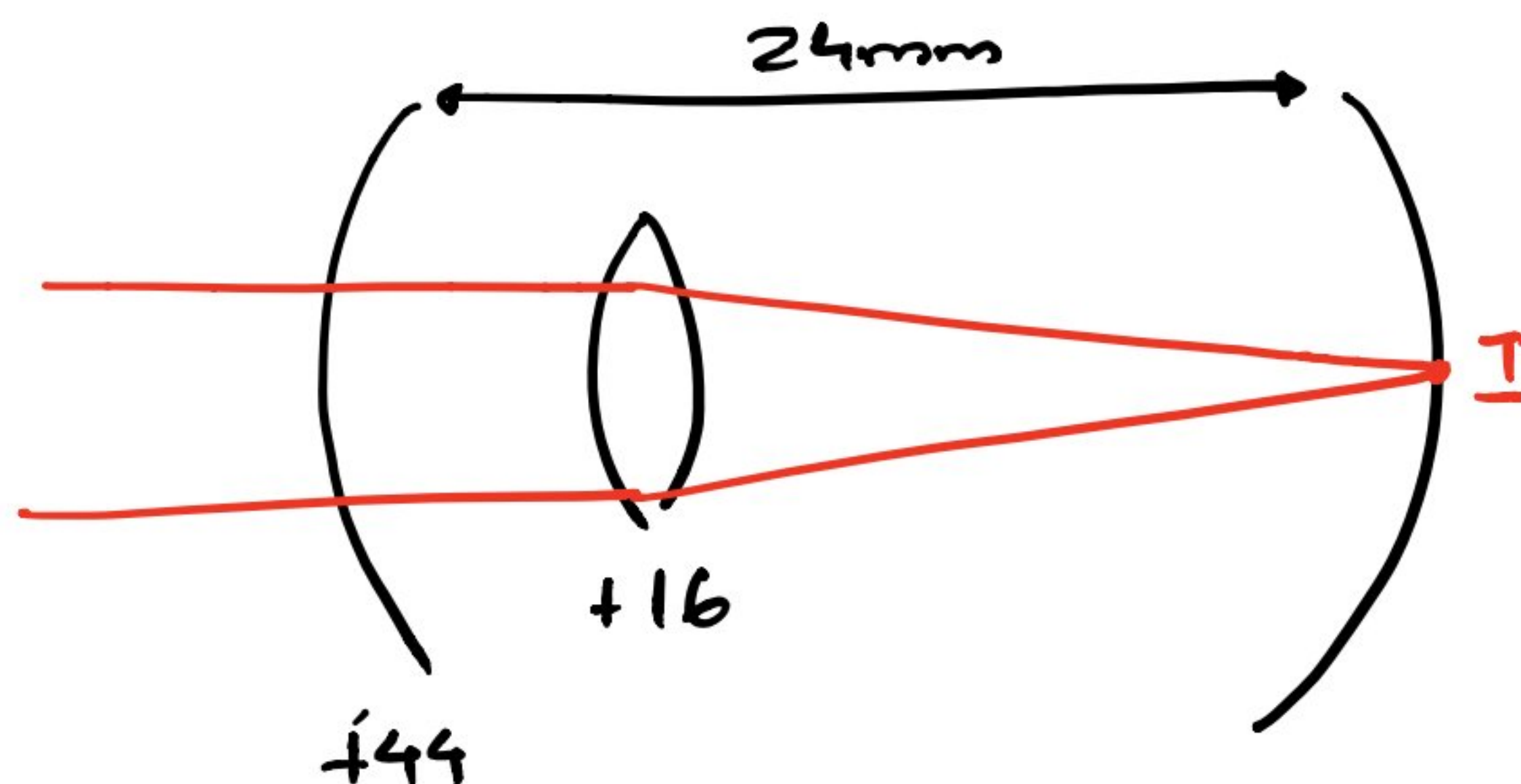
- 1) Thick in periphery and thinner in center
- 2) Minification of Image
- 3) Image moves in the same direction of that of lens movement
- 4) Divergent Lens (C.T.)

Types of Myopia:

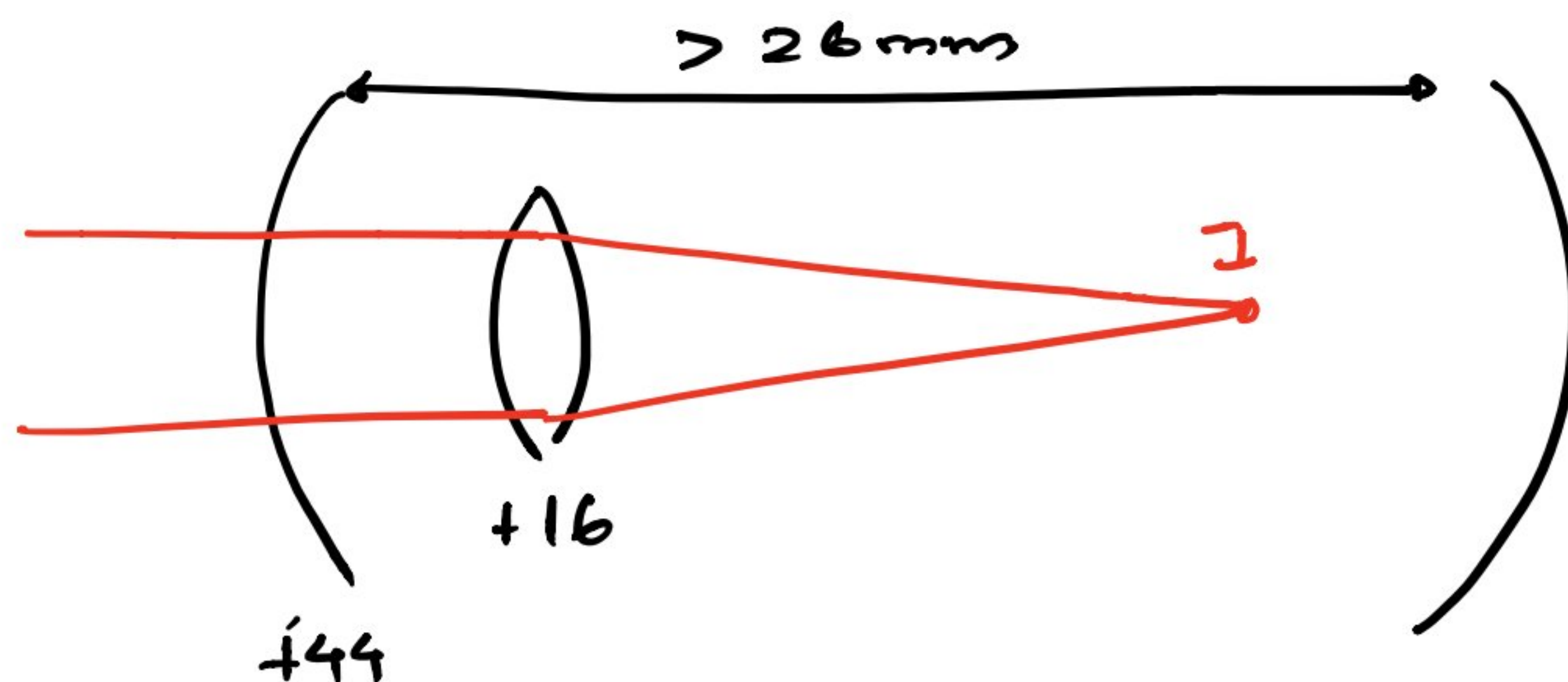
1) Axial Myopia = due to ↑ axial length of eyeball

(N) adult axL = 24 mm

Pathological axL ≥ 26 mm



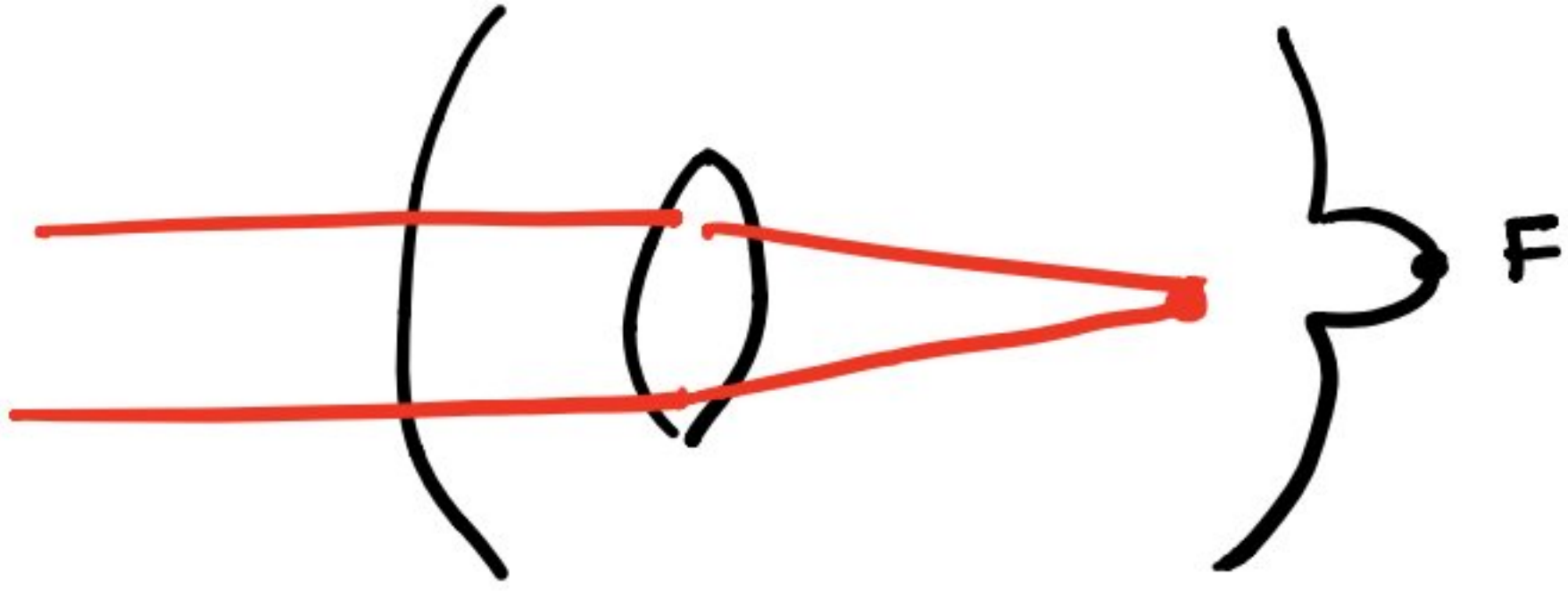
+60D, $f = 1/60$



+60D, $f = 1/60$

Example:

- * Pathological / high myopia
- * Buphthalmos {congenital glaucoma}
- * Post. Staphyloma



2) Curvatural Myopia

↑ sed curvature → cornea
→ lens

Eg: Cornea [steep]

i) Keratoconus
→ progressive

$$\left\{ \begin{array}{l} P \geq 47.2D \end{array} \right.$$

ii) Keratoglobus
→ non-progressive
→ congenital
→ AR

overall
↑ se in curvature

Lens:

1) Lenticonus

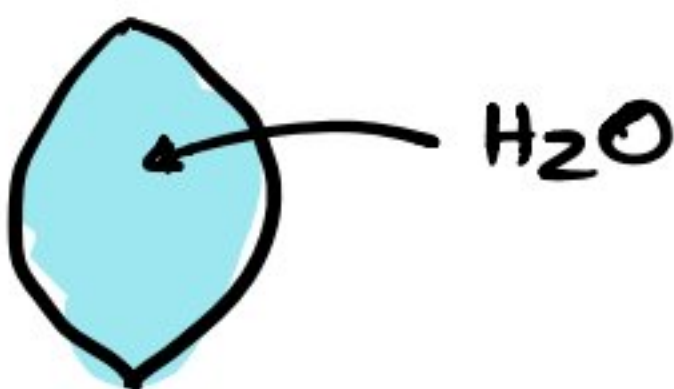
Ant. → Alport's syndrome
Post. → Lewis syndrome



2) Intumescent Cataract

→ Freq. change in refractive error

3) Diabetic Cataract

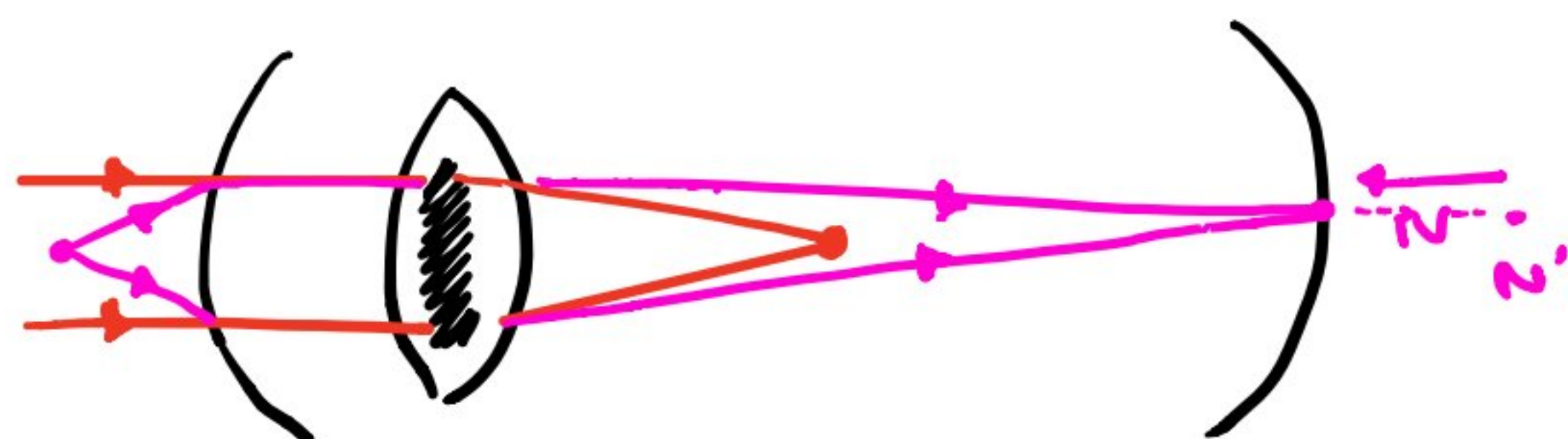


4) Spher o phakia

3) Index Myopia : d/t ↑sed 'n of the lens nucleus

→ (N) lens nucleus $\mu = 1.406$

→ Nuclear sclerosis $\uparrow$ Albumin $\uparrow$ = $\mu \uparrow$ sc



Presbyopic

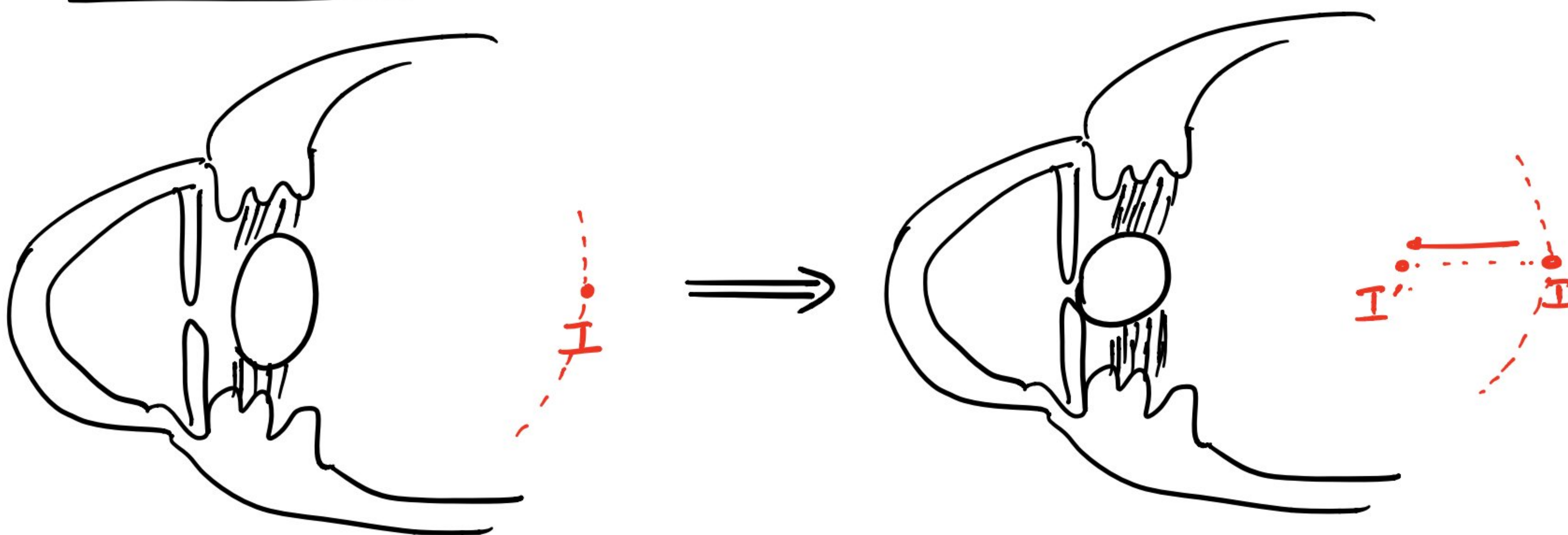
48yr
Glasses

55yr
Nuclear sclerosis
No glass

↓
"second sight"

Second sight: Improved near vision in a presbyopia s/i Nuclear sclerosis is d/t index myopia.

4) Positional Myopia: d/t Ant. dislocation of lens



WEIL MARCHSANI SYNDROME

- Anterior dislocation of lens
- Positional Myopia
- Microspherophakic lens

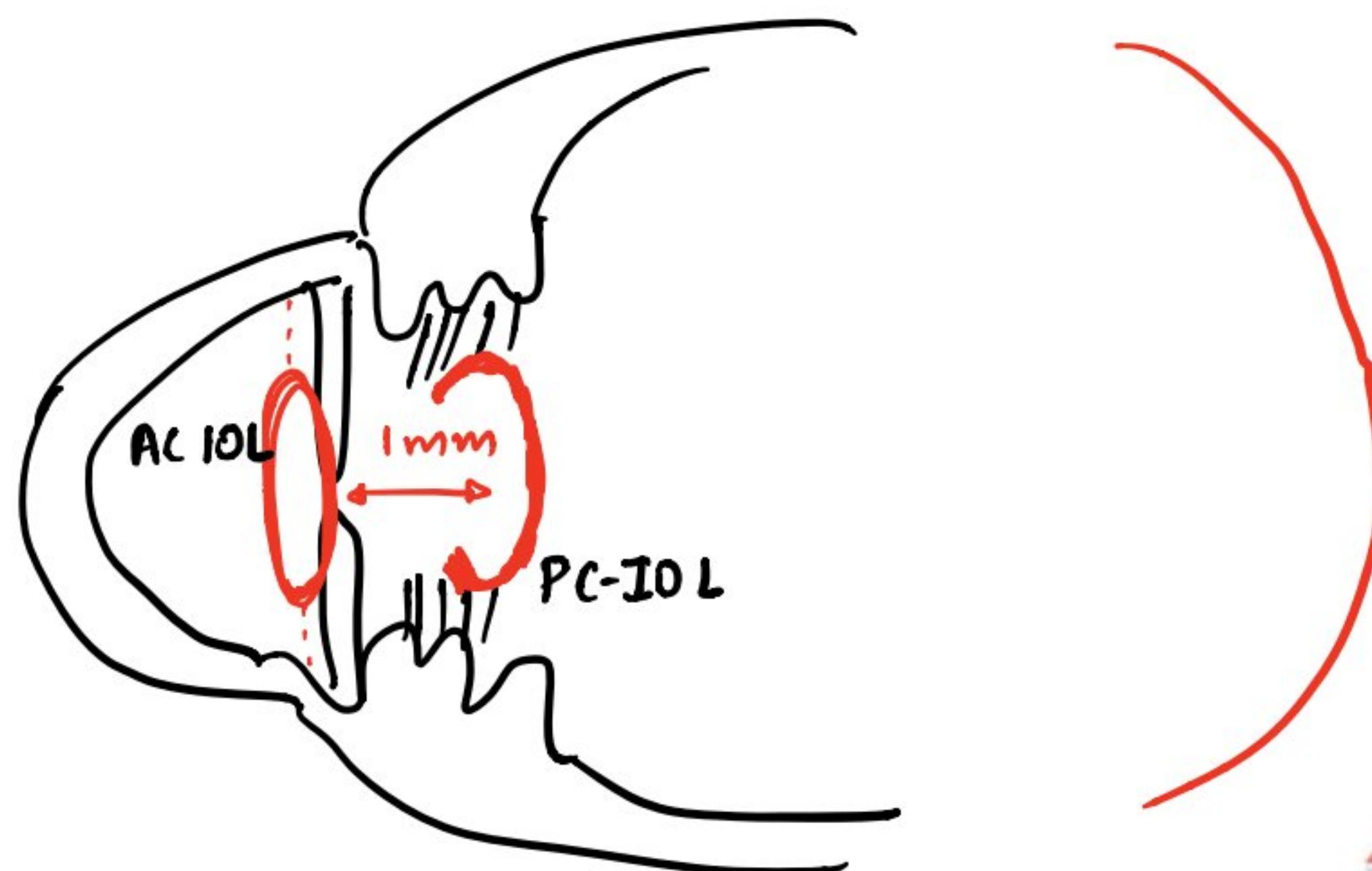
4) Consecutive Myopia: Every 1mm change = 3D change in IOL Power

PC-IOL = +23D
{calculated}

AC-IOL = +23 - 3D
= +20D

Management

- Glass = concave / minus sphere
- Contact lens



ii) LASIK = Laser Assisted In Situ Keratomileusis (Sculpting)

Laser used = **Excimer Lens**

- Excited dimer
- ArF (Argon Fluoride)
- $\lambda = 193 \text{ nm}$

Principle :- **Photoablation** { non thermal vapourisation of tissue }

Layer photoablated $\Rightarrow$ corneal stroma

C/I:

- < 18 yrs of age
- Unstable refraction
- Dryness of eye
- collagen vascular d/s
- Diabetic pt / Diabetic Retinopathy
- Herpes Simplex Keratitis $\rightarrow$ Excimer laser reactivates the keratitis.

Relative C/I $\rightarrow$ Keratoconus

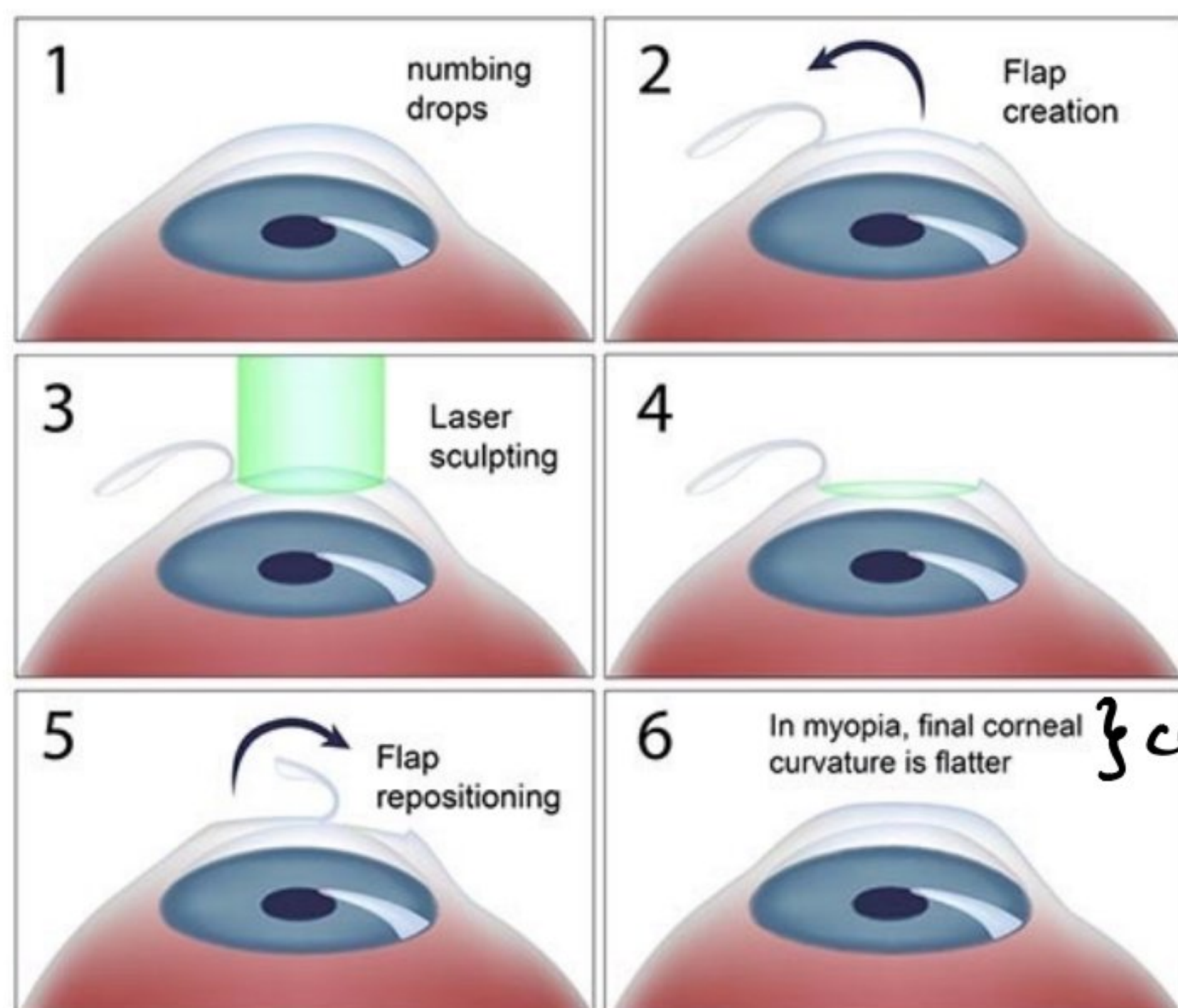
Pre-op $\Rightarrow$ i) No exposure keratitis

ii) Tear amount

iii) Pachymetry (Thickness of cornea)

Residual stromal thickness should be at least **$\approx 250 \mu$**

LASIK EYE SURGERY



• Power of myopia upto -10D

} curvature $\downarrow \rightarrow D \downarrow \rightarrow f \uparrow$

iv) **SMILE**: Small Incision Lenticular Extraction

- Bladeless
- Flapless

FEMTOSECOND LASER

$\downarrow$

Lenticulate removed

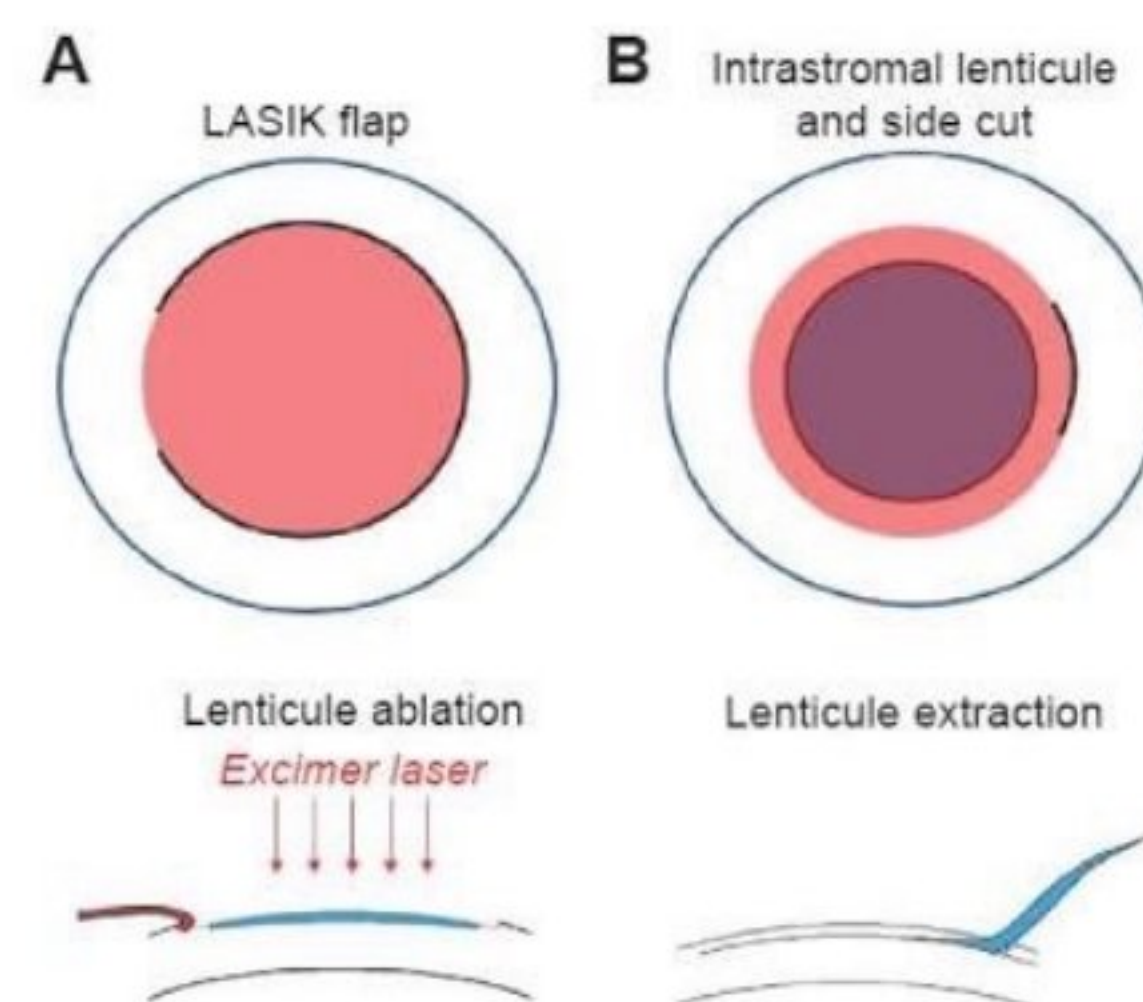


Figure 1 Schematic drawings of LASIK (A) and SMILE (B).
Abbreviations: LASIK, laser-assisted in situ keratomileusis; SMILE, small-incision lenticule extraction.

v) ICL / Phakic IOL $\rightarrow$ used for $Mx \Delta_0 -20 D$
{Implantable collamer lens}

vi) Radial Keratotomy :-

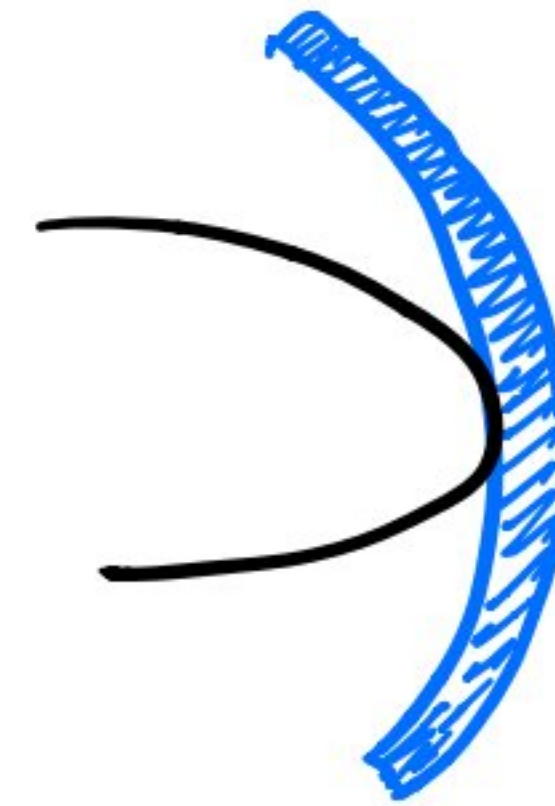


• Flattens the central cornea steepen the spherical cornea

Disadvantage

- i) Unpredictable
- ii) Astigmatism
- iii) Regression

vii) Orthokeratology :- Using reverse geometry contact lens flatten the central cornea
used in :- Keratoconus.



viii) Clear lens Extraction :- Removal of lens $\&$ $\downarrow$ se in power of eye by $-16 D$

a.k.a FUKALA'S SURGERY

Disadv. :- i) Retinal Detachment
ii) Young Pt. = accommodation lost

$\downarrow f/b$

Refractive Lens Exchange

Pathological Myopia / High myopia

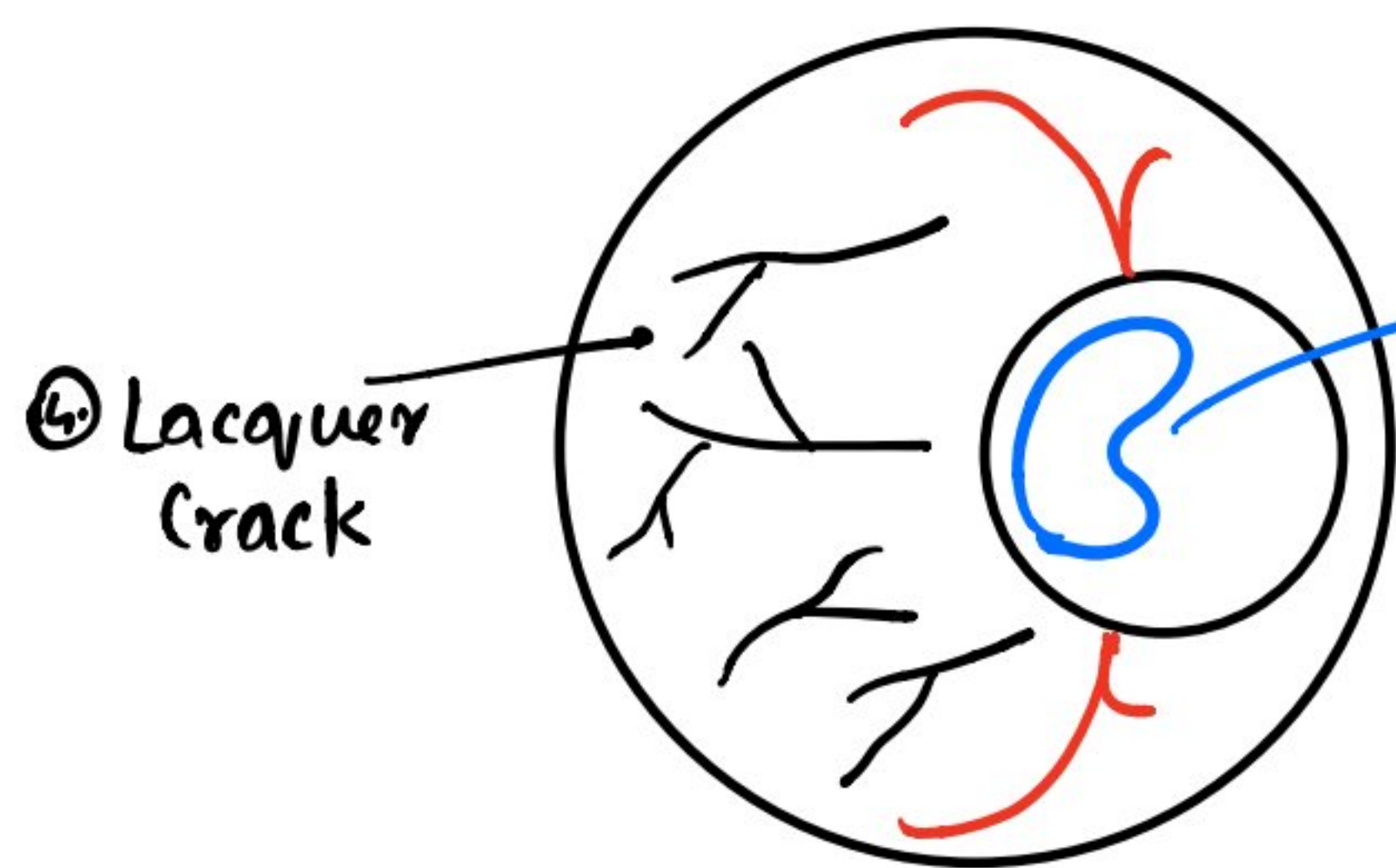
High myopia $\Rightarrow -6$ to $-8 D$
 $26mm \leq AxL < 32mm$

Pathological Myopia $\Rightarrow > -8 D$
 $AxL \geq 32mm.$

Ocular finding:-

- 1) Blue Sclera $\&$ uveal tissue visible
- 2) AC $\Rightarrow$ deep
- 3) POAG {primary open angle glaucoma} $\rightarrow$ TM gets collapsed on stretching of sclera.
- 4) Complicated cataract $\rightarrow$ Post. subcapsular in location
 $\hookrightarrow$ Rainbow Cataract

Fundus Finding:



① Temporal crescent on optic disc

↳ d/t absence of choroid & retina whitish color
 ↳ sclera visible.

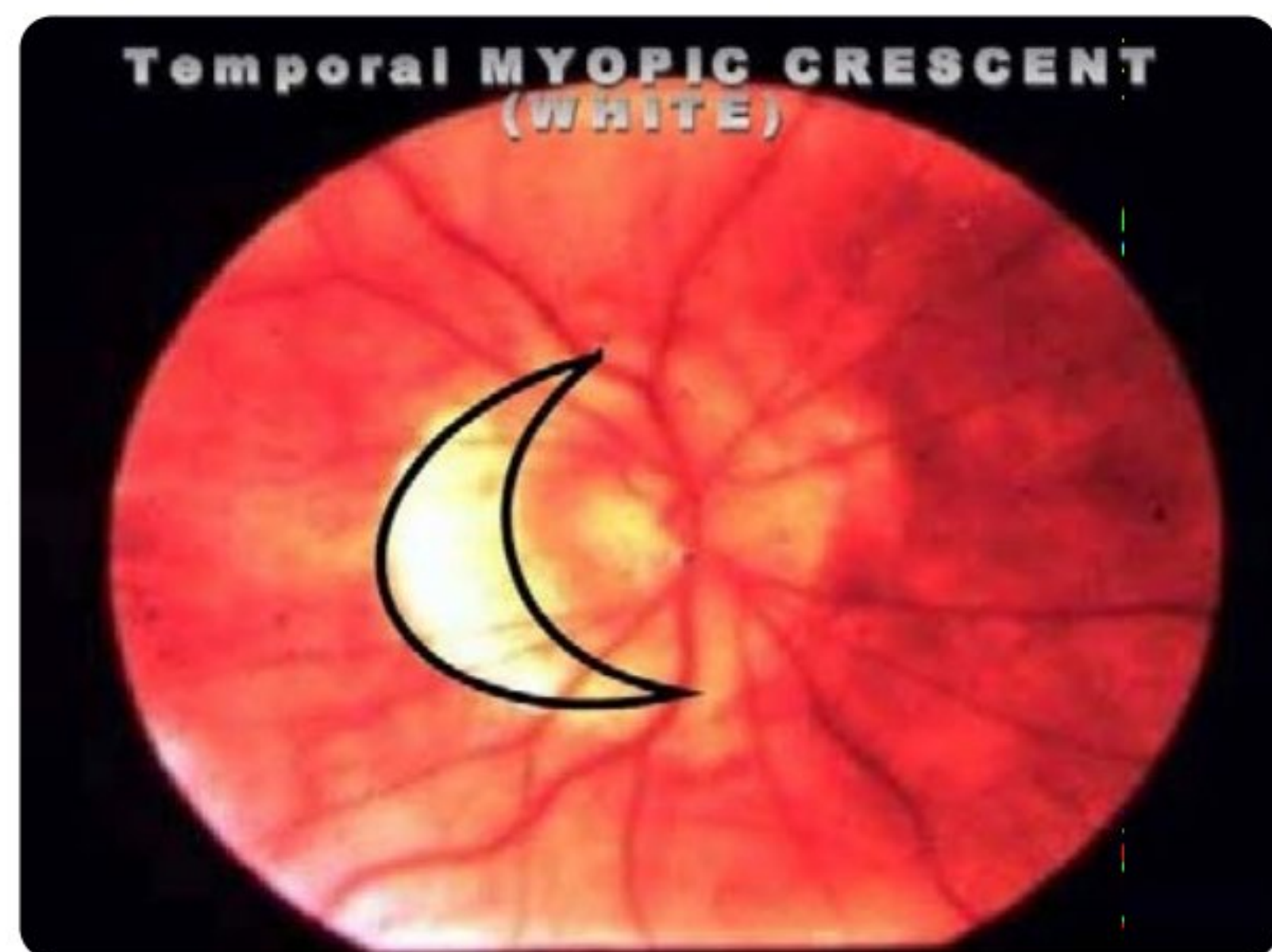
② Tessellated appearance of fundus

③ Tigroid appearance of fundus

④ Lacquer cracks of Bruch's membrane

⑤ Foster-Fuch's spot $\Rightarrow$ RPE hyperplasia in macular region

⑥ Choroidal Neovascular membrane $\rightarrow$ in macula.



Peripheral Retina finding:

1) RETINAL BREAK

↳ { Crack: RHEGMA

also known as **RHEGMATOGENOUS DETACHMENT**

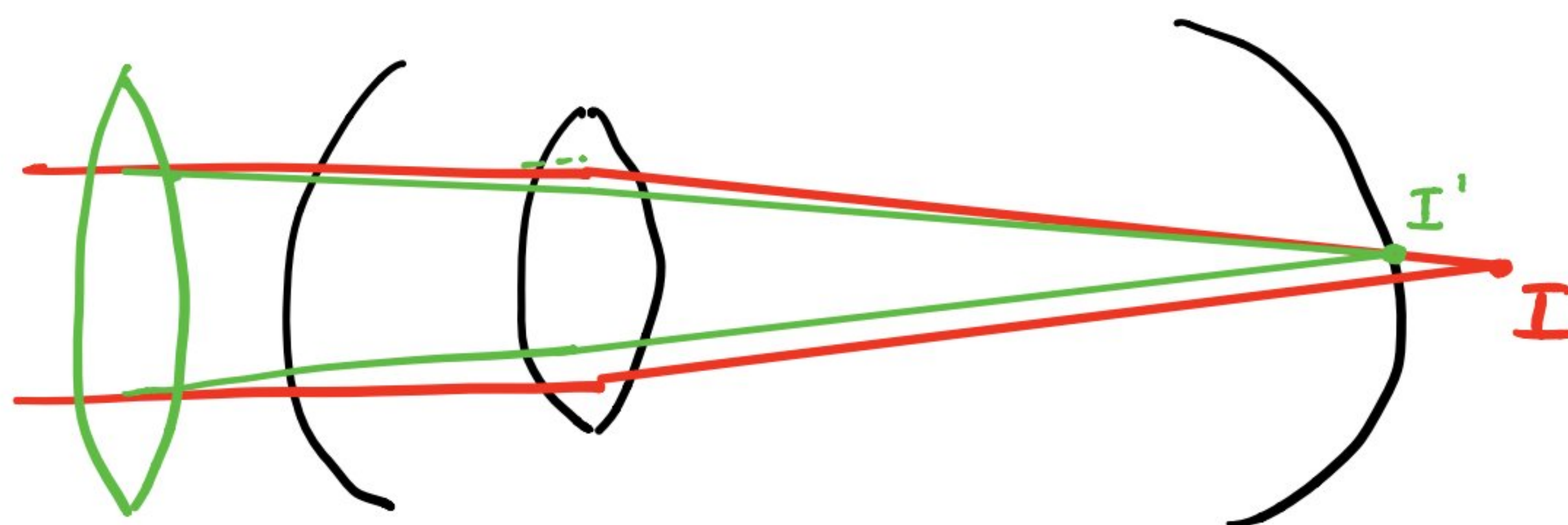
2) LATTICE RETINAL DEGENERATION

3) Posterior staphyloma

$\downarrow$
 BULGING OF THE RETINA

4) Posterior Vitreous Detachment $\rightarrow$ "Weiss Ring"
 ↳ spider web floating in eye
 described by a patient?

HYPERMETROPIA / far sightedness / Long sight ::



convex / + lens

Grading of Hypermetropia

Low H.M : $\leq +2D$

Moderate H.M : $+2.25D - +5D$

High H.M : $> +5D$

1) Thicker at center; thinner @ periphery

2) Magnified images

3) Opp. movement of images

↳ Converging lens

Types of Hypermetropia

1) Axial H.M : d/t ↓ se AxL of eyeball

• Physiological AxL HM

Newborn AxL : 16.5 mm

↓

H.M = +2.5 D

@ 14yrs = 24mm

emmetropization

q. 1mm shortening of A-P diameter of eye results in 3D of hypermetropia.

eg. 2mm shortening of axial length of eyeball → +6 D H.M

• Pathological AxL H.M:

1) Nanophthalmos : AxL ≤ 20mm

• sclera thick high

↓

Blocks venous blood outflow through vortex veins

↓

Uveal effusion & ↑ fluid in choroid → choroidal detachment

2) Micro - ophthalmos : AxL ≤ 20mm

Ab(N) eye structure

s/i is Congenital Rubella syndrome

ii) CMV

iii) PHPV.

3) Curvatural H.M : d/t ↓ sed curvature

Eg: i) Cornea Plana

ii) Sclerocornea

q. 1mm ↑ se in radius curvature results in 6 D of hypermetropia

eg. 2mm increase in radius of curvature of lens → +12 D H.M

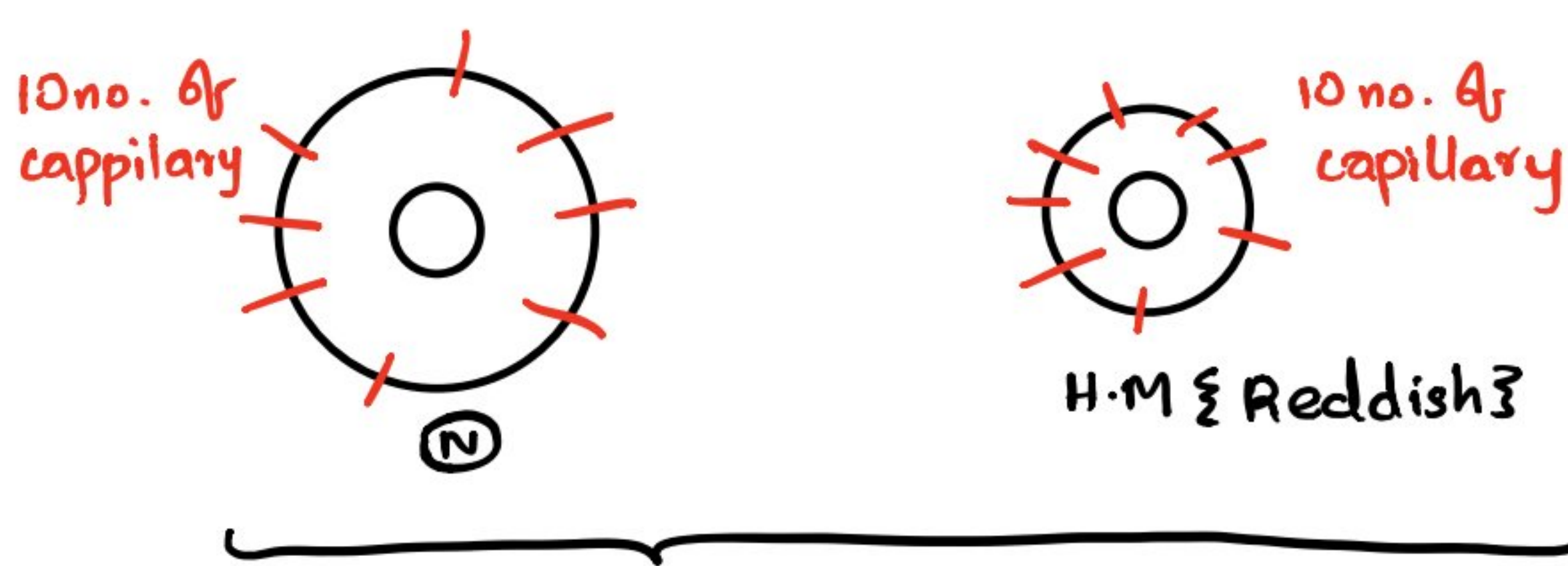
3) Index H.M : Cortical cataract

4) Positional H.M : posterior dislocation of lens.

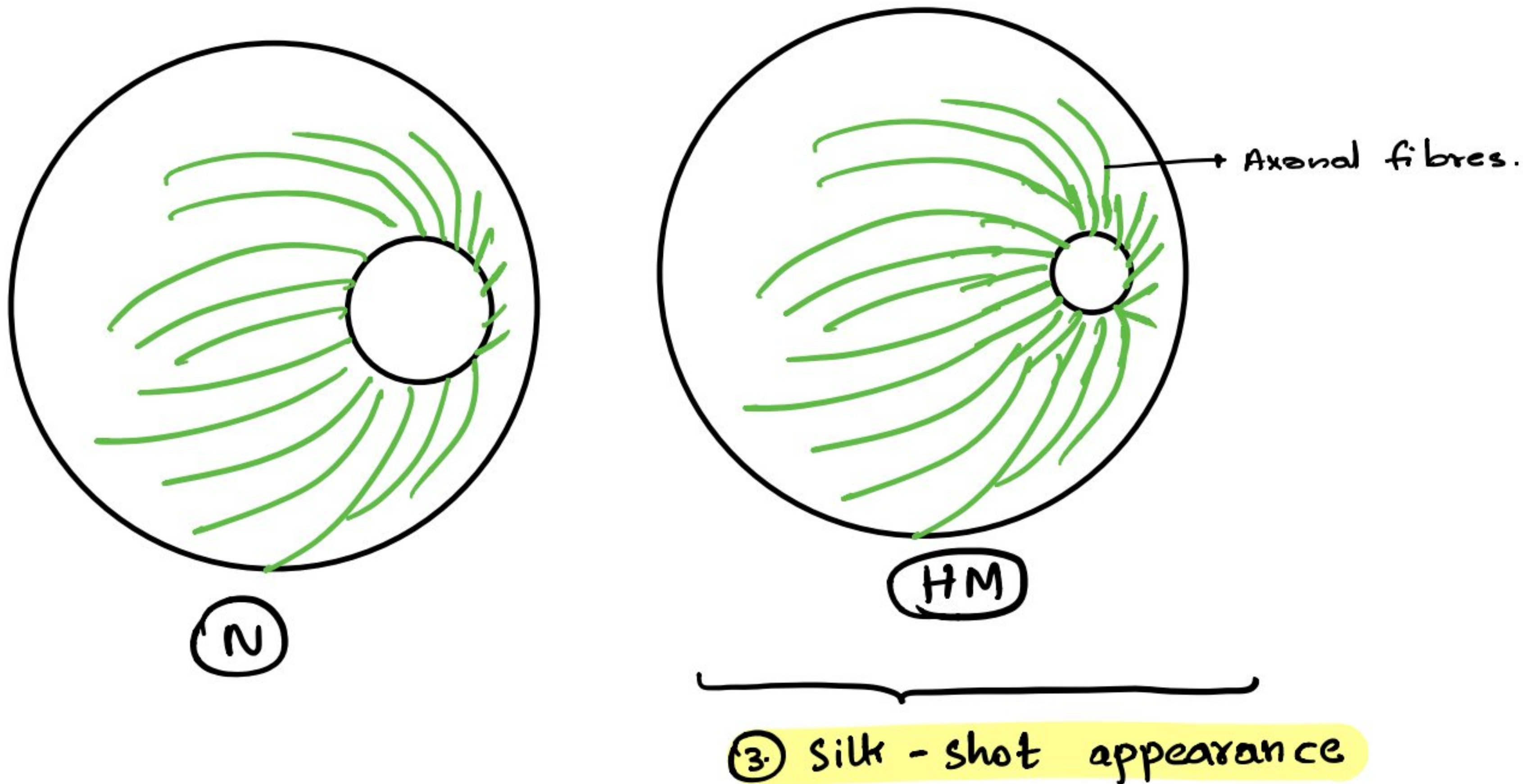
Aphakia ⇒ Extremely HM

Fundus finding of H.M:

i) Optic disc → small
cup → small



2) Psuedo papillitis



③ Silk - shot appearance

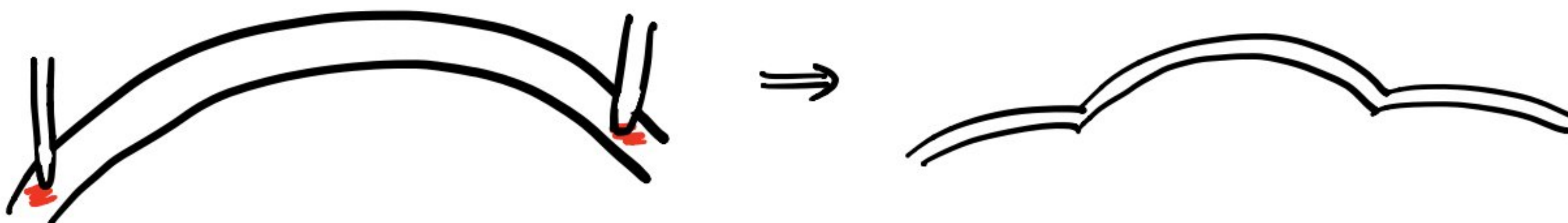
Eye finding in AxL H.M.:

- 1) A.C → shallow
- 2) Angle → Narrow
- 3) Angle → PACG
- 4) AION & Ant. ischemic optic neuropathy

Clinical Presentation:

- 1) Asthenopia → Tiredness of eye
Soreness of eye
Pain of eye } Eye accommodation → Convergen Squint to ↑se D

- Management ⇒
- 1) Glasses ⇒ Convex sphere
 - 2) Contact lenses
 - 3) LASIK : Laser → **HOLMIUM LASER**



- 4) Thermokeratoplasty
- 5) Conductive Keratoplasty

APHAKIA & Extreme HM

Surgical (ICCE)

Traumatic (Blunt)

- 1) Superior limbal scar {surgical Aphakia}
- 2) A.C $\Rightarrow$ deep & Iris move back
- 3) Iridodonesis & tremulousness of Iris
- 4) 2 purkinje image
- 5) Pupil jet black colour
- 6) Disc appears small

Symptoms $\Rightarrow$ $\downarrow$ se in Vision

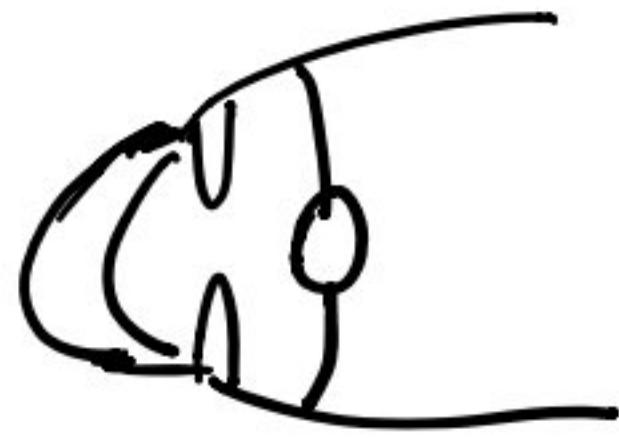
Mx: 1) Glasses {convex}

2) Contact lens

3) 2^o IOL Implantation (Best)

 i) AC-IOL

 ii) Scleral fixated PC-IOL



iii) Iris-fixated PC-IOL



APHAKIC GLASSES:

• Convex spheres $\Rightarrow$ +10 to +12 D

Disadv. $\rightarrow$ i) Cosmetic issue

ii) 30-33% magnification of image

iii) Problem τ hand-eye coordination

in U/L aphakic glasses $\rightarrow$

(R)

(N) size image

(L)

30% magnified

} ANISEIKONIA

x



Brain can only tolerate 5% difference in Retinal image size.

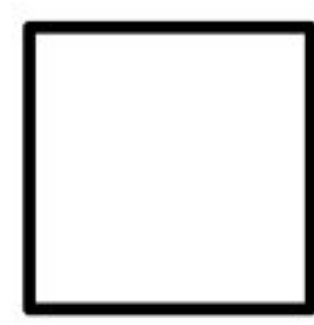
Recent NEET

iv) Roving ring scotoma



Jack in box phenomenon

v, Pincushion effect.



square



Visible with
aphakic glasses

Magnification:

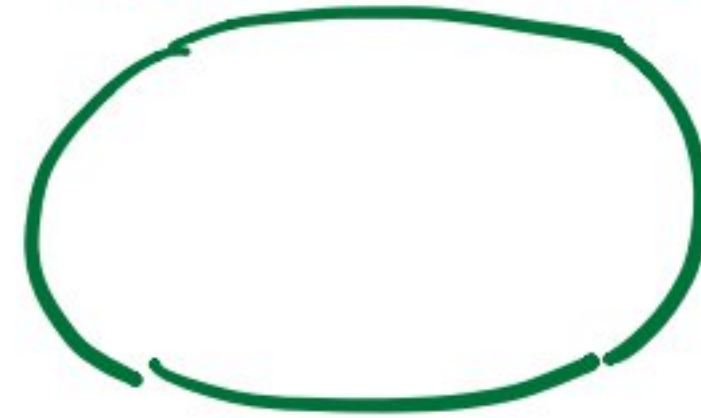
- 1, Aphakic glasses = 30-33%.
- 2, Contact lens = 7-8%.
- 3, IOL (Best) = 1-2

Pincushion Effect



High convex lens

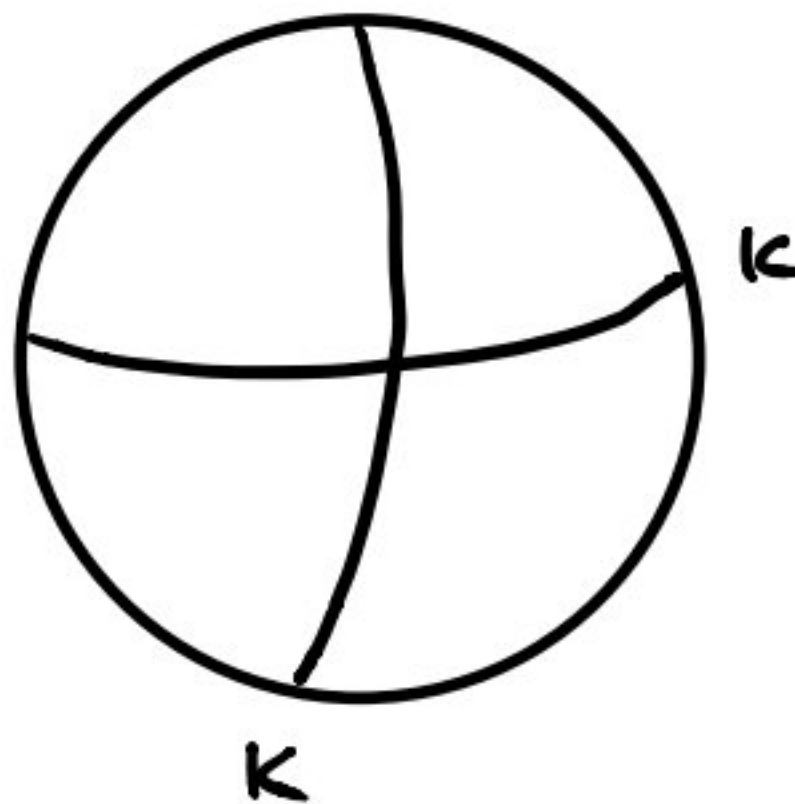
Barrel distribution



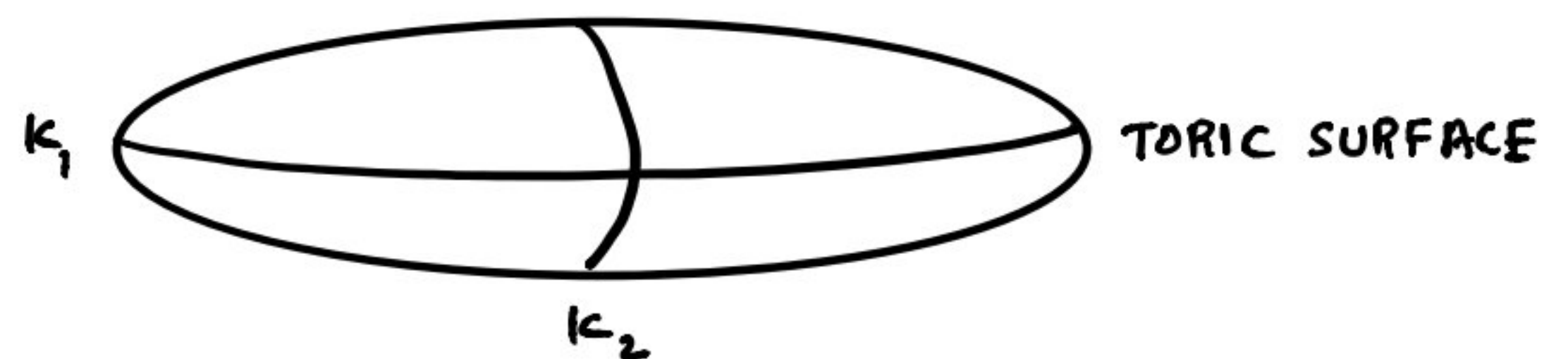
High concave lens.

ASTIGMATISM: Absence of a single point focus / presence of more than one point focus.

Spherical Surface



Astigmatism



$$\begin{array}{l|l} K_2 > K_1 & \text{curvature} \Rightarrow K \\ D_2 > D_1 & K \propto D \\ f_2 < f_1 & \end{array}$$

Type of Astigmatism

Regular

- upto 2 foci

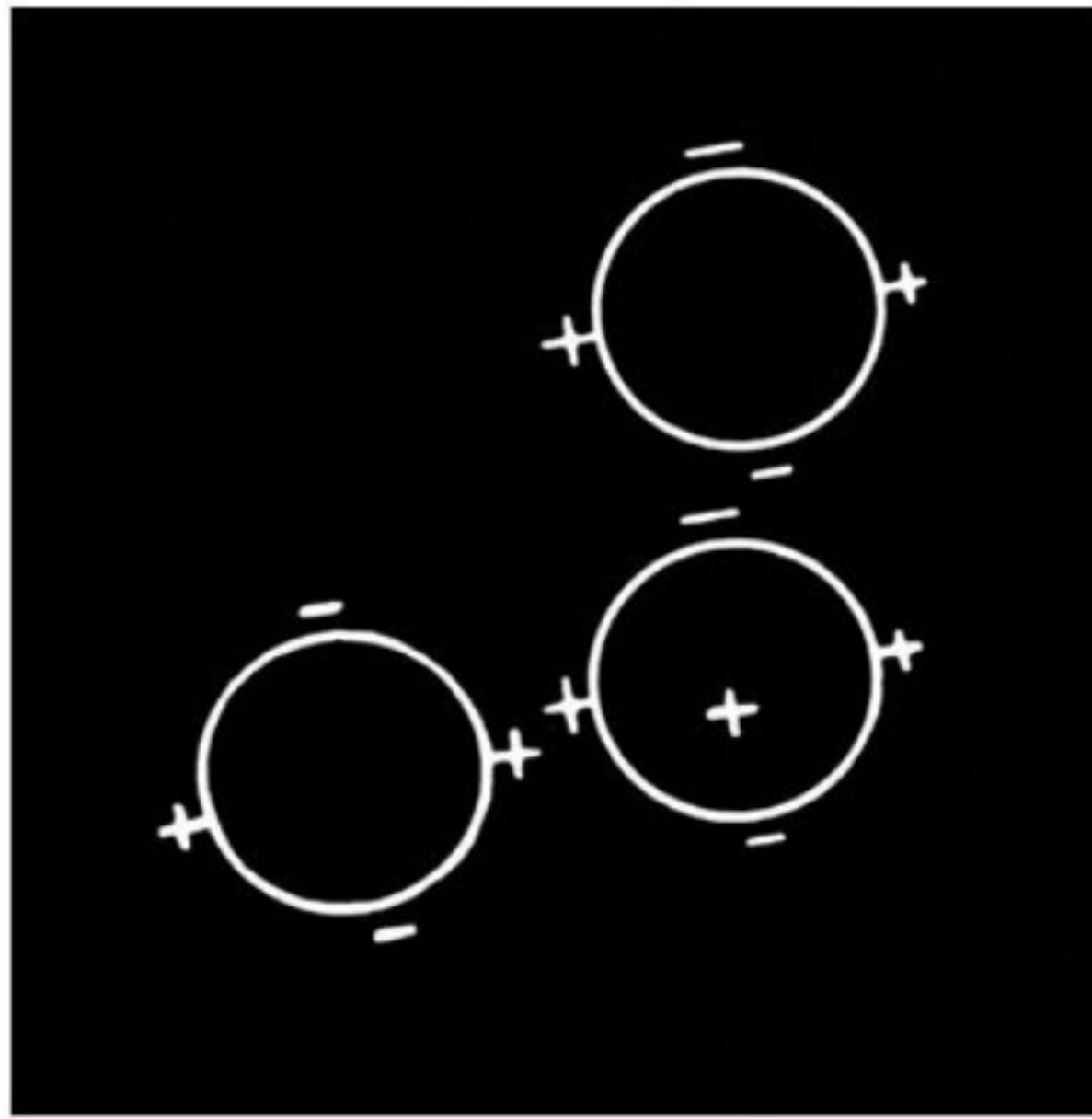
Irregular

- more than 2 foci
- any geometrical analysis is impossible
- causes :
 - i) Corneal transplant (suture)
 - ii) Keratoconus
 - iii) Highly scarred cornea

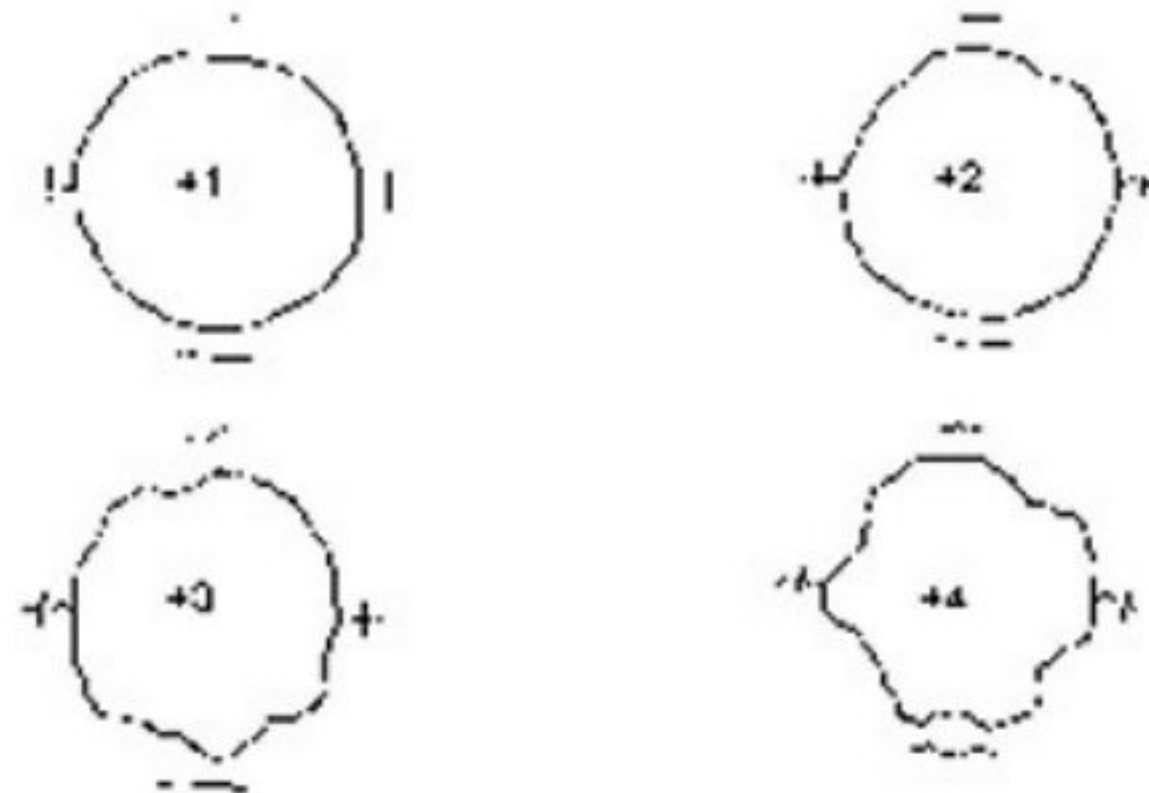


- in irregular astigmatism cannot be fully be corrected by cylindrical lens

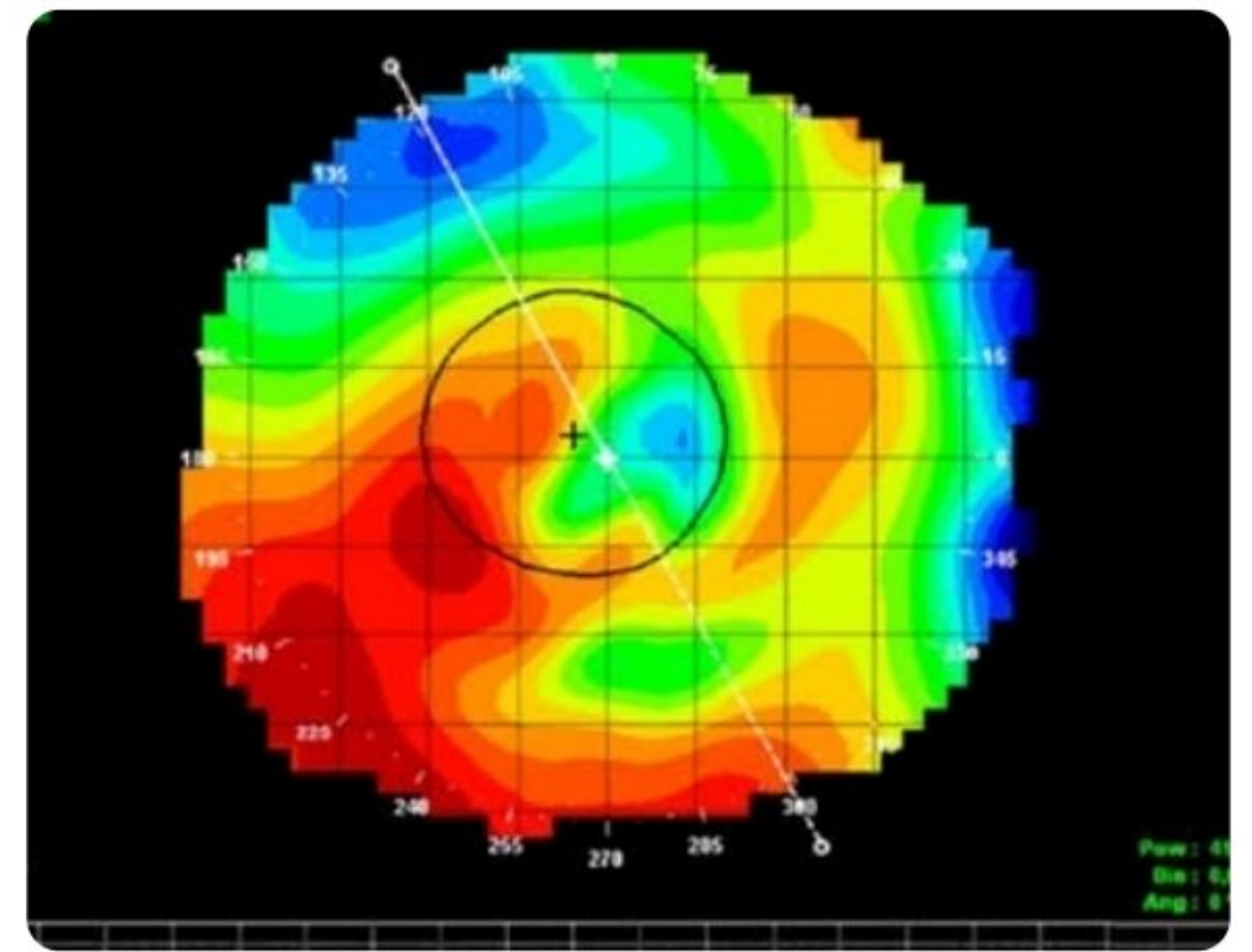
- Corneal Topography → To assess the surface irregularity of cornea
- Keratometry mires



Normal spherical lens



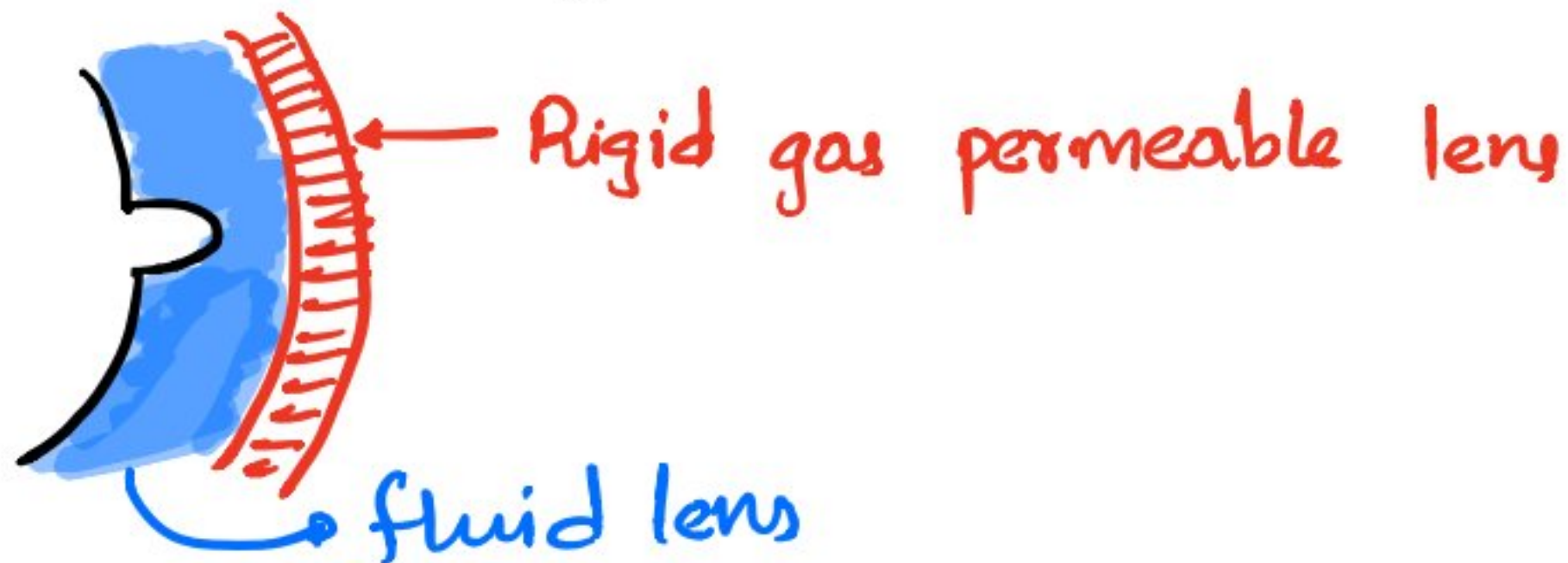
Irregular astigmatism



Irregular astigmatism

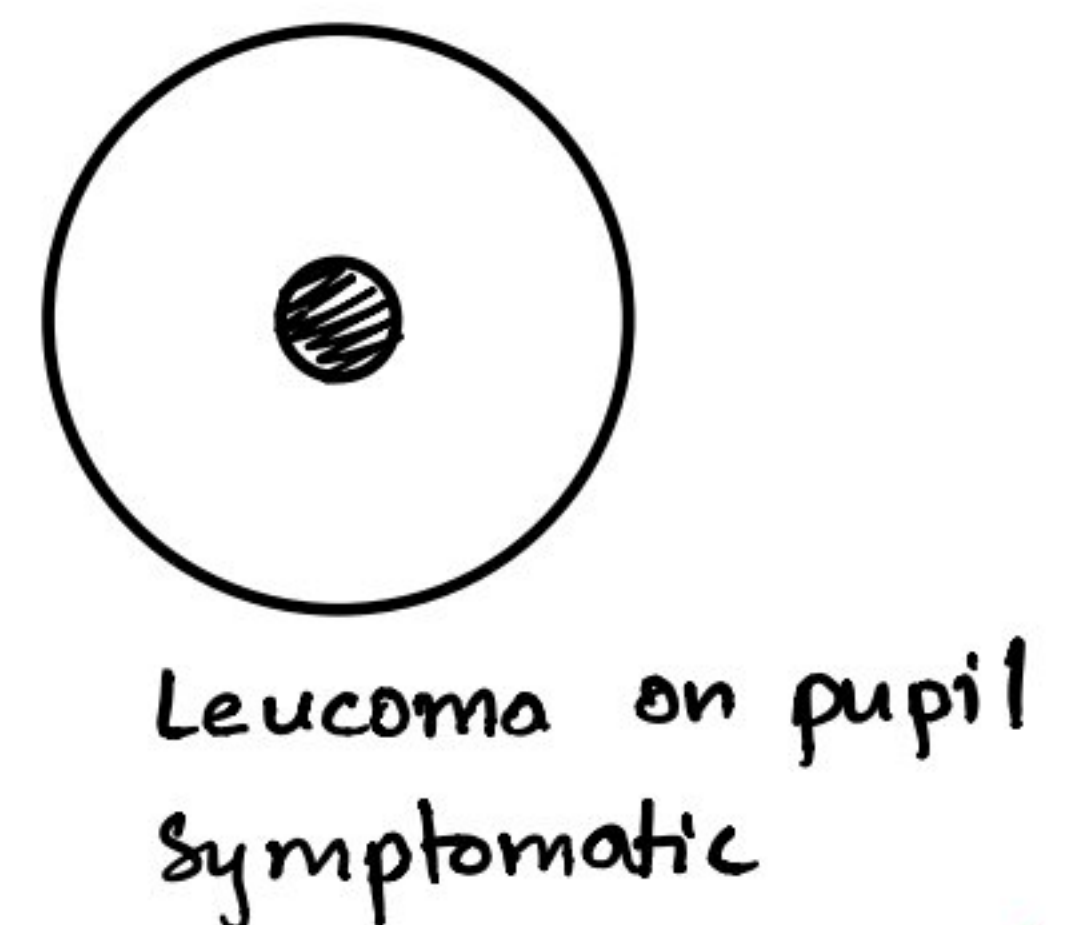
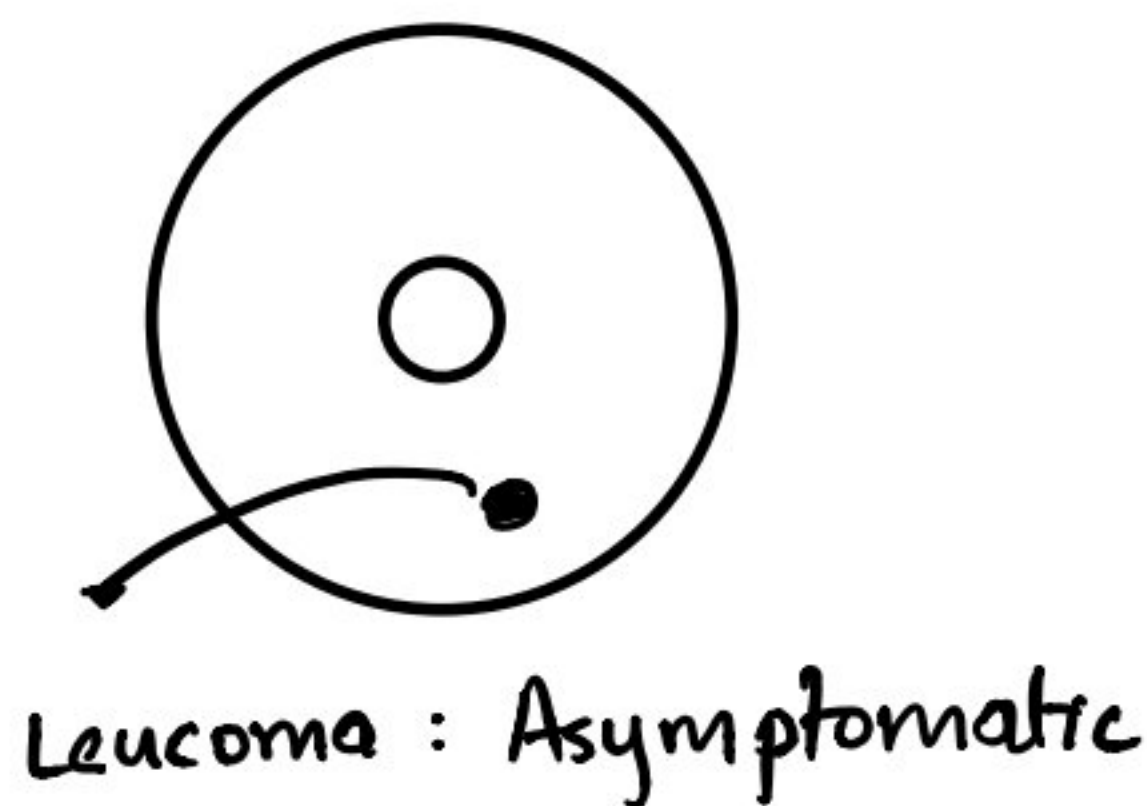
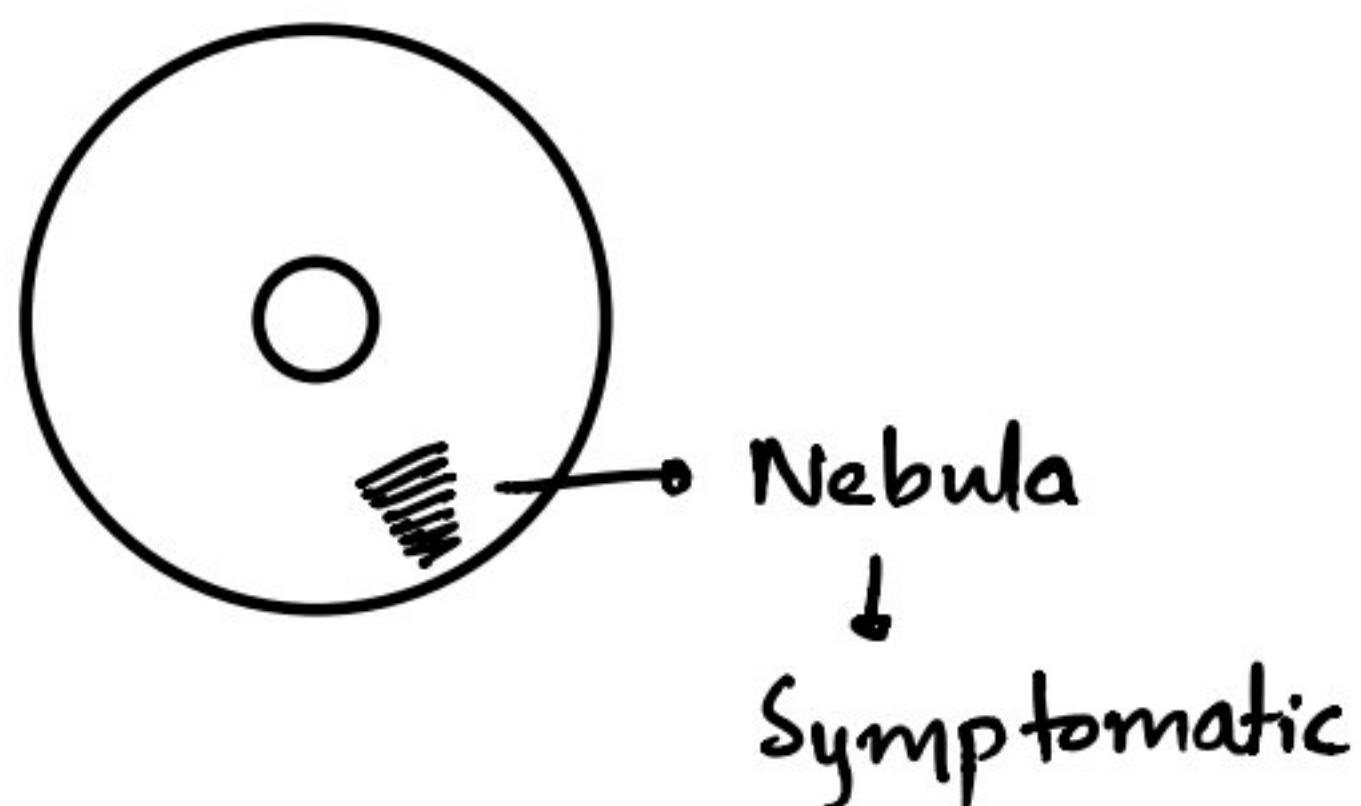
Management

- i) Corneal transplant → LASIK
- ii) Keratoconus → Rigid contact lens



- iii) Scarred Cornea →

• Nebula ⇒ PPK & Photorefractive Keratectomy? done by **EXCIMER LASER** (193nm)



Ⓢ MOST VISUALLY HANDICAPPING CORNEAL OPACITY



NEBULA

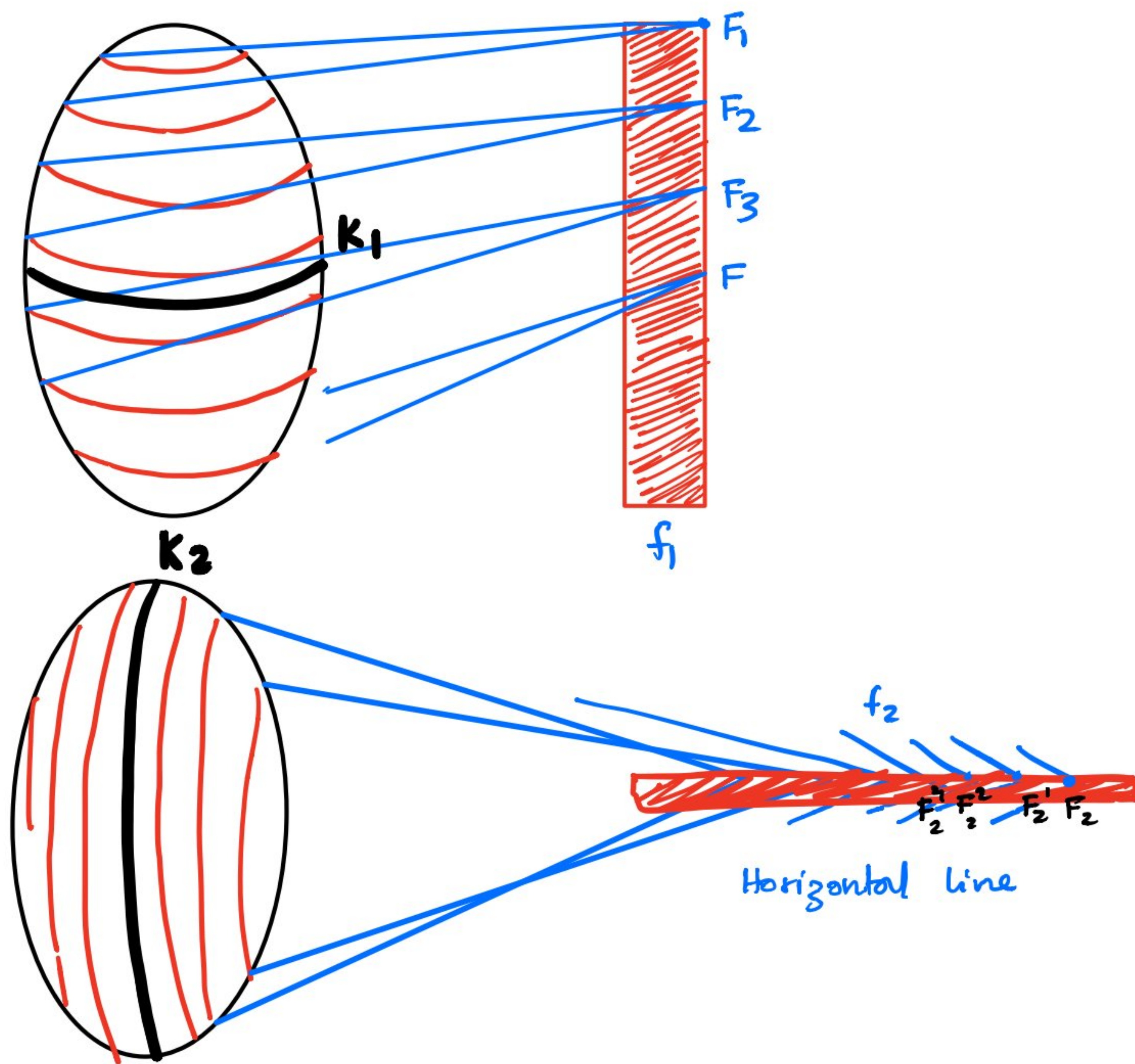
Rx: Corneal Transplant

Regular Astigmatism : • Only 2 principle meridian
i.e one steep meridian
one flat meridian
• 2 foci.

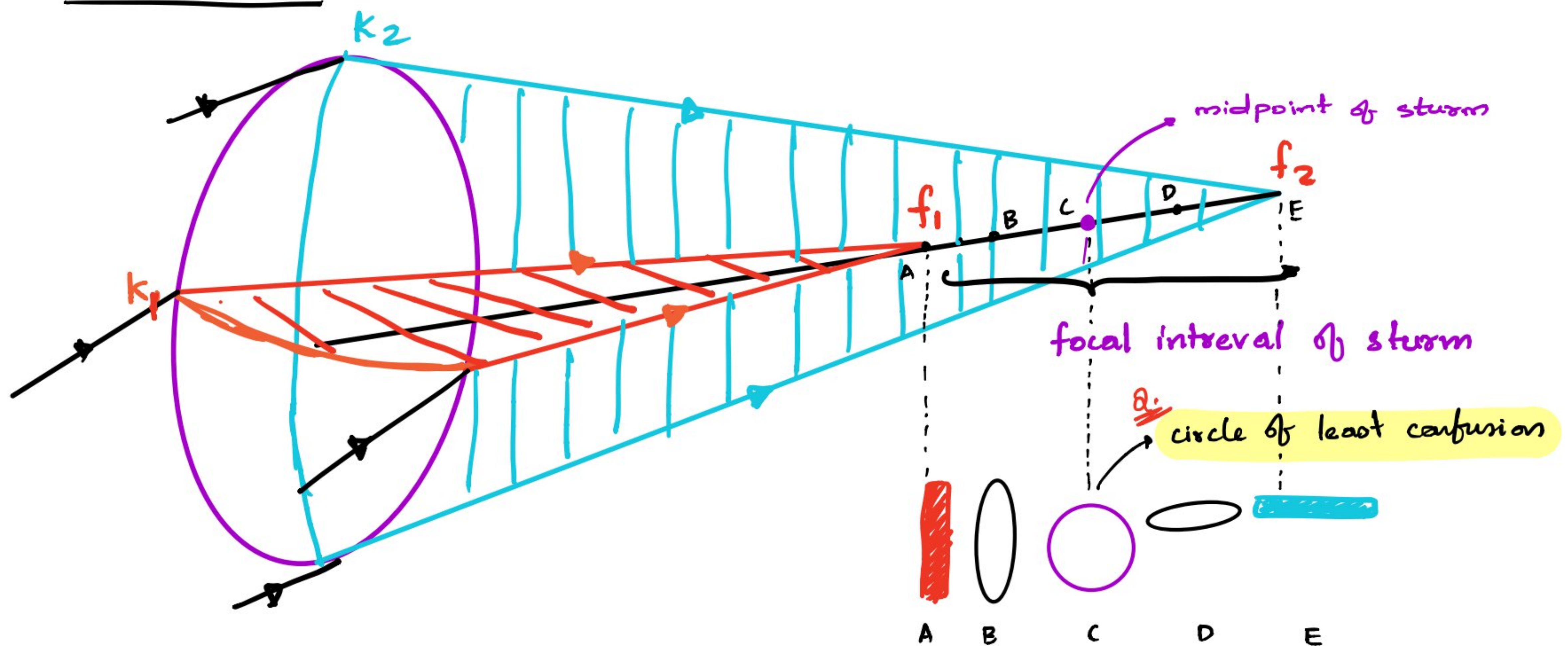
$$K_1 > K_2$$

$$D_1 > D_2$$

$$f_1 < f_2$$



STURMS CONDID:

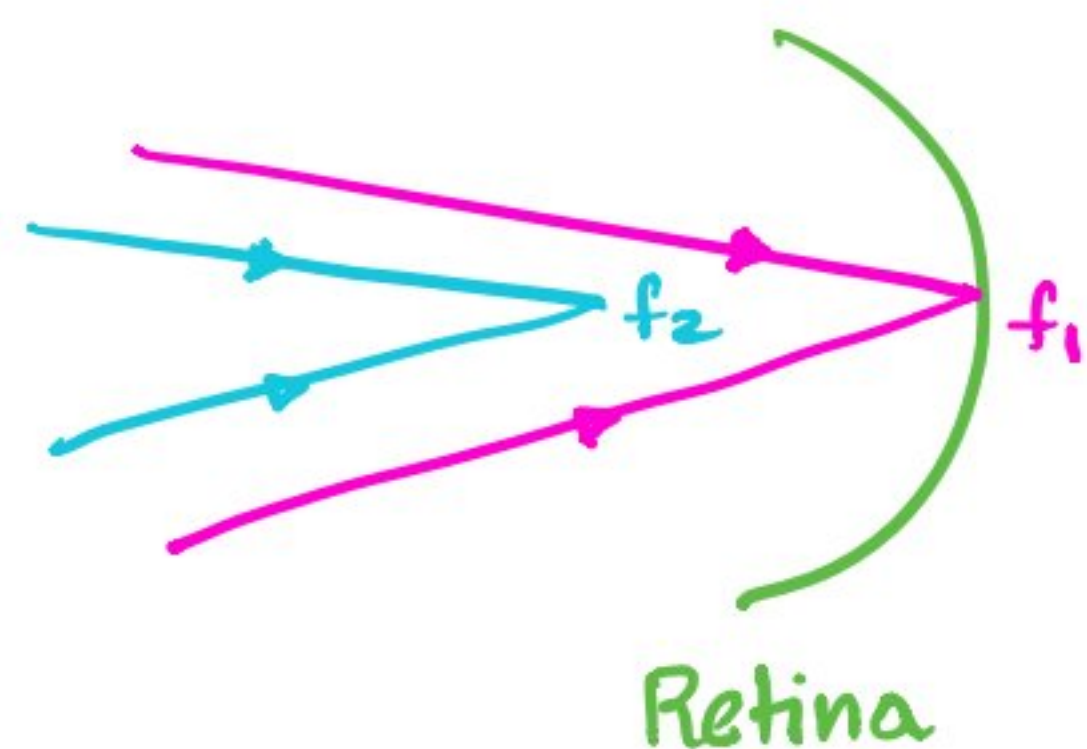


Based on Regular Astigmatism:

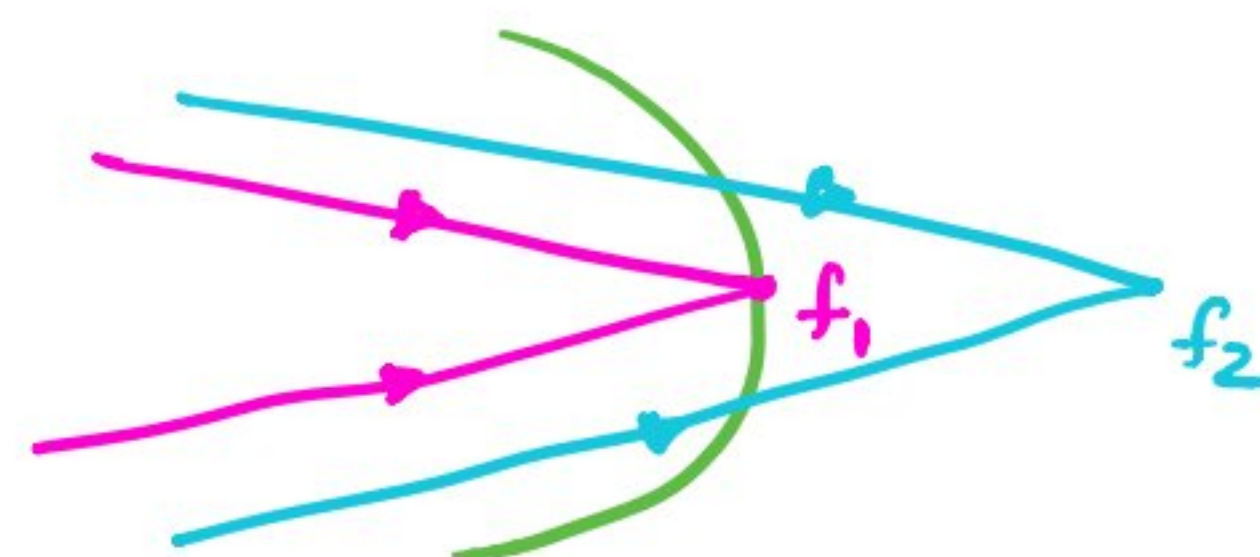
a) Based on the location of the two focal lines w.r.t retina

I) Simple Astigmatism:

. One focal line on the Retina



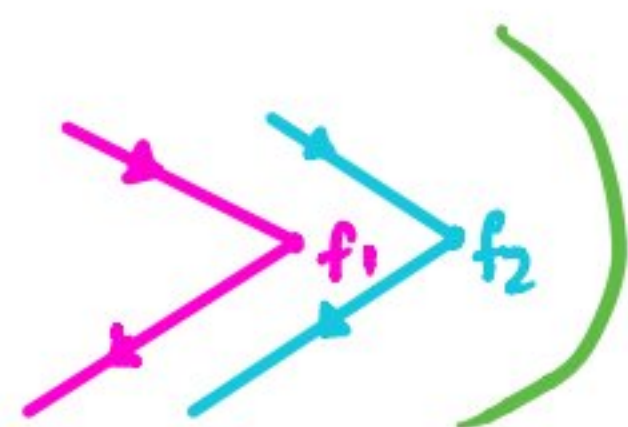
Simple Myopic
Astigmatism
(S.M.A)



Simple Hypermetropic Astigm.
(S.H.A)

II Compound Astigmatism

when both focal lines on same side of retina

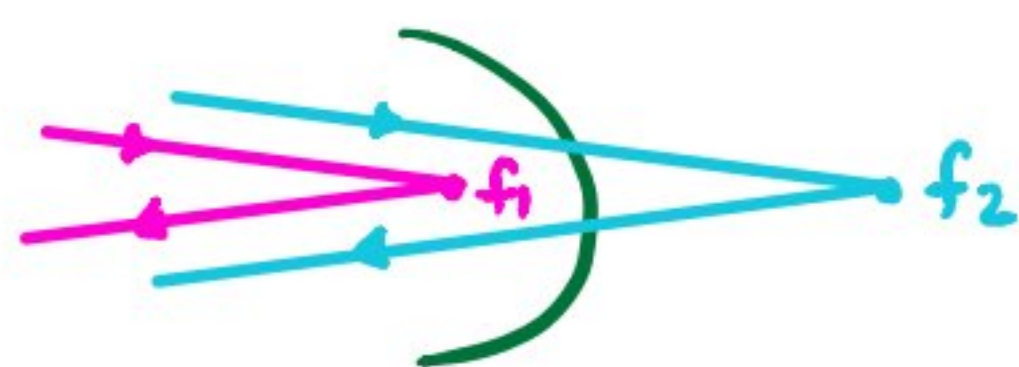


Compound myopic astigmatism
(C.M.A)



Compound Hypermetropic Astigmatism
(C.H.A)

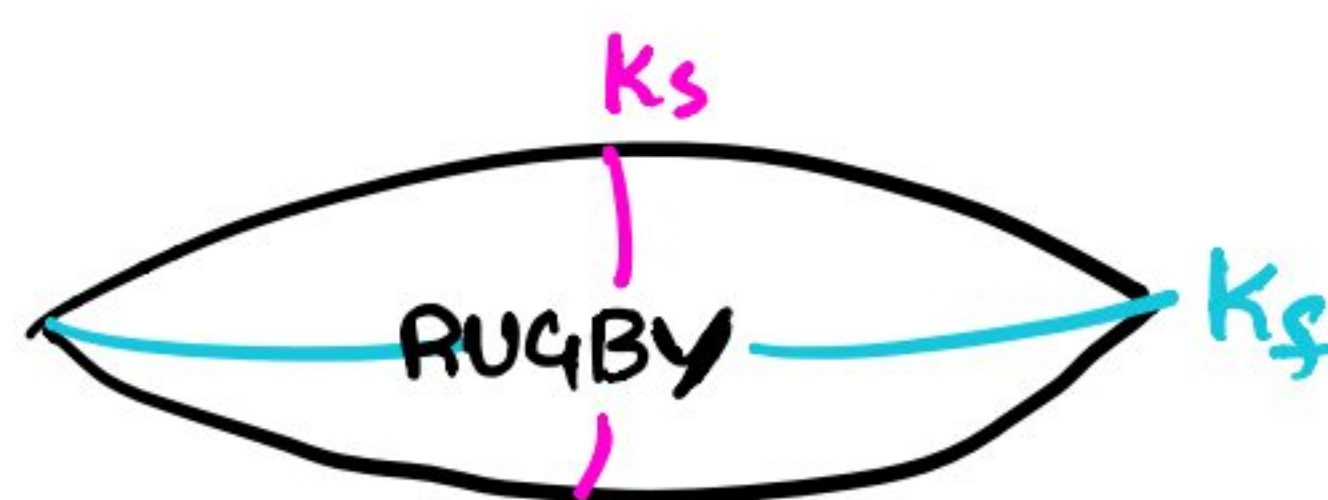
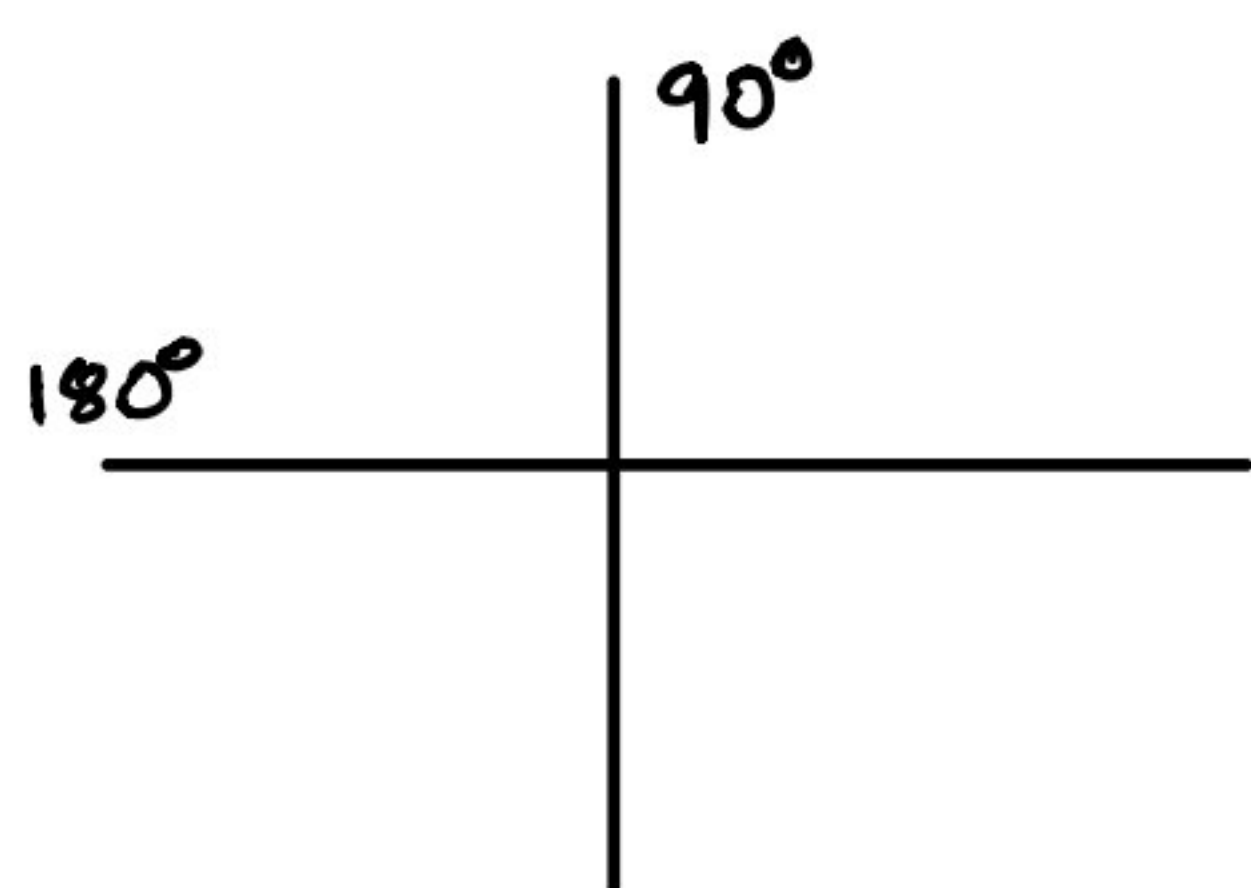
III Mixed Astigmatism: One focal line in front of retina & the other focal line is behind the retina.



∴ Visually least significant ⇒ Becoz the circle of least confusion @ retina.

B) Based on the orientation of the 2 principle meridian

I) With the Rule:



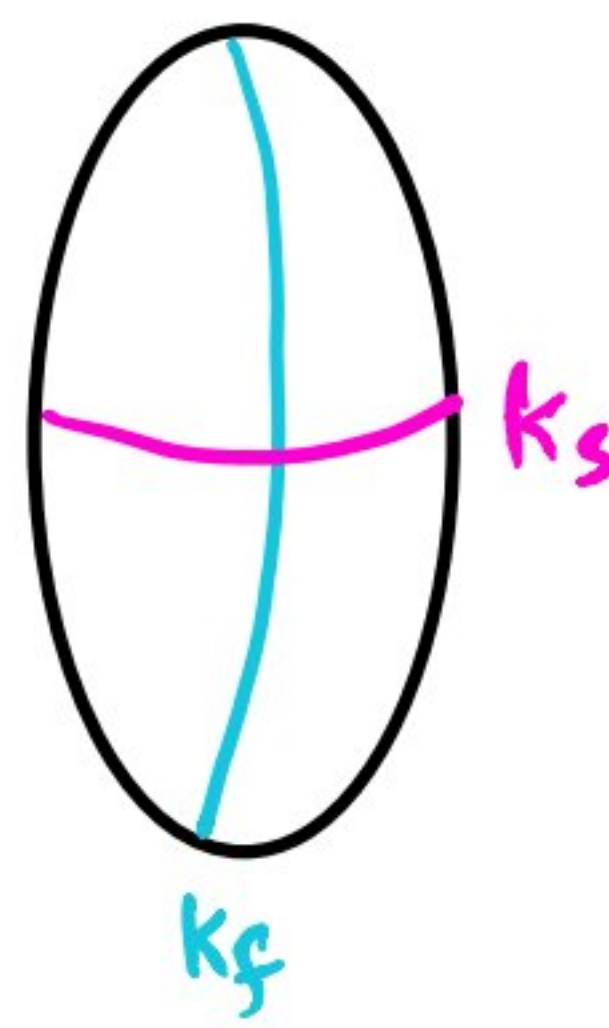
K_s (step) = vertical line, i.e. 90°

K_f (flat) = horizontal line, i.e. 180°

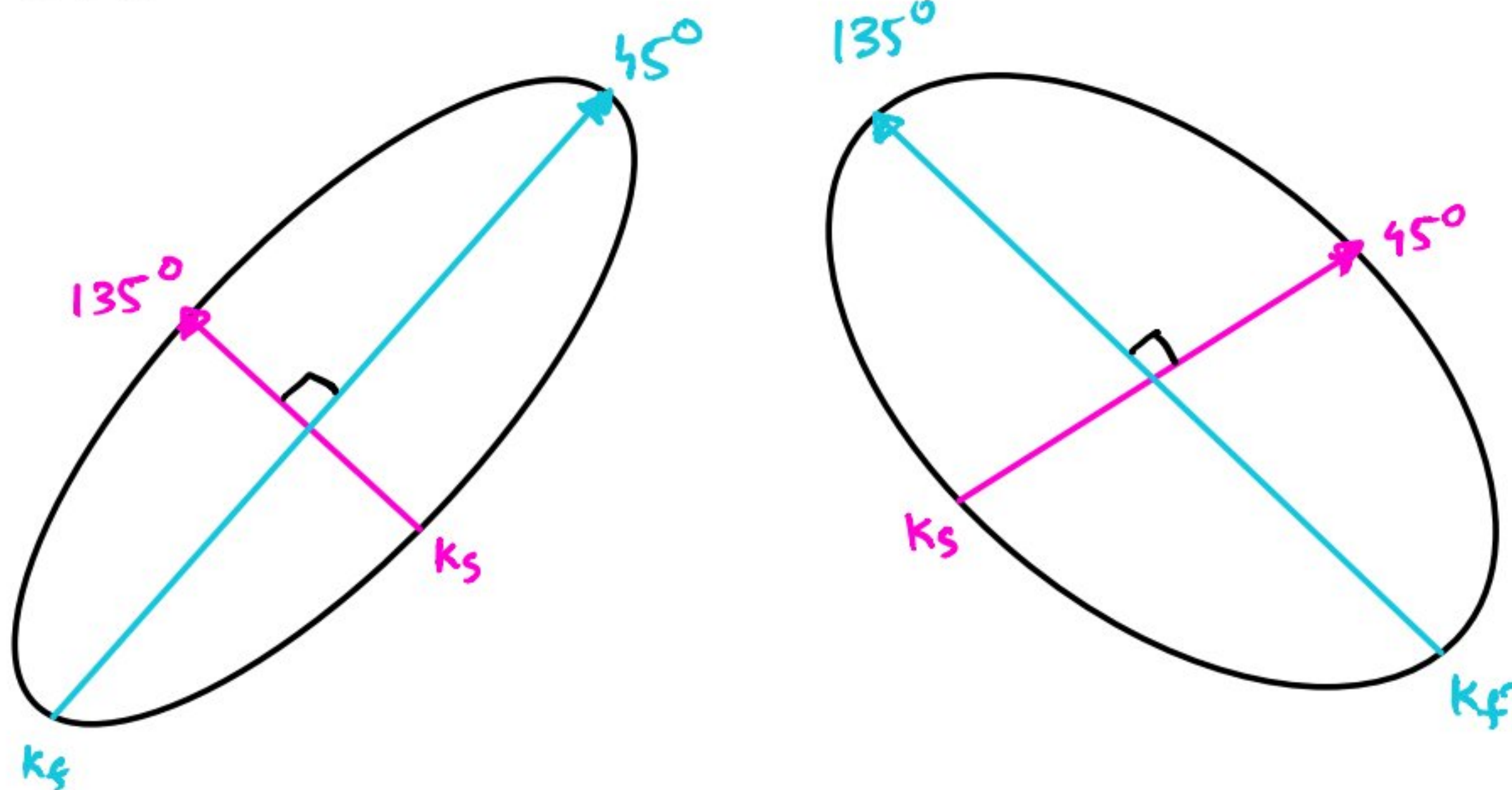
II, Against Rule:

K_s (steep) = Horizontal line i.e 180°

K_f (flat) = Vertical line i.e 90°

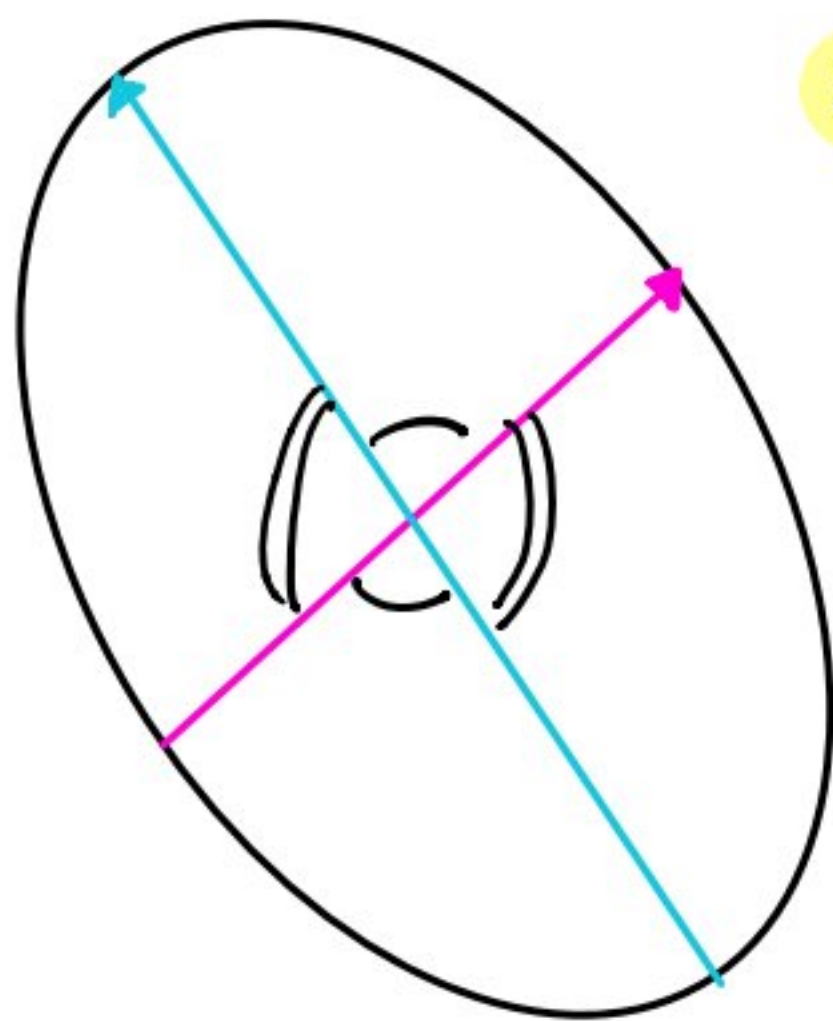


III Oblique Astigmatism:

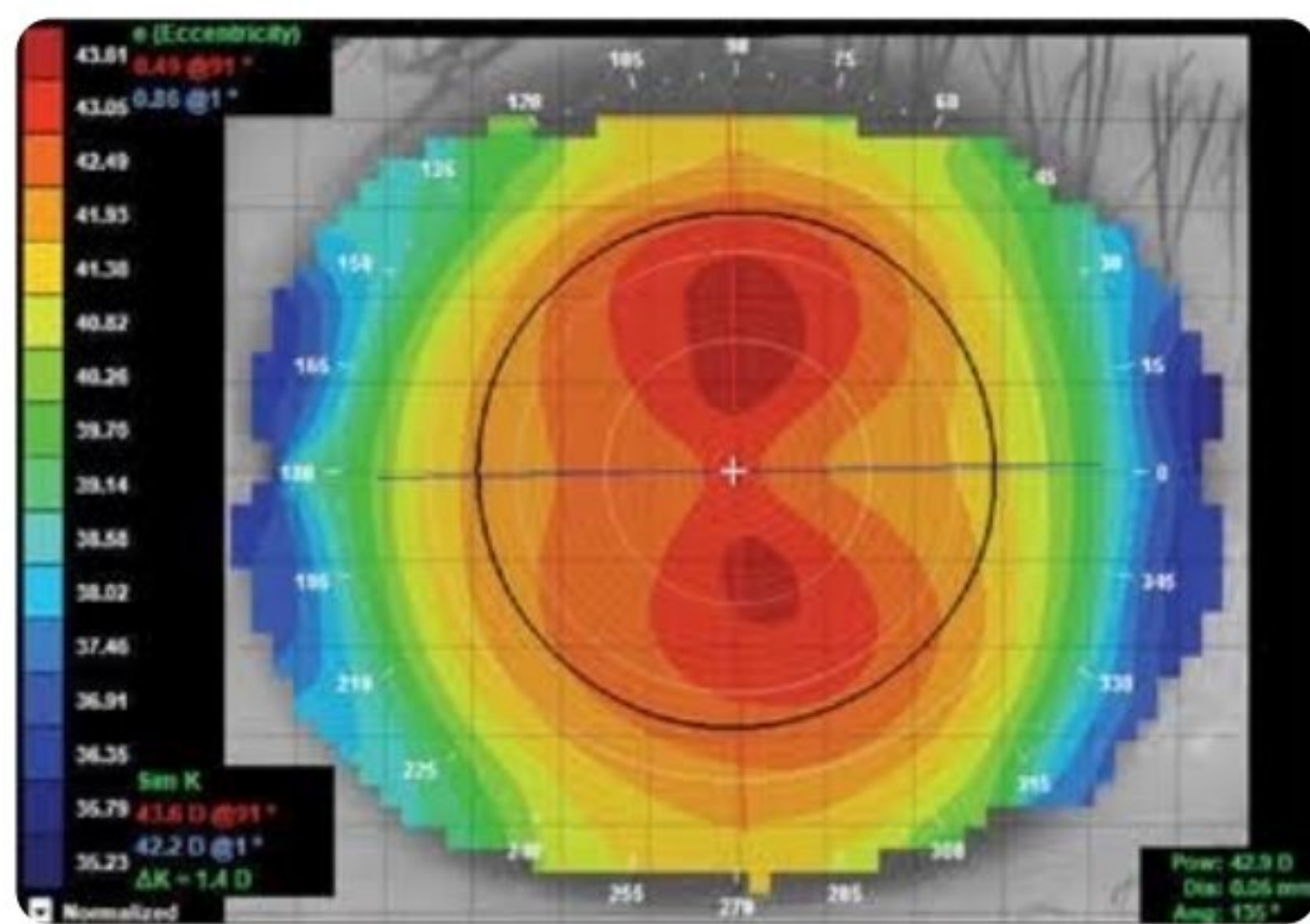


Bi-Oblique Astigmatism:

2 principle meridian are not at right angle to each other



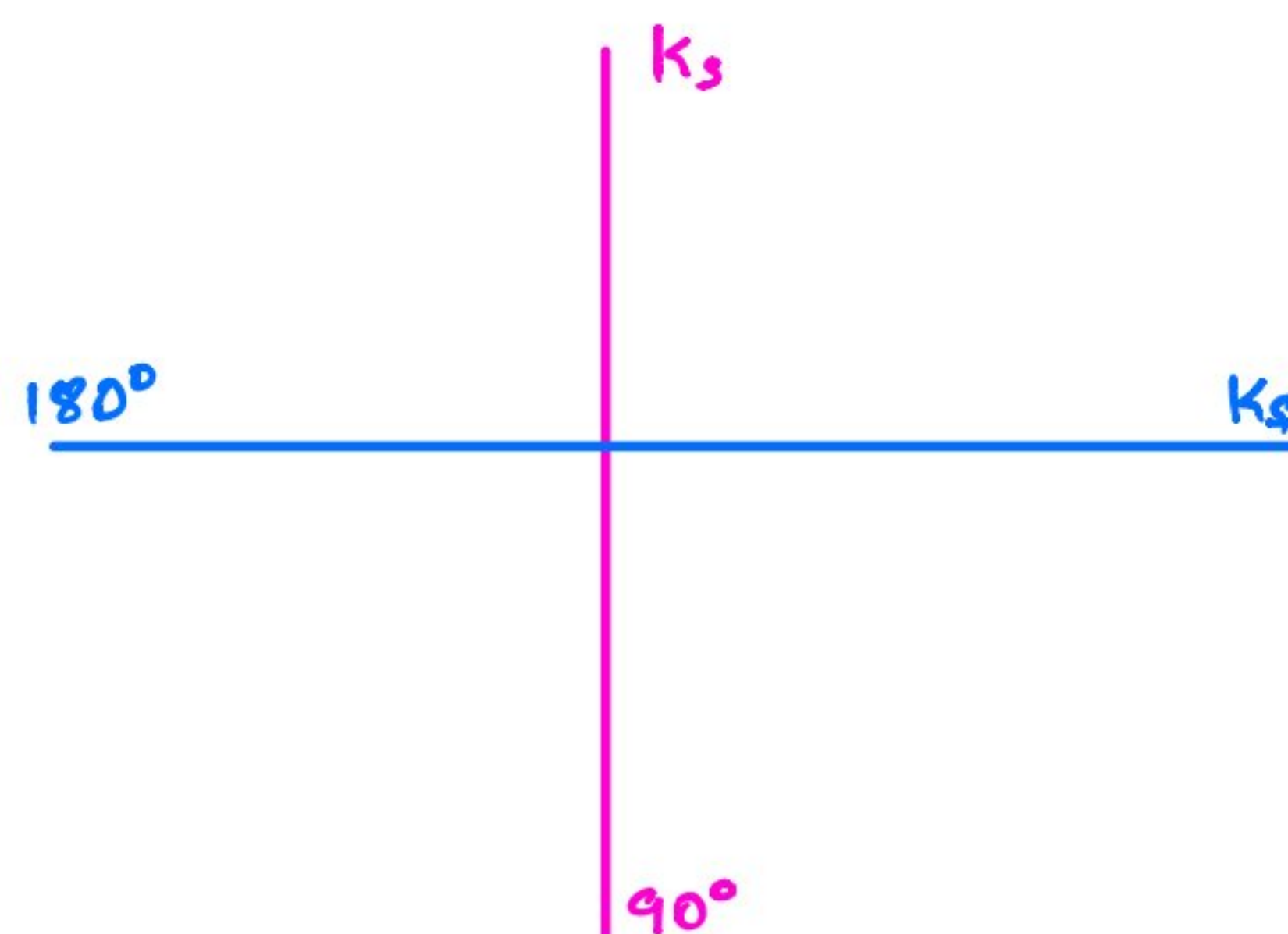
Corneal Topography of Regular Astigmatism:

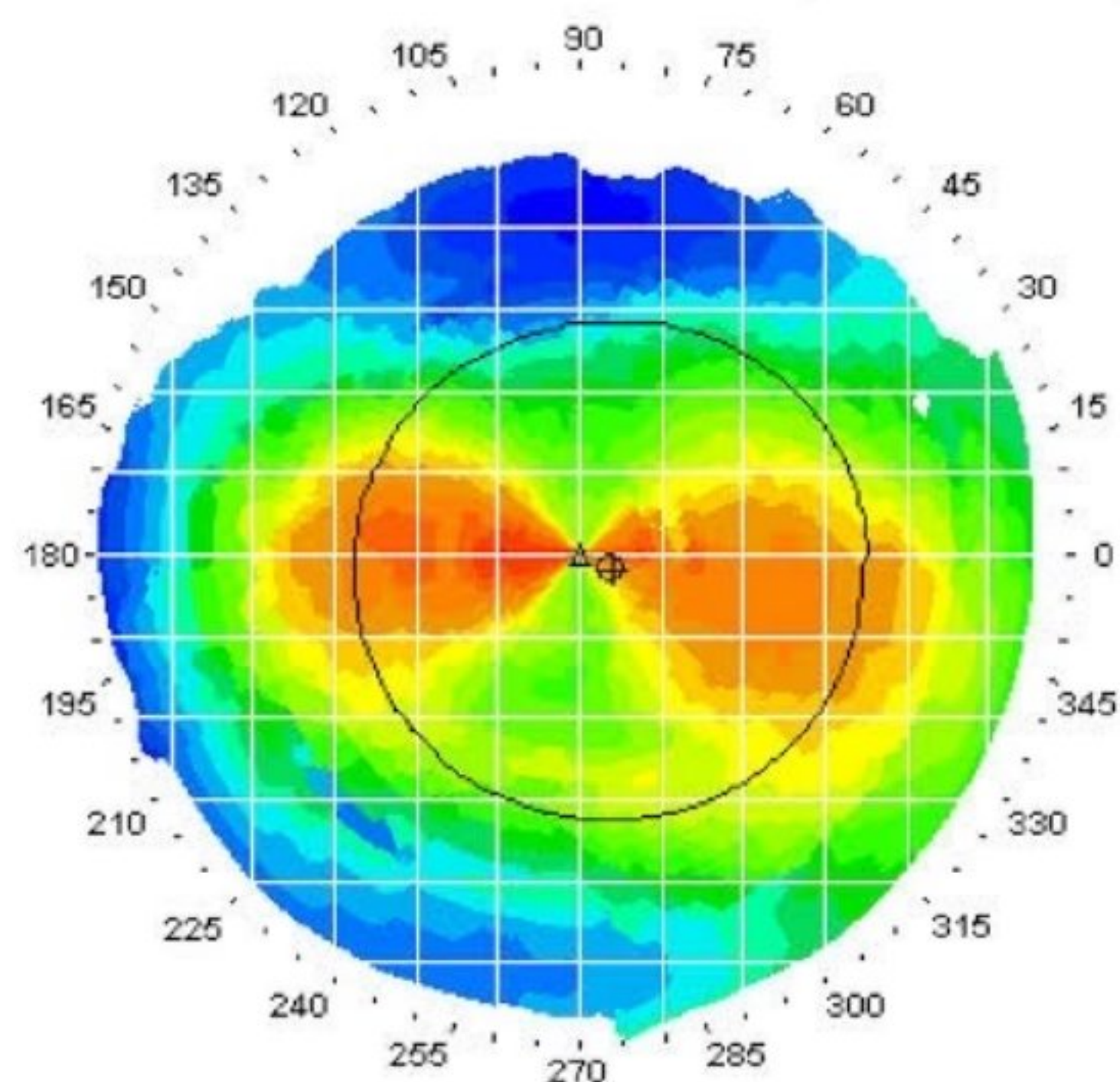


With the rule

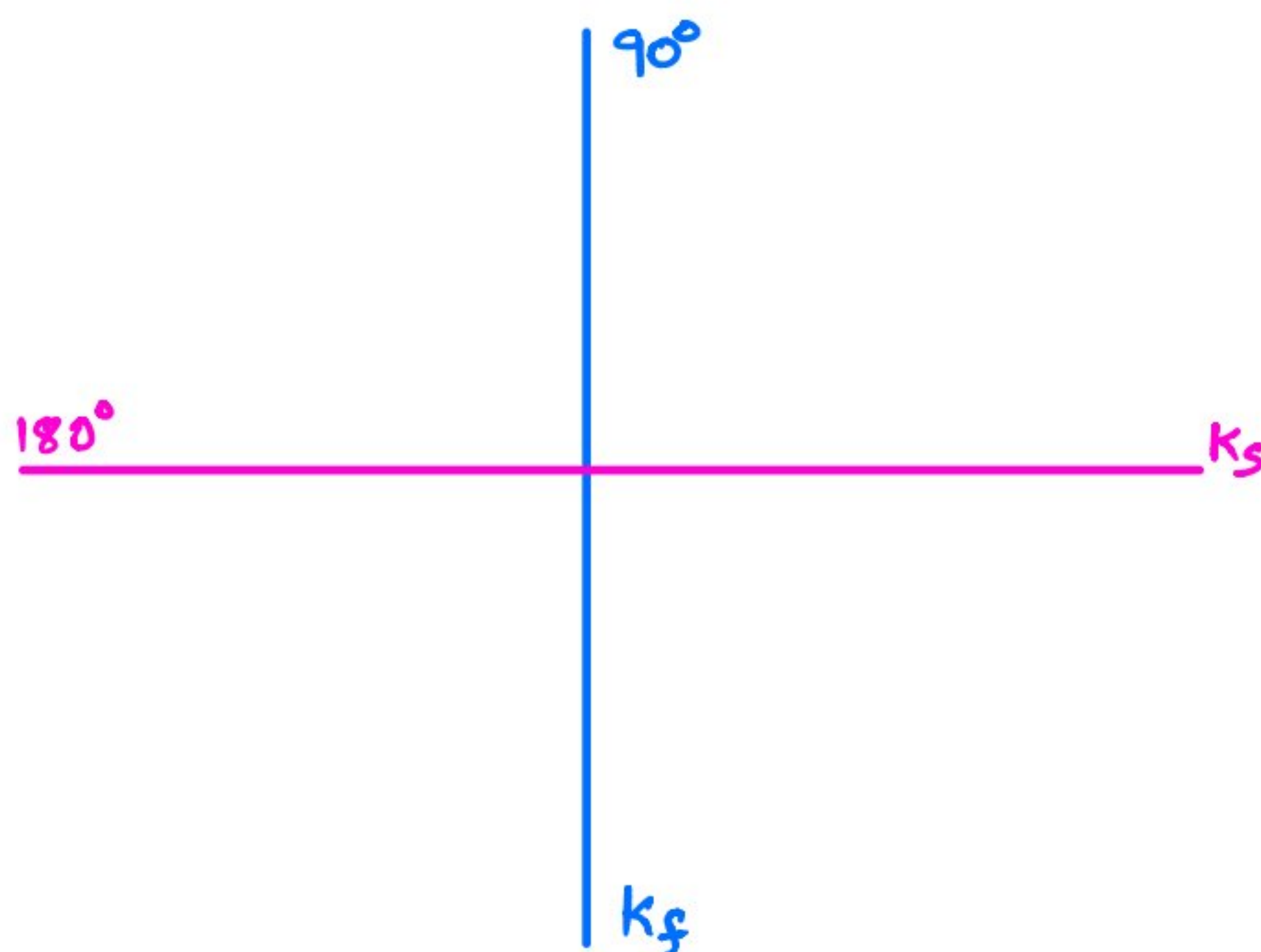
Red $\rightarrow$ Steeper Meridian

Blue/Light $\rightarrow$ Flat meridian





With out rule



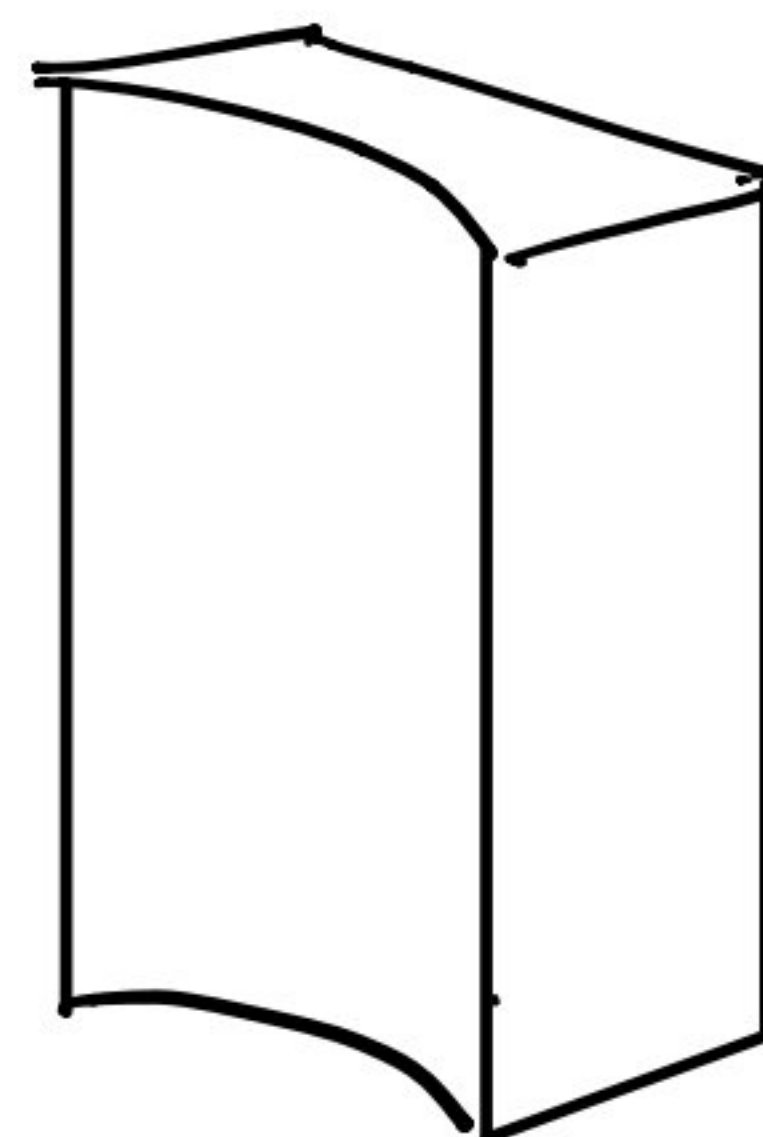
Correction of Astigmatism

i) CYLINDRICAL LENS

Convex : along the steep axis $\Rightarrow$ positive



Concave : arranged along the flat axis $\Rightarrow$ negative



ii) Contact lens

- Toric Contact lenses
- Rigid gas permeable (RGP) lens

During surgery $\longrightarrow$ • Toric IOL
• Corneal incision on the steep axis.

iii) LASIK : **OTHER PROCEDURE** : ① PARK (photo astigmatic refractive keratotomy)
② SMILE
③ Astigmatic keratotomy.

Types of Contact lens :: (Based on material)

- 1) Hard lens →
 - PMMA & polymethyl methacrylate?
 - Poor patient compliance
- 2) Rigid gas permeable lens / semi-soft lenses → made of silicone acrylate and cellulose acetate butyrate
- 3) Soft lens → Hydroxy Ethyl methyl acrylate (HEMA)

RETINOSCOPY: Objective assessment of Refractive status of patient.

/skiascopy

→ Emmetropia

/shadow test

→ Ametropia.

Prestley - Smith Retinoscope :: (outdated) → **Spot retinoscopy**



2 mirror

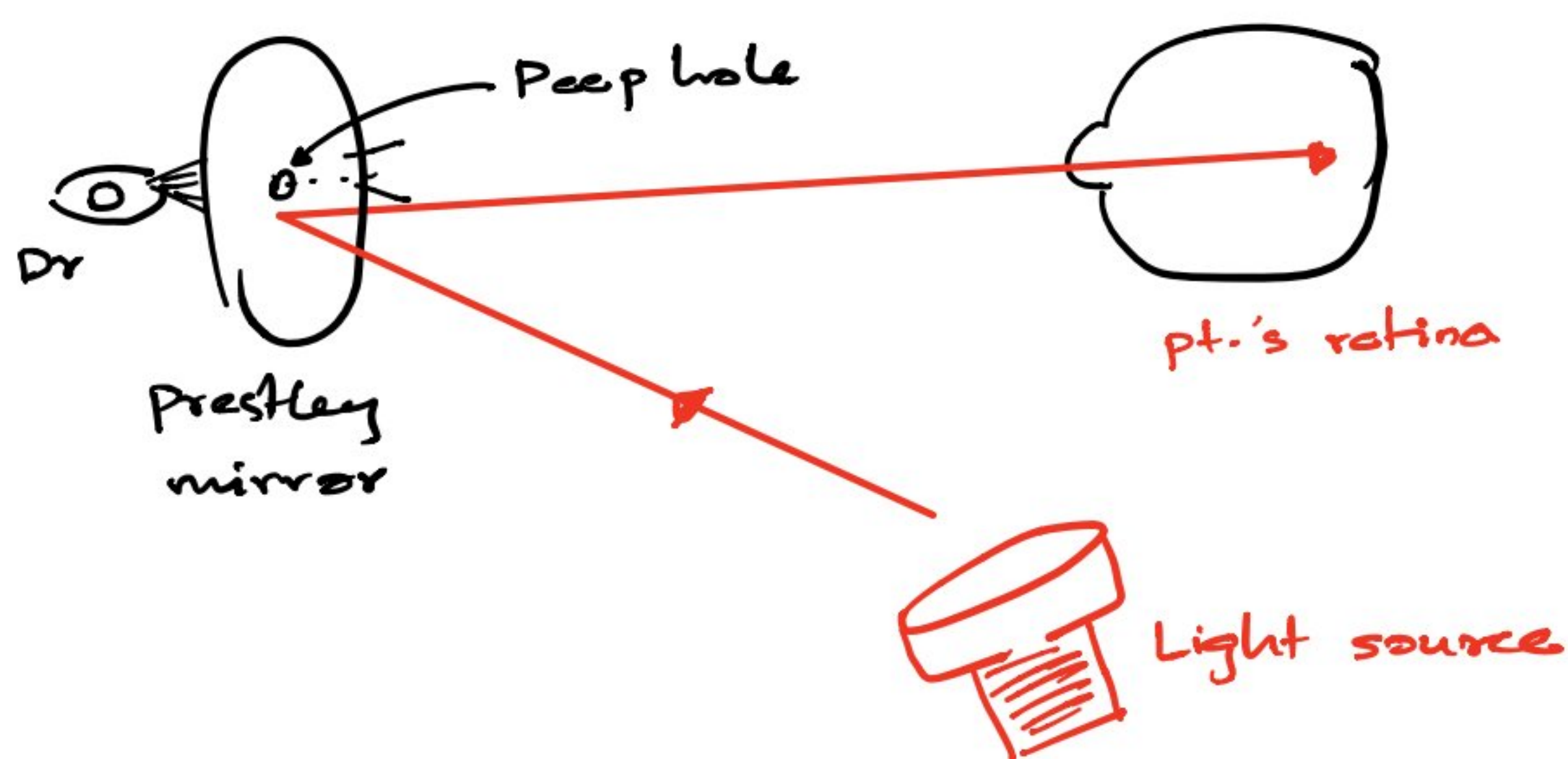
Plano mirror

MC used

Concave mirror

only used in {
• Hazy media
• High refract error.
some exception

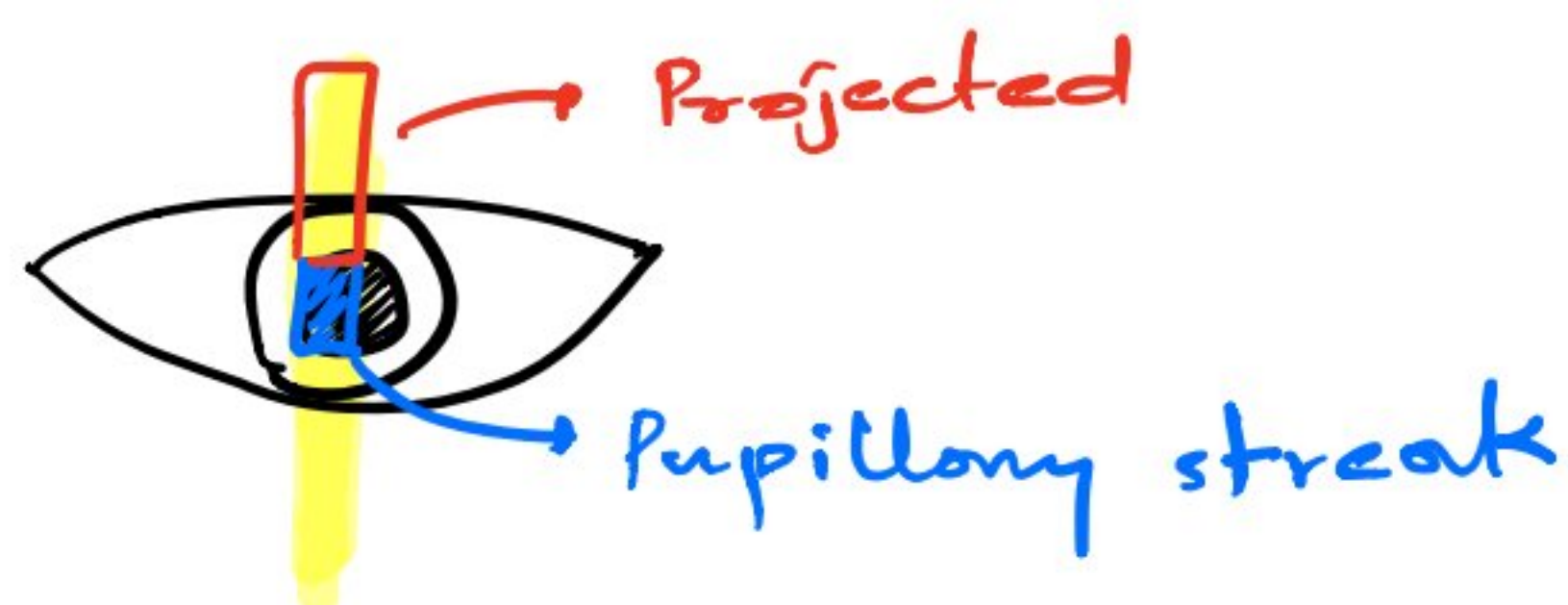
Prestley Smith Retinoscope



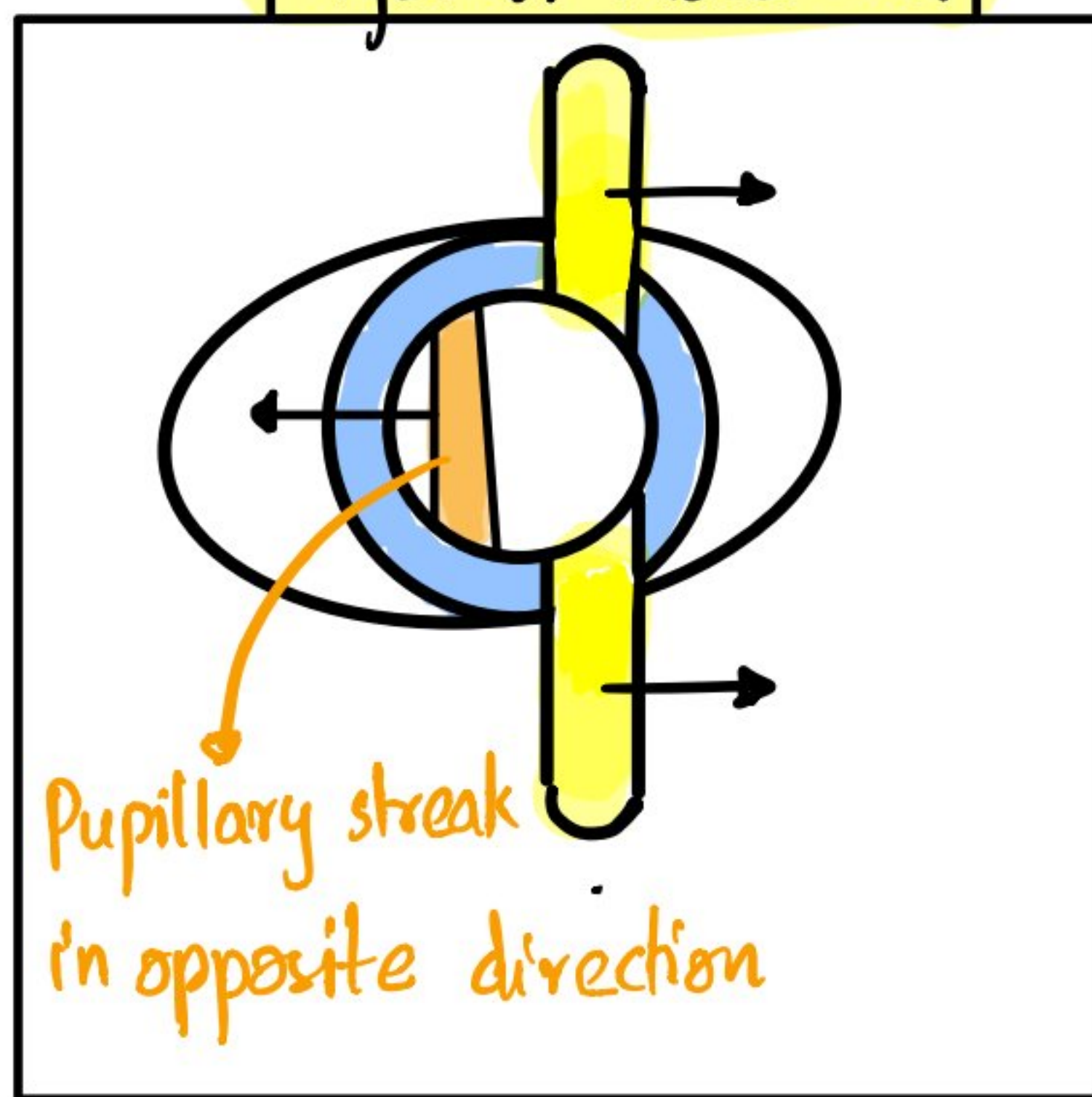
Welsh - Allyn Retinoscope → **Streak Retinoscope**



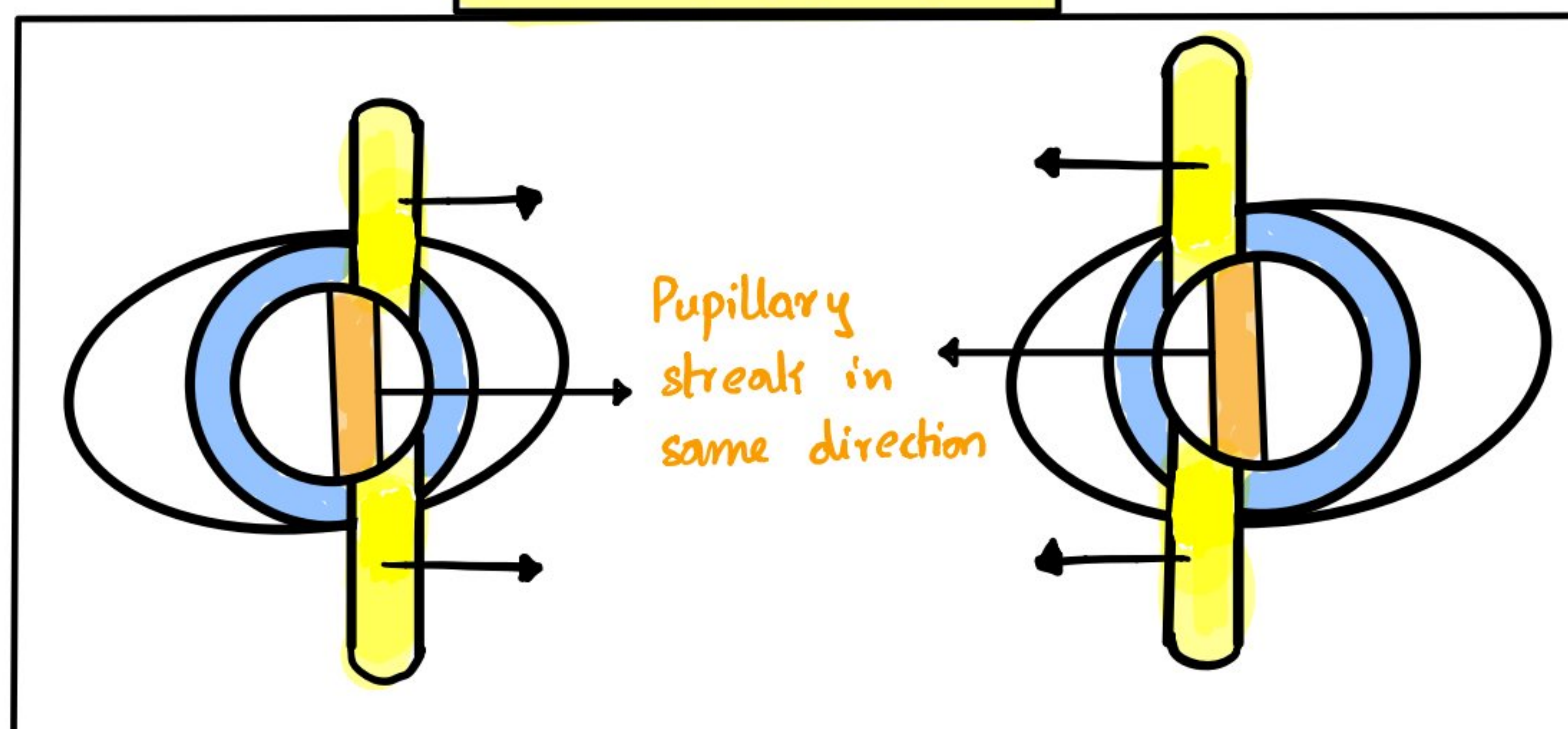
streak of light projected



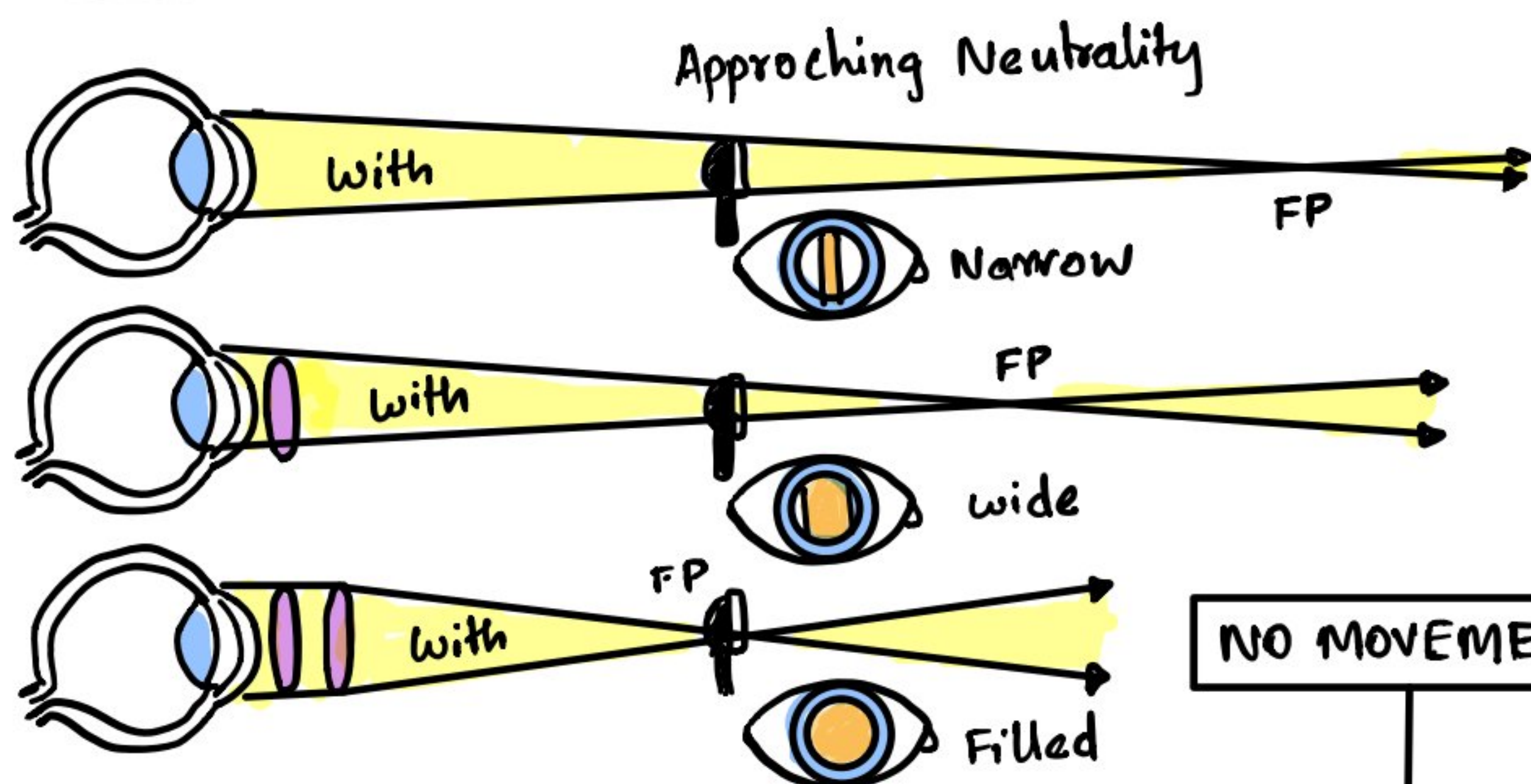
Against movement



With the movement



NEUTRALITY:



NO MOVEMENT

- filling up the entire area
- Power of the lens used to attain neutrality + $\frac{1}{\text{distance of observer}}$

CASE SCENARIO

If we get neutrality with retinoscope using +6D lens and the distance the Dr. is standing is 1D. and with the movement.

sol. Power should be < -1D or hypermetropia

$$\therefore P = +6D + \frac{1}{f} = +6 + 1 = +7D$$

is Power req. by the patient.

Retinoscopy done @ 1m under no cycloplegia

using a Plano mirror

• Against movement $\Rightarrow$ Myopia $> -1D$

(-1.25, -1.5, -1.75, ...)

• No movement = Myopia = -1D

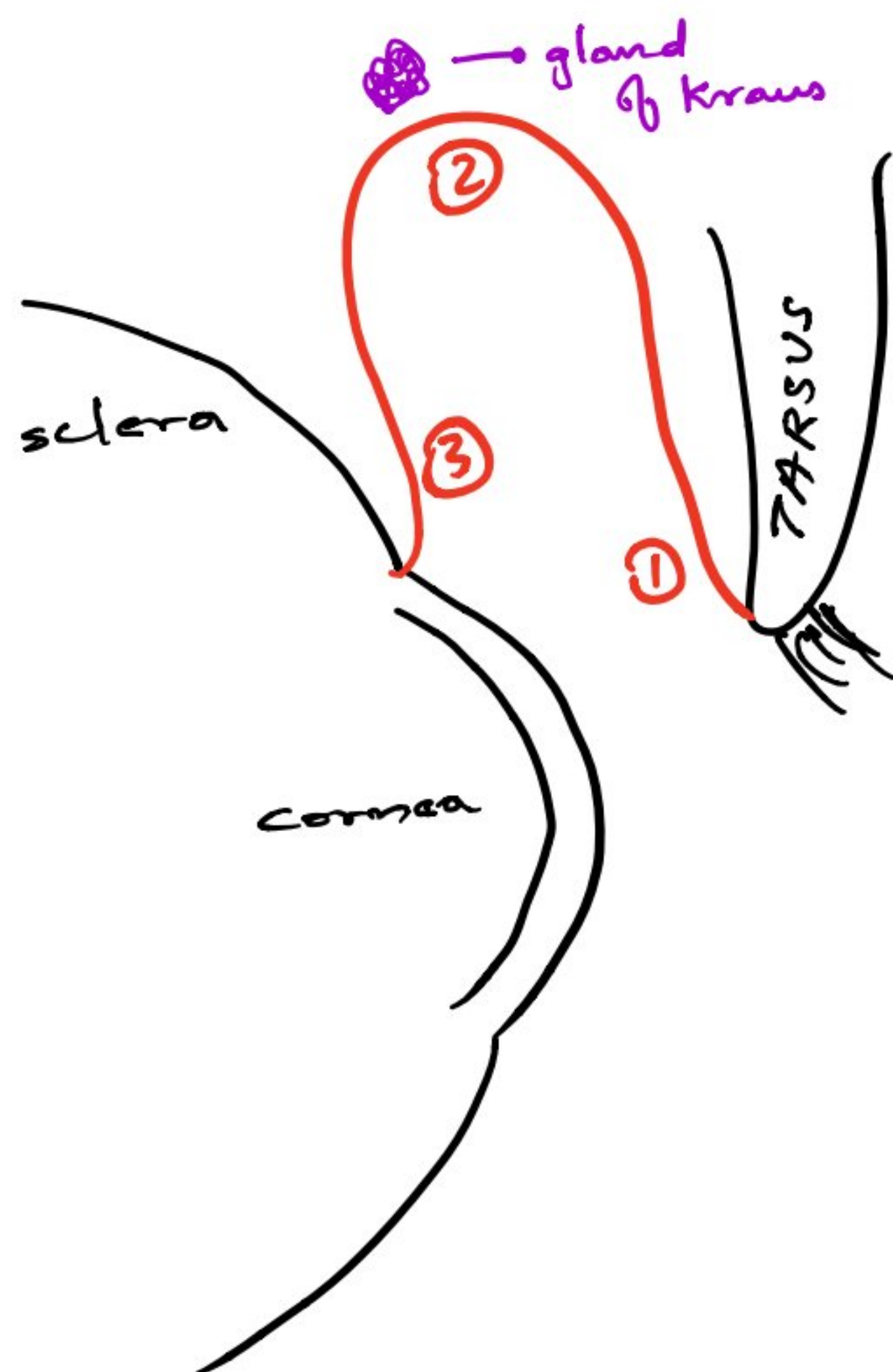
• With movement $\Rightarrow$ i, Myopia $< -1D$

(-0.25, -0.5D, -0.75D)

ii, Emmetropia

• iii, any Hypermetropia

Conjunctiva:



Parts:

- ① Palpebral conjunctivitis
- ② Forniceal conjunctivitis
- ③ Bulbar conjunctivitis

Layers of Conjunctiva

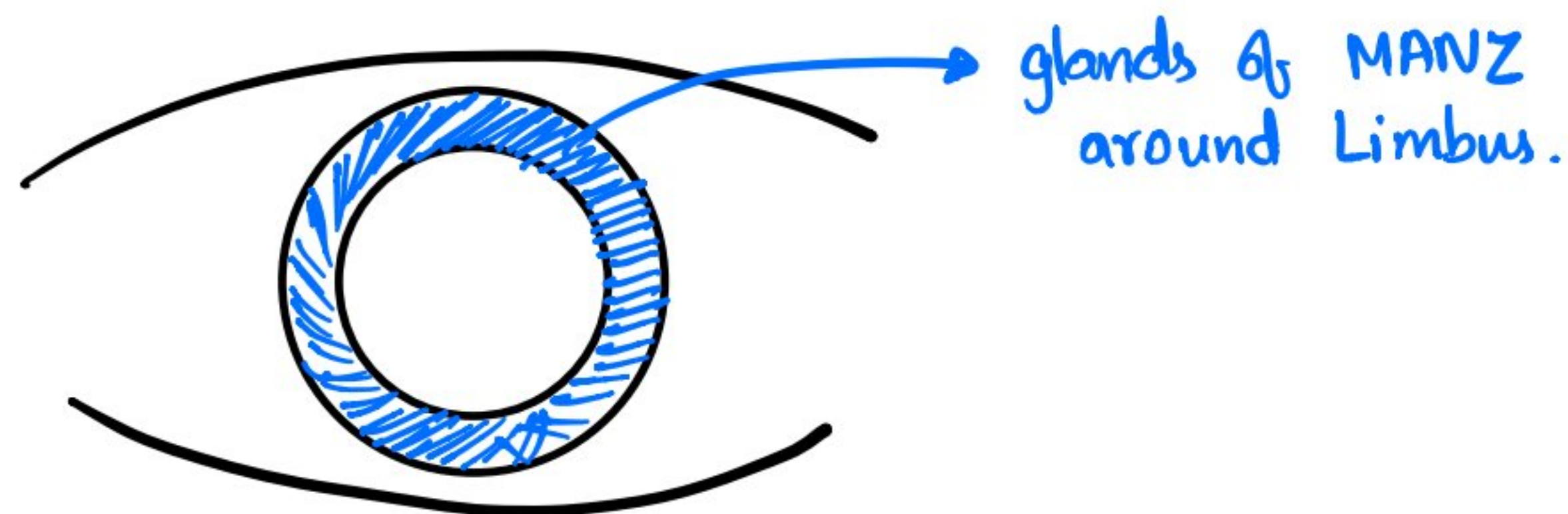
I) Epithelium

- Non keratinizing stratified squamous epithelium
- Contain **Goblet cells** → secrete **mucin**

↳ max. density @ infero nasal side of Bulbar conjunctiva

- Contain **Melanocytes**

- Langerhan's cells: Non phagocytic
fx: Ag presentation, lymphokine & prostaglandin production.



II) Stroma / Lymphoid layer / Adenoid Layer

- ↳ contains **Lymphoid tissue**
 - drain to submandibular & preauricular nodes
 - Absent in neonates.

- Conjunctiva is a part of **MALT**: Mucosa-associated Lymphoid tissue.

III) Fibrous layer → vessels & nerves (innermost)

STROMAL LAYER } substantia
FIBROUS LAYER } propria

- Accessory lacrimal gland of Krause → in fornices
 - Wolfring (along the tarsal border)
- } located deep to the substantia propria

TRACHOMA ⇒ a.k.a Egyptian Ophthalmia

- Infections d/s
- spread by fomites & flies.
- Causative organism: **CHLAMYDIA TRACHOMATIS**

Serotypes : A, B, Ba, & C ← **(MCC)** of Trachoma
≡ Serotypes : D → K ⇒ Inclusion conjunctivitis / swimming pool conjunctivitis / ophthalmia neonatorum.

WHO FISTO STAGING

Ⓕ → Follicle stage & lymphoid hyperplasia

≡ "Boiled sago grain" app.

- Contain multinucleate Giant cell $\xrightarrow{\text{k/a}}$ **LEBER CELLS**
- Presence of necrosis

within it
intra cytoplasmic
inclusion → **HP Bodies**

Δsis : > 0.5mm diameter
> 5 no.s of follicle } **upper tarsal conjunctiva**

* **Herbert's follicle @ Bulbar conjunctiva** } Pathognomic ⇒ **Corneal sign**



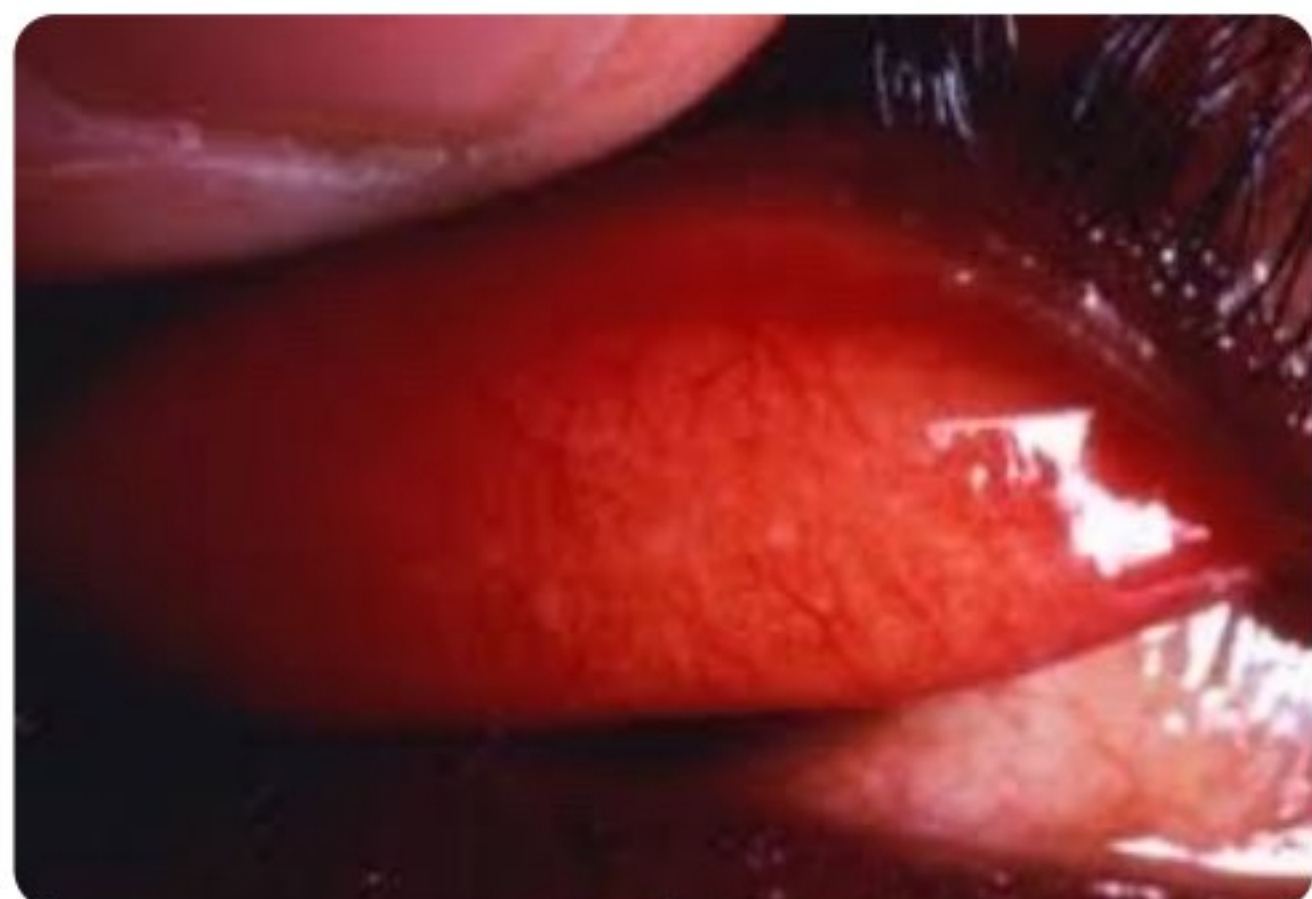
≡ Both follicles occur only in Trachoma
⊕ Papillae

Ⓖ → Intense inflammation

↳ "Papillae" ⇒ epithelial hyperplasia



Reddish appearance



Conjunctival Signs

- 1) Follicle with Leber cells
- 2) Papillae
- 3) Scarring → **ARLT'S LINE**
- 4) Congestion

Corneal Signs

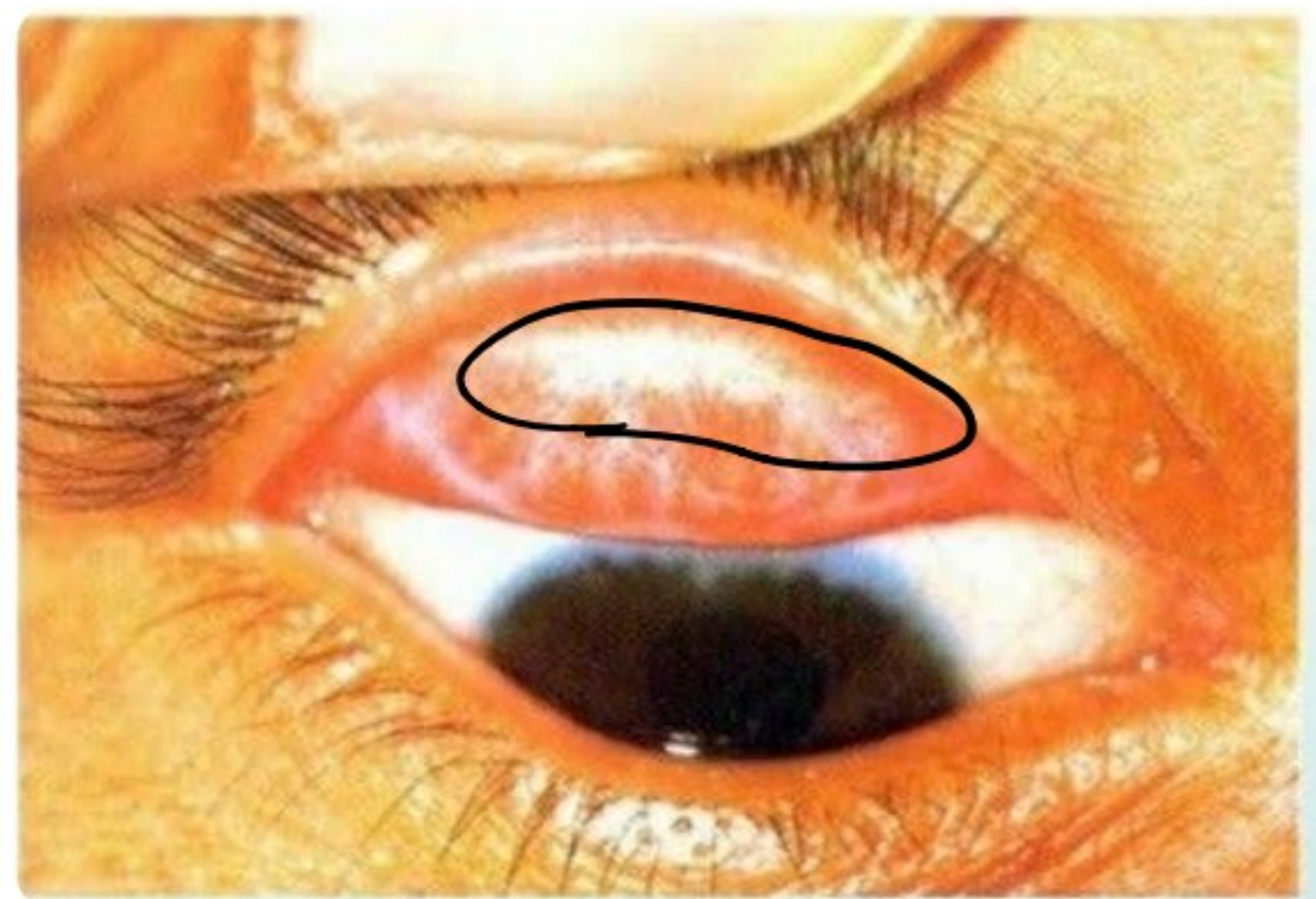
- 1) Herbert's follicles
↳ Pitted scar after healing = Herbert pits
- 2) Pannus
- 3) Superficial keratitis
- 4) Corneal Ulcer & Opacity

⑤ → Scarring

↳ On palpebral conjunctiva of U.L

ARLT's line = Horizontal white scar on palpebral conjunctiva of L.L

↳ U/L cicatricial entropion



ARTS LINE

⑦ → Trichiasis & misdirected eyelashes

↳ ⊙ = Corneal Opacities

d/t ulcer healing

or

Pannus form = cellular infiltr. + vascularisation

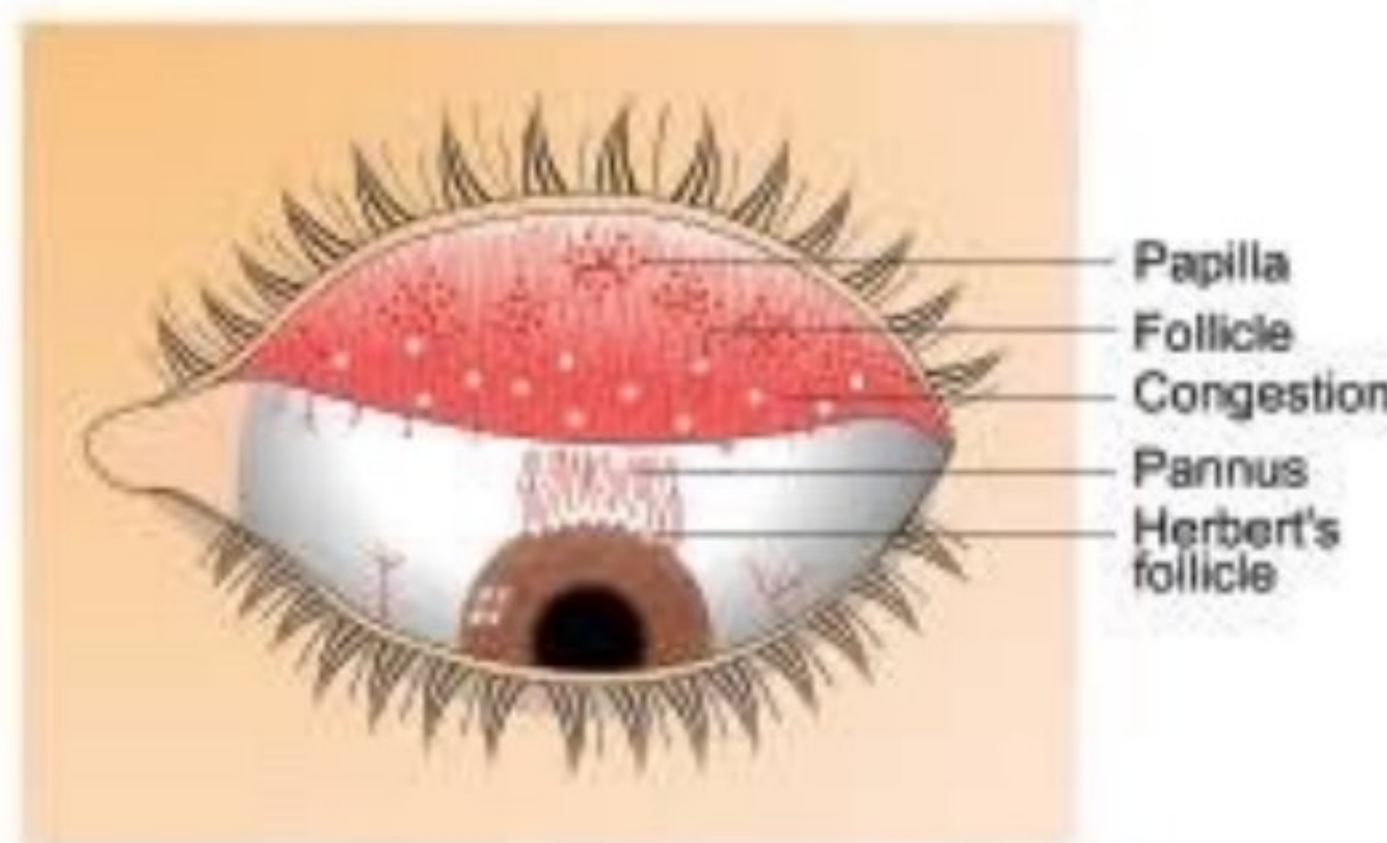
* Superior pannus

{ b/w epithelium & Bowman's layer }

Pannus crassus : Thick & fleshy ⇒ causes of blindness in Trachoma



Trichiasis



⊙ (MC) infection cause of preventable blindness (world & India) → TRACHOMA

WHO → **SAFE STRATEGY**

S ⇒ Surgery for trichiasis

A ⇒ Antibiotic : Azithro 1g OD

F ⇒ Face washing

E ⇒ Environmental cleanliness

CHILDREN AGED : 0-10yrs

SAFE strategy → indicated when the prevalence in trachoma follicle in a community is >10%. If it is b/w 5-10% → only **F.E**, if <5% → NIL

CLASSIFICATION OF CONJUNCTIVITIS:

Based on type of exudate

- a) Serous → clear fluid →
 - virus
 - allergic
 - toxic
- b) Catarrhal → Thick or viscid secretion → allergic
resembling mucus
- c) Purulent → bacterial
- d) Mucopurulent →
 - bacterial
 - chlamydial
- e) Membranous & Pseudomembranous → [Bacterial / Diphtherial]

Based on Conjunctival Response

- a) Follicular →
 - viral
 - chlamydial
- b) Papillary → allergic
- c) Granulomatous →
 - Fungal
 - Parinaud Oculoglandular Sx
 - TB
 - Syphilis
 - Sarcoidosis
 - Tularemia
 - Parasite
 - Fungal.

Based on aetiology

1) Infectious:

- a) Viral →
 - HSV
 - Adenovirus
 - Picornaviruses
 - Myxovirus (measles)
 - Paramyxovirus (mumps, Newcastle conjunctivitis)
 - Molluscum Contagiosum
- b) Chlamydial →
 - Adult / acute inclusion conjunctivitis D → K
 - Trachoma → A, B, Ba, C

PARINAUD'S OCULOGLANDULAR L1, L2, L3

- c) Fungal →
 - Aspergillus
 - Candida
 - Nocardia
 - R. seeberi
 - Actinomyces
 - Sporothrix.
- d) Parasite

SPRING CATARRH a.k.a Vernal Keratoconjunctivitis or warm weather conjunctivitis

Ⓜ seen in boys 5-10 years of age

Ⓐ Type:

- a) Palpebral
- b) Limbal VKC
- c) Mixed

C/F :: a) Intense Itching

b) ROPY DISCHARGE - MAXWELL LYON SIGN

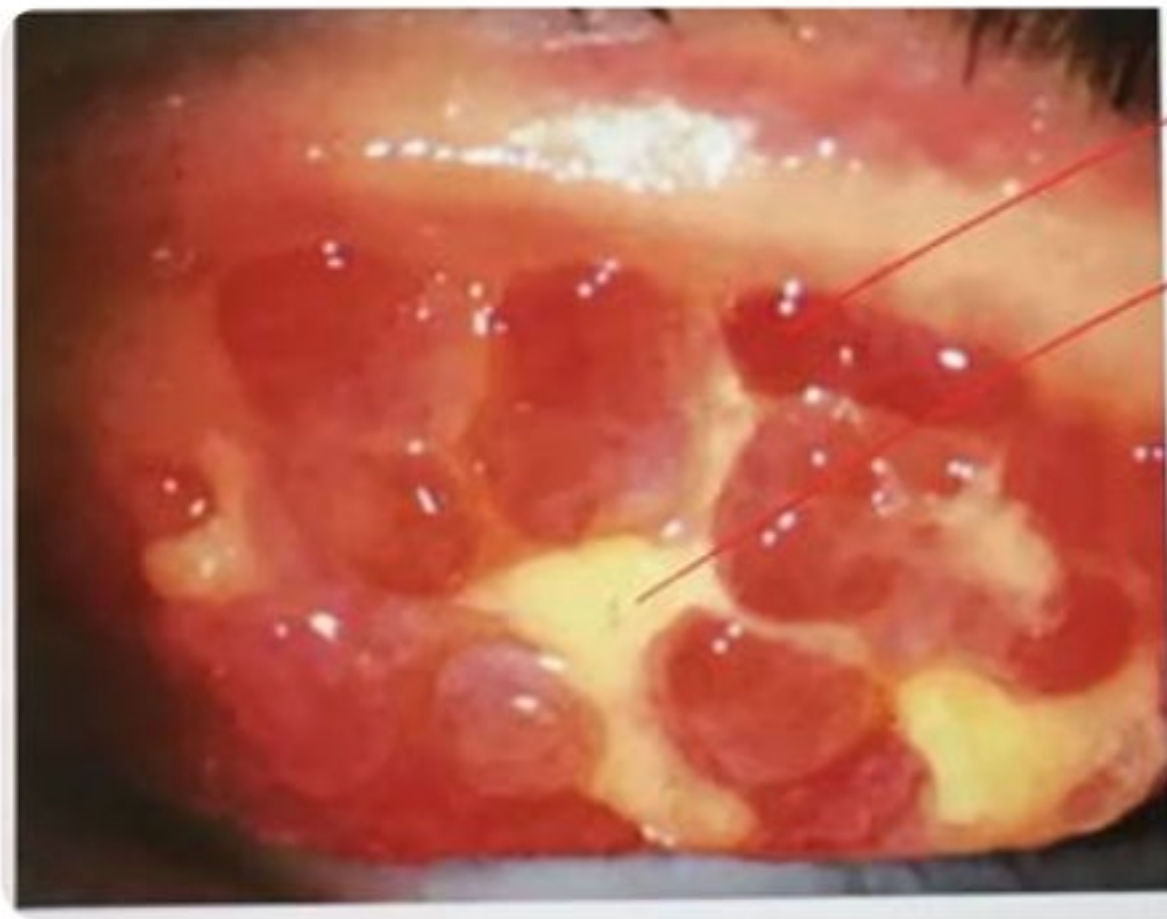
c) Palpebral VKC → cobble stone appearance (Giant papillae >1mm)

d) Limbal VKC → Horner Tranta spots

Pathology : Type I HS → IgE mediated
Type II HS → Cell mediated



Cobblestone appearance- papilla



Maxwell Lyon sign

papillae

coagulated discharge



made up of Eosinophils
Epithelial debris

HORNER - TRANTA DOTS

Corneal finding in VKC

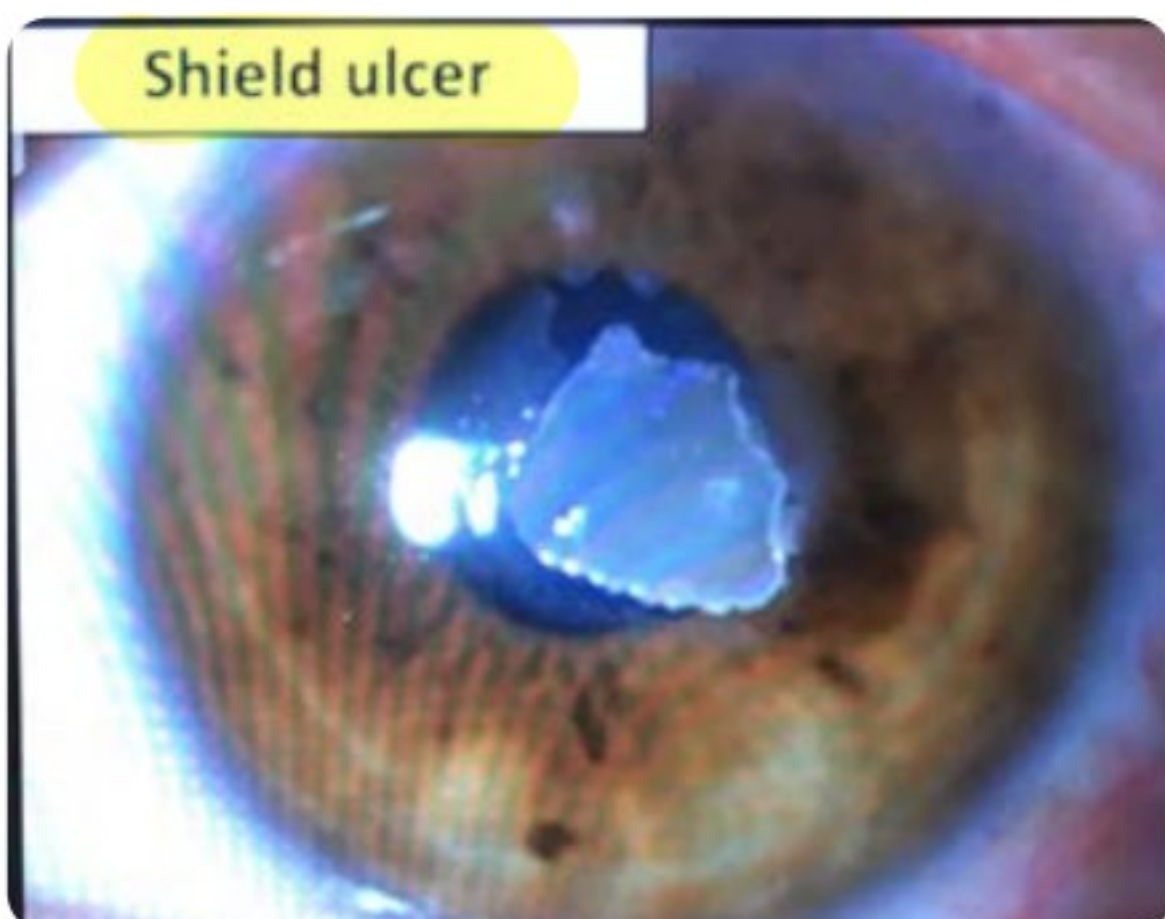
① shield ulcer

② Pseudogerontoxon



mimics: Arcus senilis

③ 2° Keratoconus → d/t repeated rubbing



Shield ulcer

Treatment:

MEDICAL:

a) Steroids - Fluorometholone and Clobetasone → Topical, supratarsal injection of

triamcinolone (Kenacort) in refractory cases

- b) DOC → Antihistaminics with Dual action (Antihistaminics + Mast) cell stabilizers
↳ Sodium Cromoglycate, Ketotifen, Olopatadine and Epinastine
- c) Refractory VKC → Cyclosporine (Topical), **TACROLIMUS^a**
- d) ACETYL CYSTEINE 0.5%.

SURGICAL

- a) Excision of papillae
- b) Amniotic membrane transplantation

c) Photo-therapeutic keratectomy

PHLYCTENULAR KERATOCONJUNCTIVITIS

• Predominantly affects children.



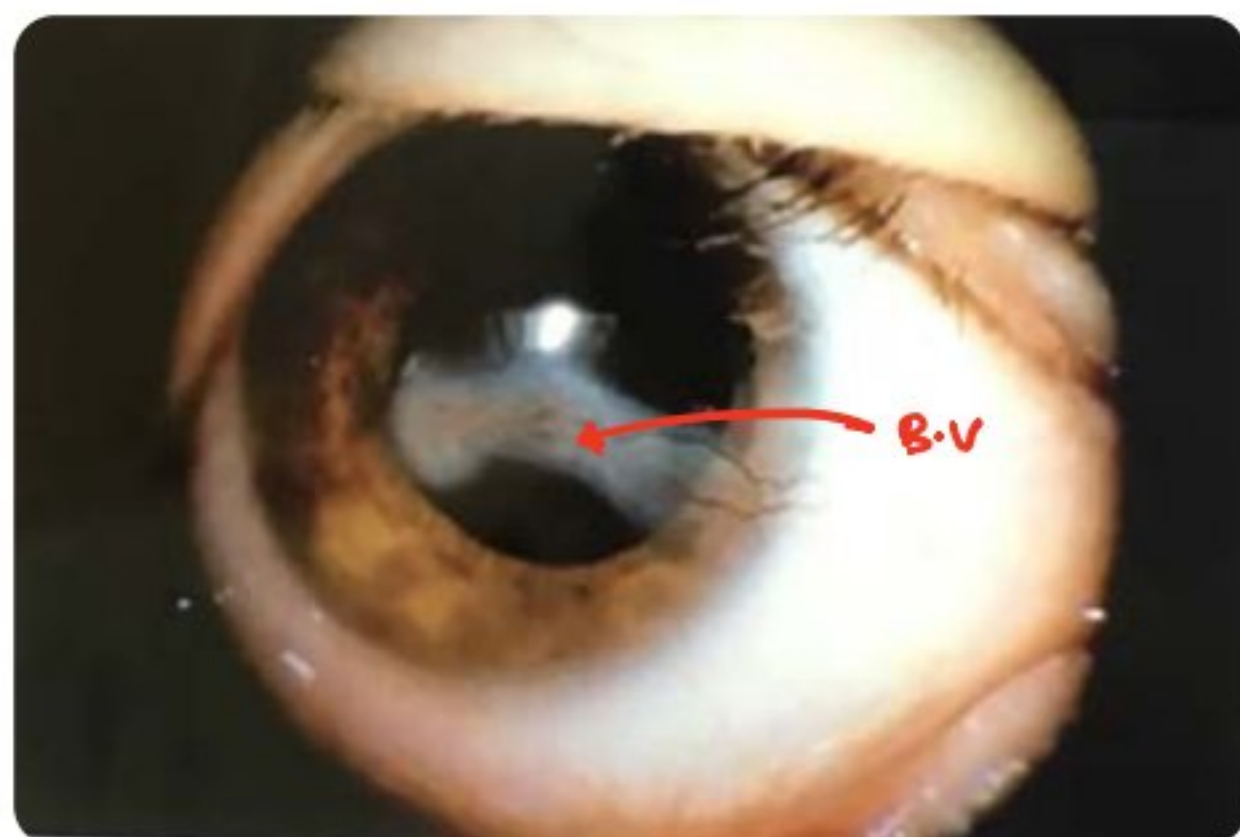
Phlycten

ETIOLOGY: Delayed hypersensitivity reaction to [Type IV]
i) Staphylococcal protein (most common)
ii) TB proteins.

CLINICAL FEATURE:
i) Limbal nodule
ii) Fascicular corneal ulcer

T.O.C = Topical Steroids

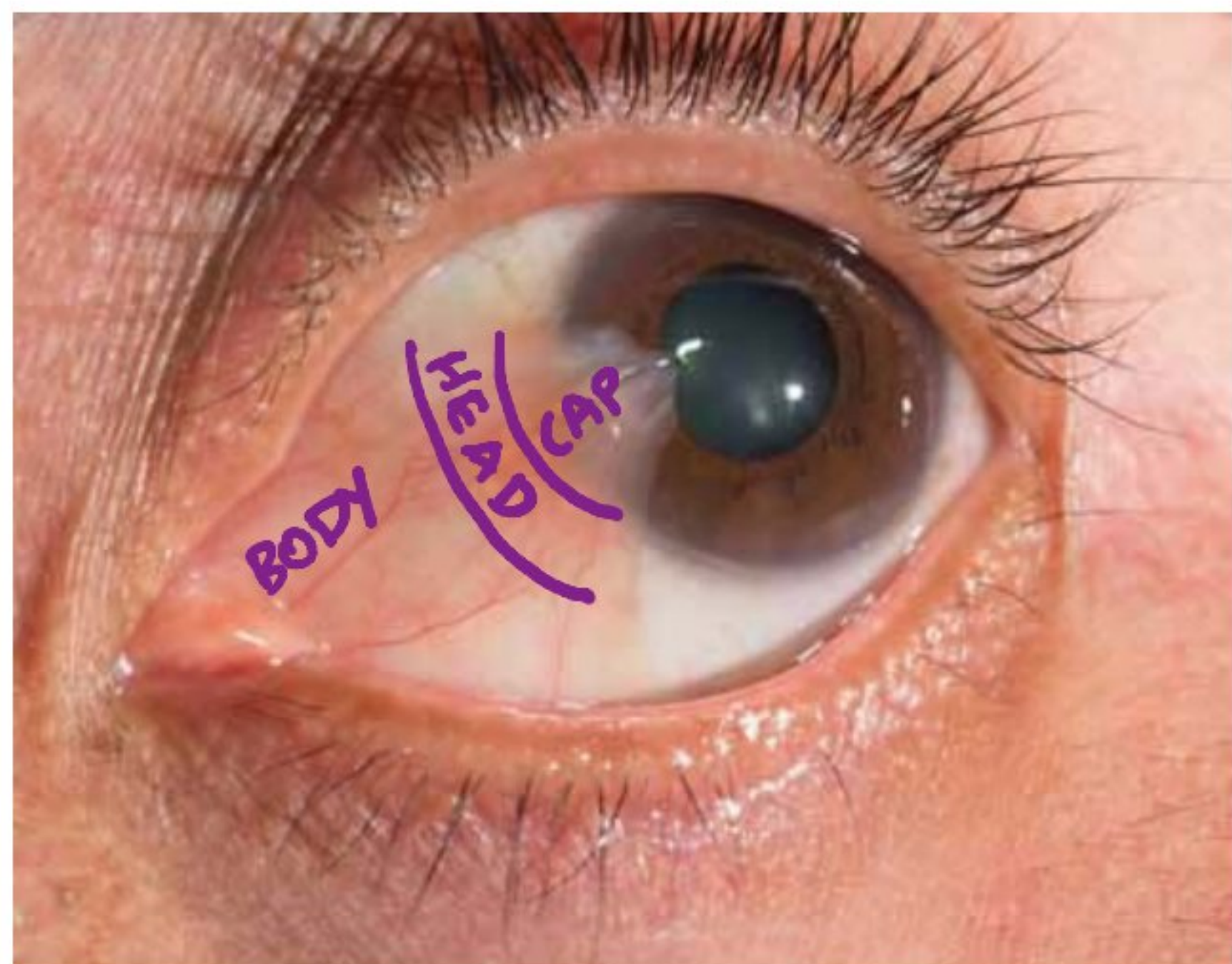
Complication:
(unique)



Fascicular Ulcer

CONJUNCTIVAL DEGENERATIONS

Pterygium:



Fibrovascular proliferation of conjunctiva & subconjunctiva tissue over cornea.

* Connective tissue d/s and is never inflammatory

Risk factors:

- 1) Sunlight (UV-B)
- 2) Smoking
- 3) p53 mutation
- 4) **HPV infection**
- 5) **LSC** deficiency
Limbal stem cell

STOCKER'S LINE

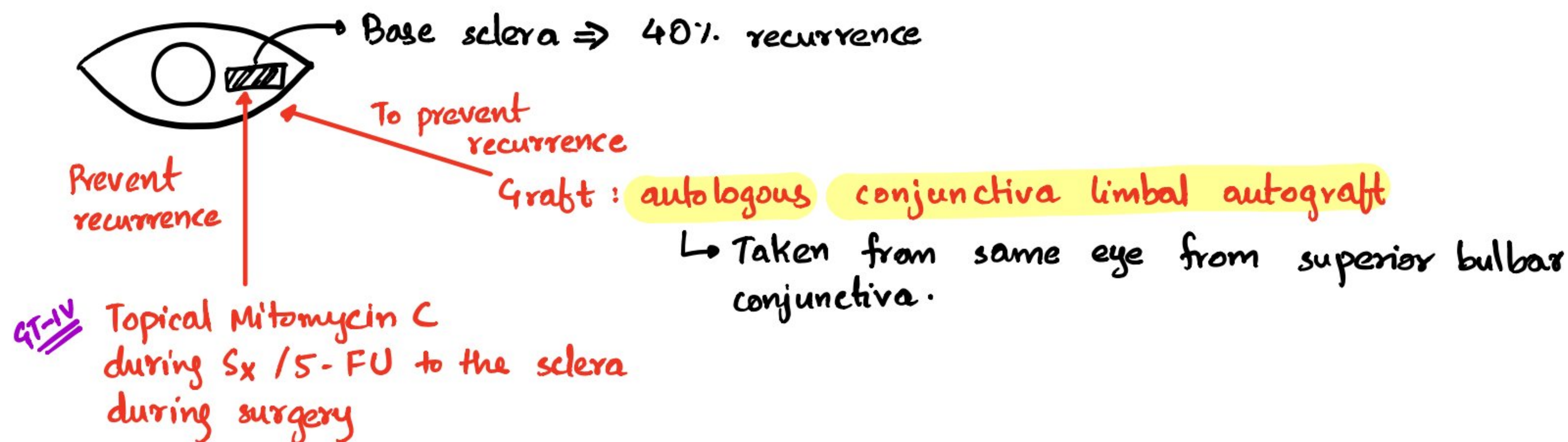
↳ deposition of iron on the advancing edge (cap)

FUCH'S ISLETS

↳ Whitish fleck with pterygial epithelial cells at the advancing edge (cap)

- C/F: 1) Triangular fibrovascular fleshy ingrowth.
 2) MC on NASAL SIDE (96%) / (4%) are Temporal.
 3) Glass rod (probe) test: to distinguish from pseudo-pterygium.
 4) Blurring of vision 5) Consist of head cap and body

Management: Surgery: excision of pterygium



CONJUNCTIVAL DEGENERATIONS

- 1) Pinguecula
- 2) Pterygium
- 3) Concretions $\rightarrow$ small yellow white deposits in palpebral conjunctiva
- 4) Retention Cyst $\rightarrow$ Asymptomatic thin wall clear fluid cyst d/t obstruction of Krause's gland.
- 1) Pinguecula $\rightarrow$ Yellowish white deposits on Bulbar conjunctiva adjoining limbus in the palpebral fissure are
 - i) Usually B/L
 - ii) More nasally > temporal
 - iii) A/w Macular age related degeneration (x3)

Rx: NSAID or Topical corticosteroids
 for cosmetic / c/c inflammation } excision may be done of pinguecula



Pingicuela

PTERYGIUM	PSUEDO-PTERYGIUM
• Degenerative Process occurs in old @ palpebral aperture • Progressive / regressive or stationary • Probes cannot be passed underneath	• Inflammatory Process at any age, at any site • Always stationary • Probes can be passed under the neck

Subconjunctival H'mage :

- (MCC) is blunt trauma → usually reduce by itself within 1-3wk

Sub-conjunctival hemorrhage



If posterior limit of SCH is not seen



Base of skull #

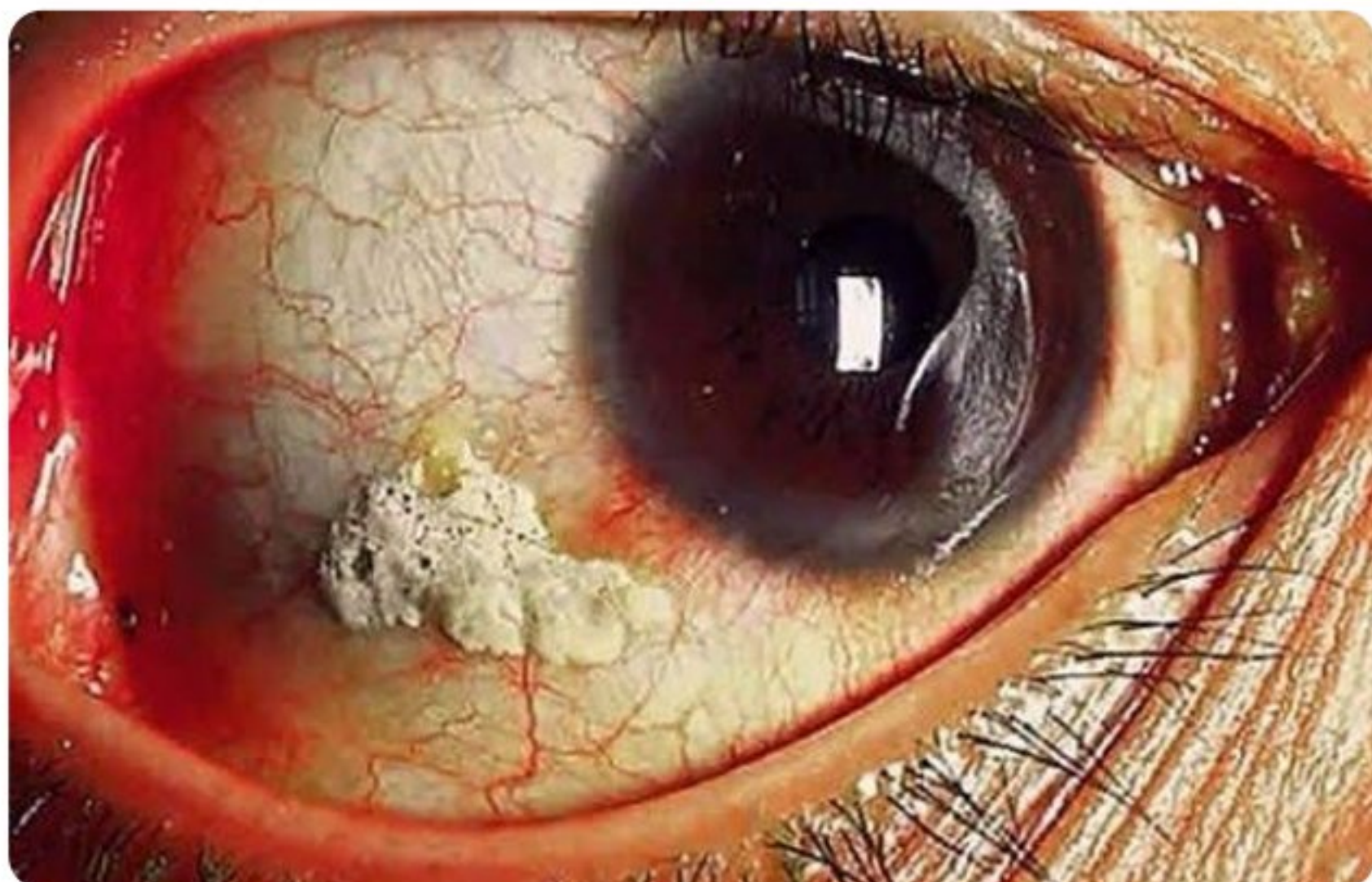
Q: It occurs in

- Trauma
- Repeated eye rubbing
- Pertussis

Q: But it does not occur in High IOP

BITOT SPOT ⇒ Vit A deficiency

Q: @ Temporal bulbar conjunctiva



- Usually B/L
- Irreversible

Membranous Conjunctivitis ⇒ Caused by Corynebacterium Diphtheriae

- Other cause :-
- β hemolytic streptococci
 - S. pneumonia
 - Haemophilus aegypticus
 - Neisseria gonorrhoea

* They can also occur as a complication of Steven-Johnson Syndrome

v) Staph aureus

vi) E. coli

C/F :: Usually affect children

Q: **Bleed on peeling**

Rx :: Anti diphtheric serum

complication :: Symblepharon

Pseudomembraneous Conjunctivitis

Cause :

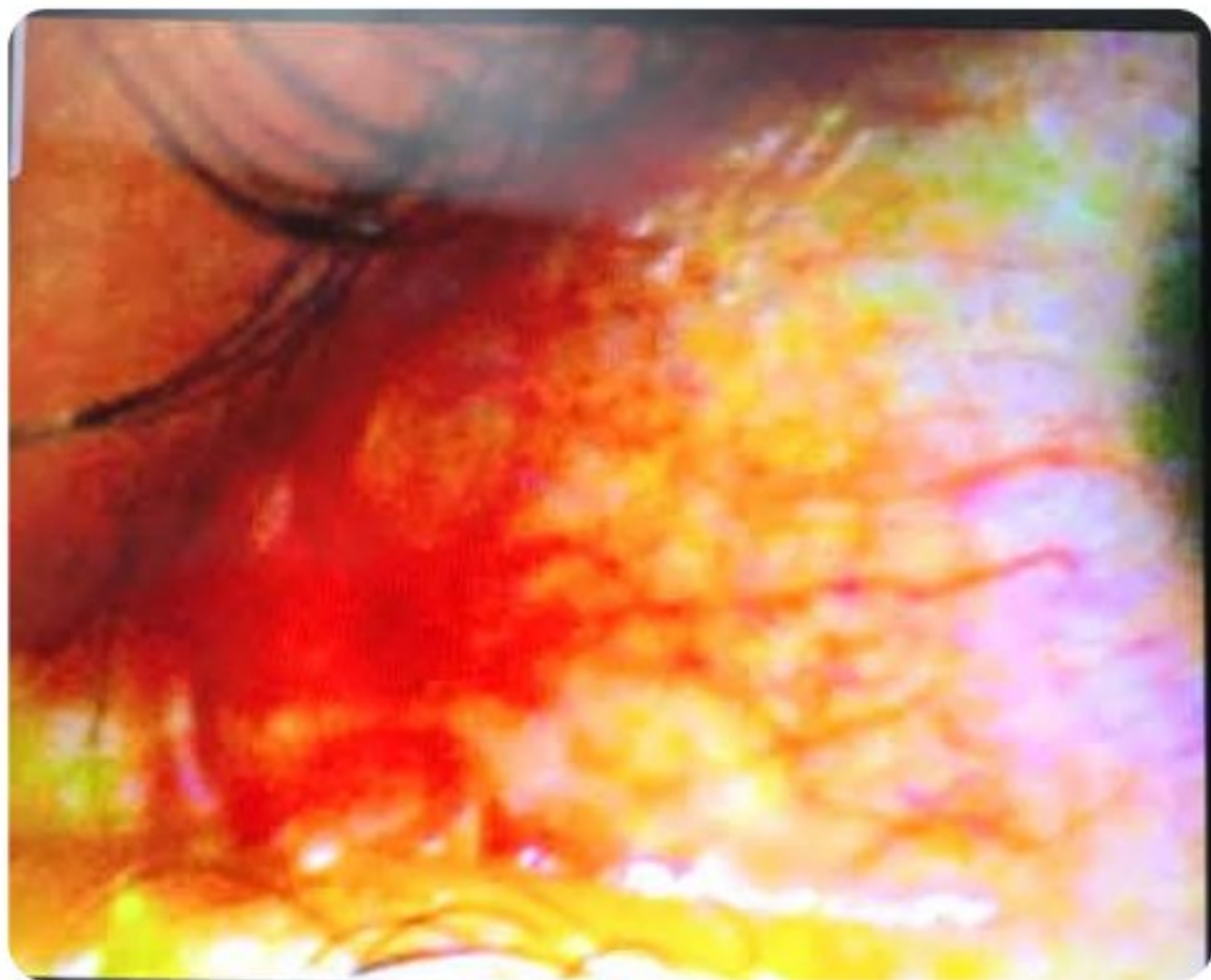
- Gonococcal Conjunctivitis
- Autoimmune conjunctivitis
- Ligneous conjunctivitis
- Severe adenoviral infection

Q: C/F : Doesnot bleed on peeling

Rx : Antibiotics

NEET-PG

ANGULAR CONJUNCTIVITIS : Inflammation of the inner and outer canthi with excoriation of skin & collection of dry white, foamy discharge at the angles.



Causes:

- i) Moraxella axenfield
 - ii) Moraxella lacunata
 - iii) Moraxella catarrhalis
 - iv) Staph aureus (Rare)
- } Gram -ve diplobacillus

Source of infection: Nasal discharge from patients with angular conjunctivitis
Nose → eye.

Rx :: Oxytetracycline x 10-14 day

• Zinc Oxide → inhibits proteolytic fermentation

Viral conjunctivitis

Hemolytic Conjunctive / Epidemic Hemorrhagic conjunctivitis / Apollo conjunctivitis

- S/i i) Enterovirus 70 (M.I) } picornavirus group & coxsackies virus A
ii) Coxsackie virus 24
iii) Echo virus 34
iv) Adenovirus 8, 19, 37

Clinical Scenario A patient complaints of pain of B/L with congestion Blurring of vision, photobia and mucopurulent discharge. Many cases have b/n reported from the same community

FOLLICULAR CONJUNCTIVITIS:

- Pharyngoconjunctivitis fever → Adenovirus 3, 4 & 7.
- Epidemic keratoconjunctivitis → Adenovirus 8 & 19 → also 37
- Newcastle conjunctivitis → contact with diseased fowls.
- Inclusion conjunctivitis → Chlamydia Trachomatis D → K

OPHTHALMIA NODOSA:



- Occurs d/t irritation by caterpillar hair
- D/D :: TB
- Microscopy :: Caterpillar hair surrounded by giant cells & lymphocytes
- Rx :: Nodule excision

Ophthalmia Neonatorum:

Neonatal conjunctivitis { affecting neonates less than 1 month old }

Causes:

1) Chlamydia : most common in developed countries

2) Gonorrhoea

3) Chemical conjunctivitis : 1% AgNO₃ [Credé's method causes chemical conjunctivitis]
↳ Was prophylaxis used for preventing ophthalmia neonatorum.

4) S. aureus

5) Herpes Simplex type 2

Table: Ophthalmia neonatorum: mode of presentation, differential diagnosis and treatment.

Time of onset after birth	Differential diagnosis	Treatment
Within the first 48 hours	Chemical (Minimal Congestion)	Wash eyes, erythromycin ointment, observe, usually improves in 24 hours
within 4-6 hours	Neisseria gonorrhoeae (Hyperacute Congestion)	Ceftriaxone injection i.m. single dose gentamicin drops. Bacitracin eye ointment
48-72 hours	Other bacteria like S. Aureus	Neomycin - bacitracin eye ointment, gentamicin, or tobramycin drops
5 - 7 days	Herpes simplex virus (HSV II)	Acyclovir 3% eye ointment, systemic acyclovir for systemic involvement in consultation with a paediatrician
> 1 weeks	Chlamydia trachomatis (D- K)	Erythromycin or chlortetracycline eye ointment, oral erythromycin for systemic infection

In the past, if a maternal infection was suspected a drop of silver nitrate (AgNO_3) solution 1% Q was instilled in to each eye (Crede's method) to prevent ophthalmia neonatorum. Chemical conjunctivitis used to be seen with in a few hours of prophylactic topical treatment with AgNO_3 , and disappeared spontaneously in 24-36 hours. It is diagnosed by watery discharge with absence of chemosis and edema which indicate towards infective pathology.

GIANT PAPILLARY CONJUNCTIVITIS:

- Severe mechanically induced papillary conjunctivitis
- Causes:
 - i) Contact lens → CL associated Papillary conjunctivitis
 - ii) Ocular prosthesis
 - iii) Corneal surface irregularity

≡ Mucus Fishing Syndrome → Patient develop or exacerbate a chronic papillary Rn. d/t a repetitive manual removal of mucus

XEROPHTHALMIA: { Epithelial Xerosis }

- ↳ covers all the ocular manifestations of Vit. A deficiency
- Xero epithelial = ↓sed no. of Goblet cell
- ≡ Hypokeratosis of squamous epithelium.

WHO Grading:

- a) XN: Night blindness ← 1st symptom (nyctalopia)
- b) X1A: Conjunctival Xerosis ← 1st sign
- c) X1B: Bitots spot ← Pathognomic feature
- d) X2: Corneal Xerosis
- e) X3: Keratomalacia
 - 3A: < 1/3rd
 - 3B: > 1/3rd
- f) X5: Corneal Scarring → Opacity of Cornea
- g) XF: Fundus changes; white spotted fundus.

CAUSES OF NIGHT BLINDNESS:

- 1) Vit. A deficiency
- 2) High Myopia
- 3) Retinitis Pigmentosa
- 4) Fove - Goldman Fove Sx
- 5) Late PDA
- 6) Oguchi d/s

Conjunctiva in Vit. A deficiency

- 1) Squamous Metaplasia
- 2) Keratosis → Hyperplasia of squamous epithelium
- 3) ↓ in goblet cell density.

Vitamin A therapy

Prevention

Immunisation :- 9 doses given
Starting from 9 months

- 1st → 9 months : 1 Lakh IU drops along with measles
- 2nd → 18 months : 2 Lakh → along with DPT booster

2yr = 2L
2.5yr = 2L
3yr = 2L
3.5yr = 2L
4yr = 2L
4.5yr = 2L
5yr = 2L

→ maximum age | maximum dose = 17 Lakh IU

Rx

3 doses
0, 1, 14 days

oral

- < 6 months ⇒ 50,000 IU
- 6m - 1yr ⇒ 1 Lakh
- > 1yr ⇒ 2 Lakh

i.m = oral/2

Oculo - Cutaneous Syndromes

- Inflammation of conjunctiva
⊕
Mucous membrane of mouth, nose, urethra
- Includes → STEVEN JOHNSON Sx
OCULAR CATERACTIAL PEMPHIGOID
- Type III H.S → Ag - Ab Rn.
- (MC) cause intake of Sulfa drugs
- cicatrization of conjunctiva

Sequence ::

- a) **SYMBLEPHARON** ⇒ Complete adhesion of the palpebral conjunctiva of the eyelid to the bulbar conjunctiva of eyeball.

Course ::

- Chemical injuries (alkali)
- Burns
- Ocular cicatricial pemphigoid
- SJS
- Erythema multiforme
- Pemphigus bulbar
- Membranous conjunctiva

Mx : Preventions

- Lubrication
- swapping of fornices
- Symplecton rings

b) Corneal ulcer

c) Pseudomembrane

d) Dry eye



Rx ::

- Symblepharone & amniotic membrane graft

Superior Limbal Keratoconjunctivitis (SLK)

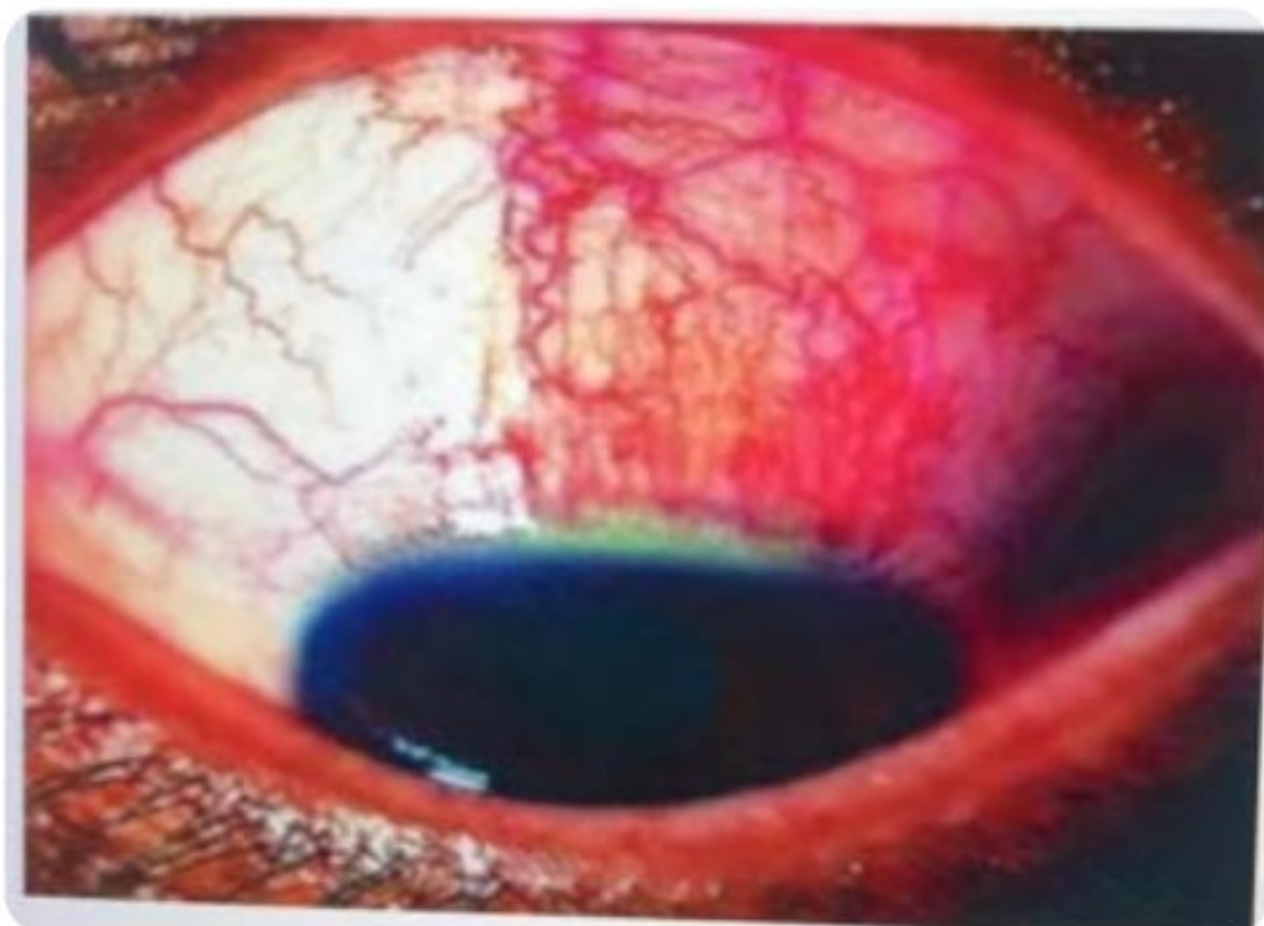
- CLC disease of superior limbus and the superior bulbar and tarsal conjunctiva
- ♀ (40-50yr), B/L
- a/w Ab (N) Thyroid function → Hyperthyroidism (50%)

Features ::

- 1) Papillary hypertrophy of the superior tarsal plate → diffuse velvety appearance
- 2) Hyperemia of superior bulbar conjunctiva → stains with Rose Bengal
- 3) Superior filamentary keratitis develops in about one third of cases.

Management ::

- Lubricants
- Topical acetylcystein 5% or 10% → breakdown filament Q6H.
- Topical Rebamipide → A.A derivative for mucous protection.



SLK

Glues use in Ophthalm

• Cyanoacrylate Glue

↳ seal a/c corneal perforations.

Fibrin glue

↳ used in corneal and conjunctival surgeries

strabismic surgeries

cataract → to fix the haptics of IOL to tissue in place of sutures in glued IOL

Ocular Surface - squamous neoplasia : (MC) malignancy of conjunctiva

Rx: TOPICAL RETINOIC ACID

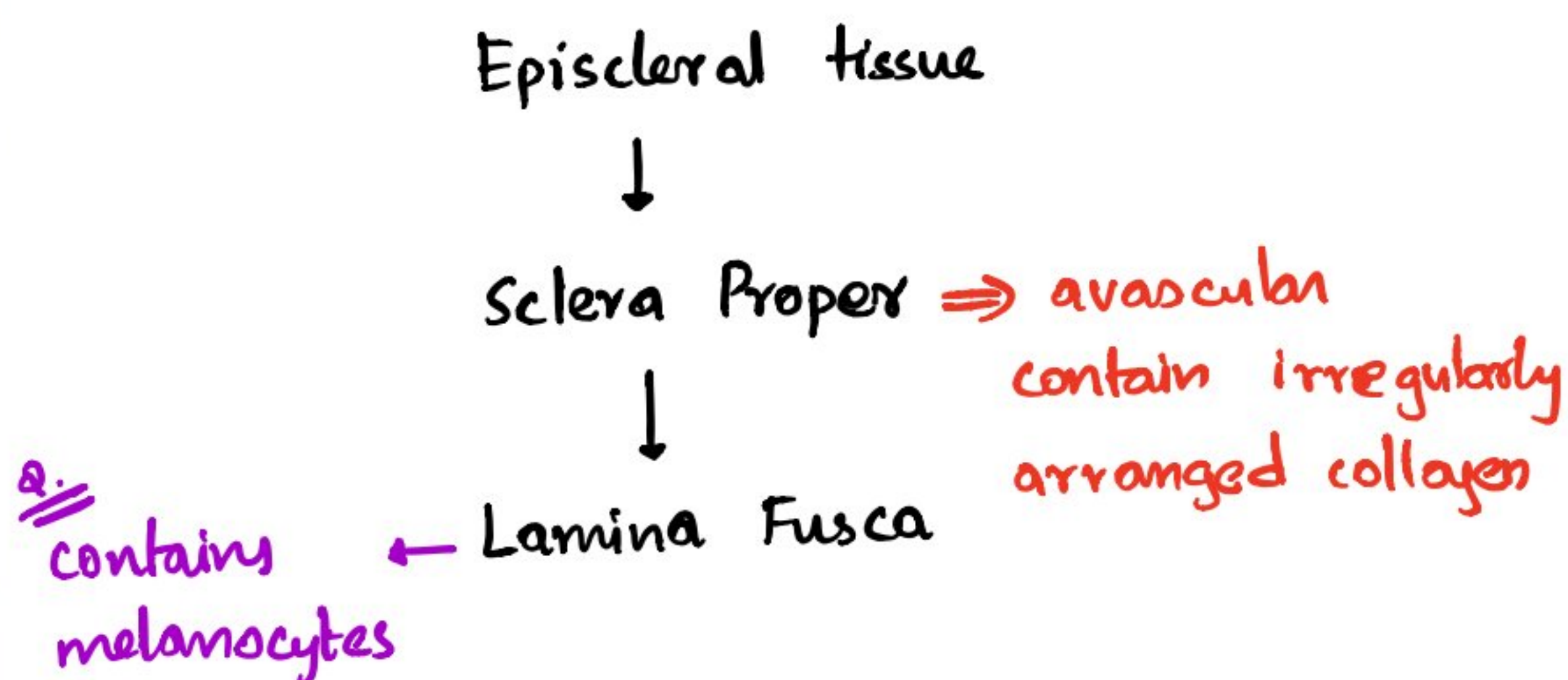
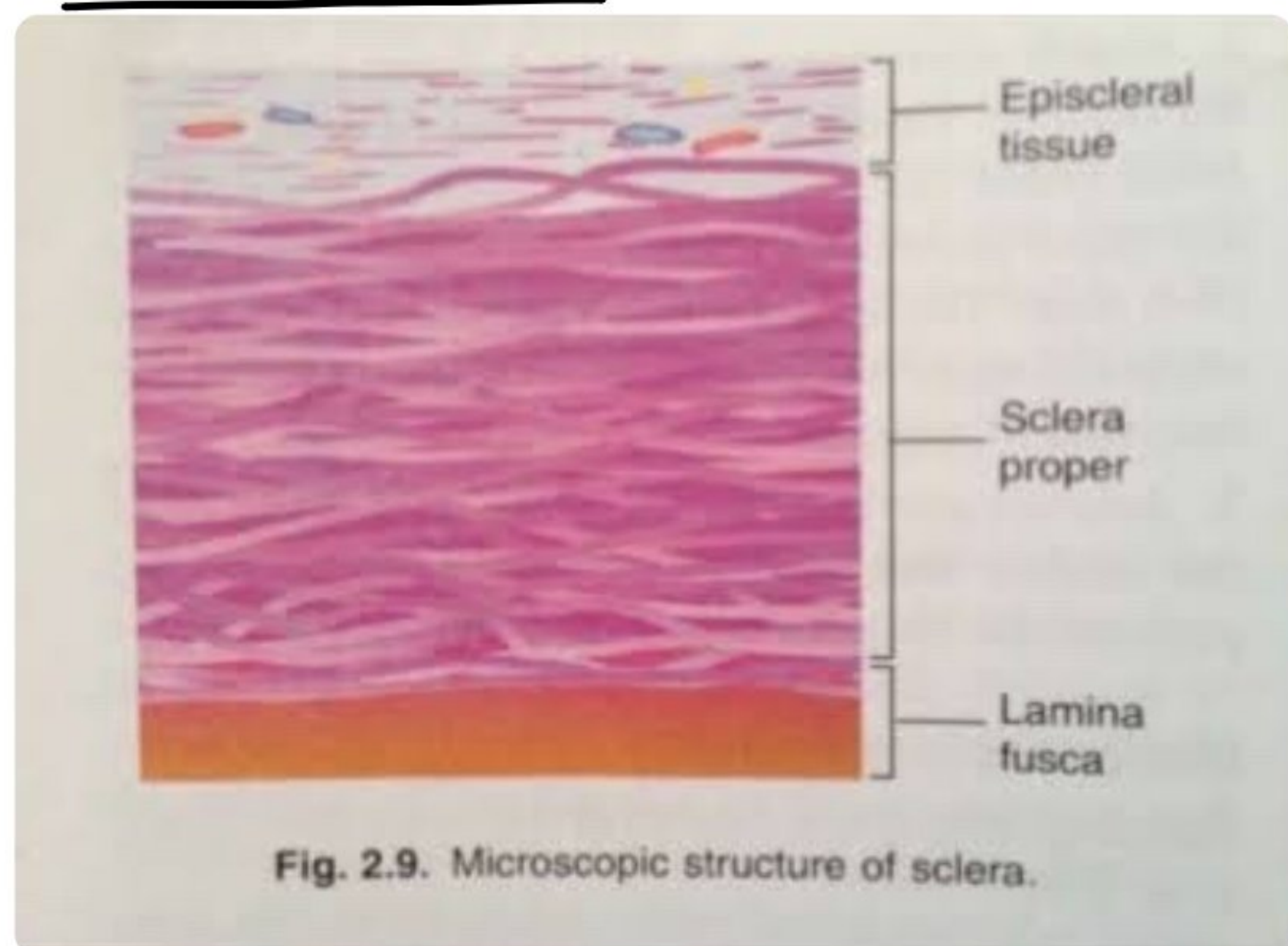
other uses of retinoic acid

- i) Manage severe, resistant, dry eye
- ii) Wound healing on ocular surface
- iii) ↓ ocular surface keratinization → d/t lined stem cell deficiency.

SCLERA:

- Outer coat of the eyeball & Posterior $5/6^{th}$
- Irregular arrangements of Collagen { Type I > 95% } ⇒ Opaque & white
Type III ≤ 5% }

LAYERS OF SCLERA:



Features:

- 1mm thick posteriorly, with maximum thickness around the optic Nerve.
- 0.6mm at the insertion of the rectus muscle
- 0.3mm posterior to attachment rectus muscle.